

STORIES OF
USEFUL
INVENTIONS



S. B. FORMAN

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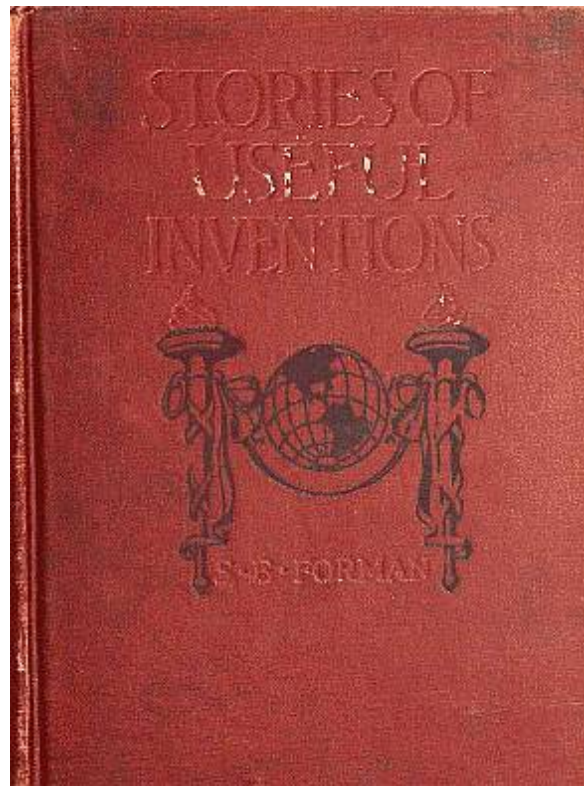
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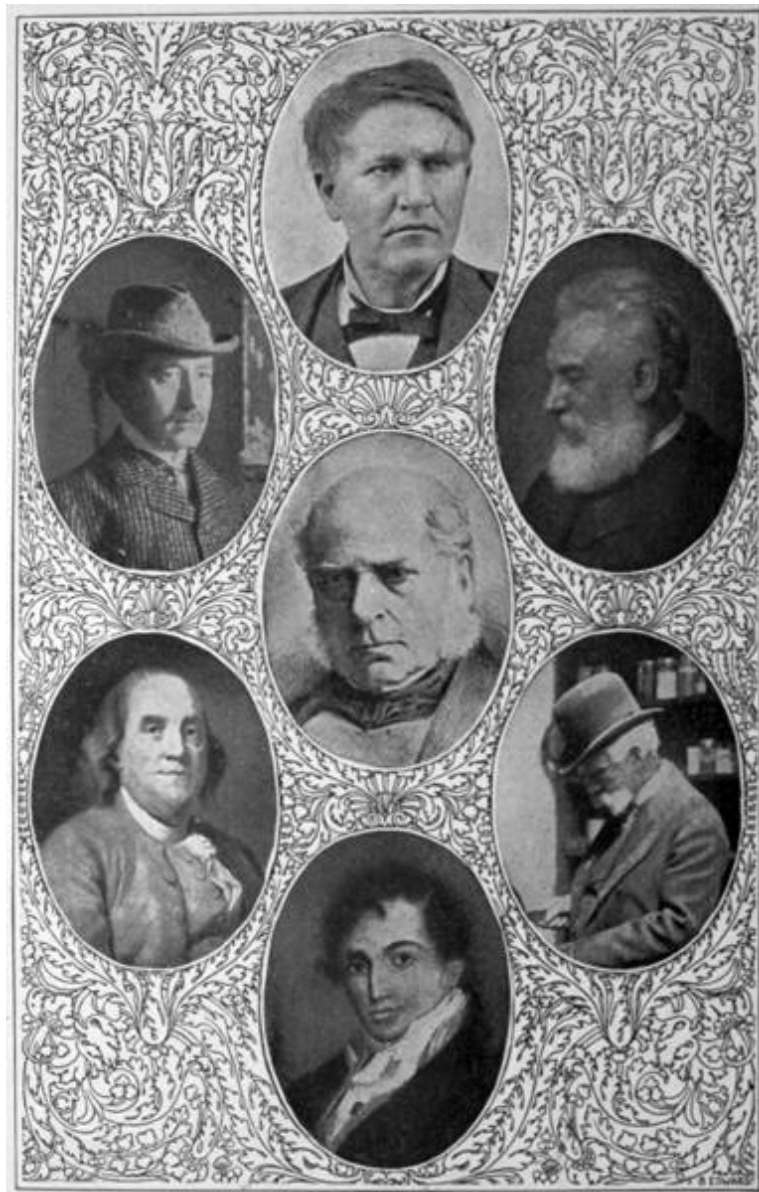
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*** START OF THE PROJECT GUTENBERG EBOOK STORIES OF
USEFUL INVENTIONS ***



STORIES OF USEFUL INVENTIONS



Thomas Edison
Alexander Graham Bell
Guglielmo Marconi
Sir Henry Bessemer
Benjamin Franklin
Hudson Maxim
Robert Fulton

A GROUP OF INVENTORS

STORIES OF USEFUL INVENTIONS

BY
S. E. FORMAN

AUTHOR OF "A HISTORY OF THE UNITED STATES,"
"ADVANCED CIVICS," ETC.



NEW YORK
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1911

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PREFACE

In this little book I have given the history of those inventions which are most useful to man in his daily life. I have told the story of the Match, the Stove, the Lamp, the Forge, the Steam-Engine, the Plow, the Reaper, the Mill, the Loom, the House, the Carriage, the Boat, the Clock, the Book, and the Message. From the history of these inventions we learn how man became the master of the world of nature around him, how he brought fire and air and earth and water under his control and compelled them to do his will and his work. When we trace the growth of these inventions we at the same time trace the course of human progress. These stories, therefore, are stories of human progress; they are chapters in the history of civilization.

And they are chapters which have not hitherto been brought together in one book. Monographs on most of the subjects included in this book have appeared, and excellent books about modern inventions have been written, but as far as I know, this is the first time the evolution of these useful inventions has been fully traced in a single volume.

While preparing the stories I have received many courtesies from officers in the Library of Congress and from those of the National Museum.

S. E. F.

May, 1911.

Washington, D. C.

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A FOREWORD¹

These stories of useful inventions are chapters in the history of civilization and this little book is a book of history. Now we are told by Herodotus, one of the oldest and greatest of historians, that when the writer of history records an event he should state the *time* and the *place* of its happening. In some kinds of history—in the history of the world's wars, for example, or in the history of its politics—this is strictly true. When we are reading of the battle of Bunker Hill we should be told precisely when and where the battle was fought, and in an account of the Declaration of Independence the time and place of the declaration should be given. But in the history of inventions we cannot always be precise as to dates and places. Of course it cannot be told when the first plow or the first loom or the first clock was made. Inventions like these had their origin far back in the earliest ages when there was no such person as a historian. And when we come to the history of inventions in more recent times the historian is still sometimes unable to discover the precise time and place of an invention.

It is in the nature of things that the origin of an invention should be surrounded by uncertainty and doubt. An invention, as we shall see presently, is nearly always a response to a certain want. The world wants something and it promises a rich reward to one who will furnish the desired thing. The inventor, recognizing the want, sets to work to make the thing, but he conducts his experiments in secret, for the reason that he does not want another to steal his ideas and get ahead of him. We can see that this is true in respect to the flying machine. The first experiments with the flying machine were conducted in secret in out of the way places and pains were taken that the public should know as little as possible about the new machine and about the results of the experiments. The history of the flying machine will of course have to be written, but because of the secrecy and mystery which surrounded the beginnings of the invention it will be extremely difficult for the future historian to tell precisely when the first flying machine was invented or to name the inventor. If it is so difficult to get the facts as to the origin of an invention in our own time, how much

more difficult it is to clear away the mystery and doubt which surround the beginnings of an invention in an age long past!

In a history of inventions, then, the historian cannot be precise in respect to dates and places. Fortunately this is not a cause for deep regret. It is not a great loss to truth that we cannot know precisely when the first book was printed, nor does it make much difference whether that book was printed in Holland or in Germany. In giving an account of an invention we may be content to treat the matter of time and place broadly, for the story is apt to carry us through a stretch of years that defies computation, a stretch that is immensely longer than the life of any nation. For our purpose these millenniums, these long stretches of time, may be thought of as being divided into three great periods, namely: the *primitive*, the *ancient*, and the *modern* period. Even a division so broad as this is not satisfactory, for in the progress of their inventions all countries have not kept equal step with the march of time. In some things ancient Greece was modern, while in most things modern Alaska is primitive and modern China is ancient. Nevertheless it will be convenient at times in this book to speak of the *primitive*, the *ancient* and the *modern* periods, and it will be useful to regard the *primitive* period as beginning with the coming of man on earth and extending to the year 5000 B. C.; the *ancient* period may be thought of as beginning with the year 5000 B. C. and ending with the year 476 A. D., leaving for the *modern* period the years that have passed since 476 A. D.

In tracing the growth of an invention the periods indicated above can serve as a time-guide only for those parts of the world where the course of civilization has taken its way, for invention and civilization have traveled the same road. The region of the world's most advanced civilization includes the lands bordering on the Mediterranean Sea, Central and Northern Europe, the British Isles, North America, South America and Australia. It is within this region that we shall follow the development of whatever invention is under consideration. When speaking of the first forms of an invention, however, it will sometimes be necessary, when an illustration is desired, to draw upon the experience of people who are outside of the wall of civilization. The reason for going outside is plain. The first and simplest forms of the useful inventions have utterly perished in civilized countries, but they still exist among savage and barbarous peoples and it is among such peoples that the first forms must be studied. Thus in the story of the clock, we must go to a

far-off peninsula of Southern Asia (p. 190) for an illustration of the beginning of our modern timepiece. Such a departure from the beaten track of civilization does not spoil the story, for as a rule, the rude forms of inventions found among the lowest races of to-day are precisely the same forms that were in use among the Egyptians and Greeks when they were in their lowest state.

When studying the history of an invention there are two facts or principles which should ever be borne in mind. The first principle is this: *Necessity is the mother of invention*. This principle was touched upon when it was said that an invention appears as a response to a want. When the world wants an invention it usually gets it and makes the most of it, but it will have nothing to do with an invention it does not want. The steam-engine was invented two thousand years ago (p. 55) but the world then had no work for steam to do, so the invention attracted little attention and came to naught. About two hundred years ago, however, man did want the services of steam and inventors were not long in supplying the engine that was needed. About a hundred years ago the broad prairie lands of the United States began to be tilled but it was soon found that the vast areas could not be plowed and that the immense crops could not be harvested by the old methods. So improvements upon the plow and the reaper began to be made and in time the steam gang-plow and the complete harvester were invented. When the locomotive first came into use a simple handbrake was used to stop the slow-going trains, but as the size and the speed of trains increased the handbrake became more and more unsatisfactory. Sometimes a train would run as much as a half mile beyond a station before it could be stopped and then when "backed" it would again pass beyond the station. The problem of stopping the train promptly became fully as important as starting it. The problem was solved by the invention of the air-brake. And thus it has been with all the inventions which surround us: necessity has been the mother of them all.

The other principle is that a mechanical invention is a *growth*, or, to state the truth in another way, an invention nearly always is simply an improvement upon a previous invention. The loom, for example, was not invented by a particular person at a particular time; it did not spring into existence in a day with all its parts perfected; it *grew*, century by century, piece by piece. In the stories which will follow the steps in the growth of an

invention are shown in the illustrations. These pictures are not for amusement but for study. As you read, examine them carefully and they will teach you quite as much about the growth of the invention as you can be taught by words.



STORIES OF USEFUL INVENTIONS

THE MATCH

Did you ever think how great and how many are the blessings of fire? Try to think of a world without fire. Suppose we should wake up some bitter cold morning and find that all the fires in the world were out, and that there was no way of rekindling them; that the art of kindling a fire had been lost. In such a plight we should all soon be shivering with the cold, for our stoves and furnaces could give us no warmth; we should all soon be hungry, for we could not cook our food; we should all soon be idle, for engines could not draw trains, wheels of factories could not turn, and trade and commerce would come to a standstill; at night we would grope in darkness, for we could use neither lamp nor gas nor electric light. It is easy to see that without fire, whether for light or heat, the life of man would be most wretched.

There never was a time when the world was without fire, but there was a time when men did not know how to kindle fire; and after they learned how to kindle one, it was a long, long time before they learned how to kindle one easily. In these days we can kindle a fire without any trouble, because we can easily get a match; but we must remember that the match is one of the most wonderful things in the world, and that it took men thousands of years to learn how to make one. Let us learn the history of this familiar little object, the match.

Fire was first given to man by nature itself. When a forest is set on fire by cinders from a neighboring volcano, or when a tree is set ablaze by a thunderbolt, we may say that nature strikes a match. In the early history of the world, nature had to kindle all the fires, for man by his own effort was unable to produce a spark. The first method, then, of getting fire for use was to light sticks of wood at a flame kindled by nature—by a volcano, perhaps, or by a stroke of lightning. These firebrands (Fig. 1) were carried to the home and used in kindling the fires there. The fire secured in this way was carefully guarded and was kept burning as long as possible. But the flame,

however faithfully watched, would sometimes be extinguished. A sudden gust of wind or a sudden shower would put it out. Then a new firebrand would have to be secured, and this often meant a long journey and a deal of trouble.



FIG. 1.—GETTING A MATCH FROM NATURE.

In the course of time a man somewhere in the world hit upon a plan of kindling a fire without having any fire to begin with; that is to say, he hit upon a plan of producing a fire by *artificial* means. He knew that by rubbing his hands together very hard and very fast he could make them very warm. By trial he learned that by rubbing two pieces of dry wood together he could make *them* very warm. Then he asked himself the question: Can a fire be kindled by rubbing two pieces of wood together, if they are rubbed hard enough? He placed upon the ground a piece of perfectly dry wood (Fig. 2) and rubbed this with the end of a stick until a groove was made. In the groove a fine dust of wood—a kind of sawdust—was made by the rubbing. He went on rubbing hard and fast, and, behold, the dust in the groove began to glow! He placed some dry grass upon the embers and blew upon them with his breath, and the grass burst into a flame.² Here for the first time a man kindled a fire for himself. He had invented the match, the greatest invention, perhaps, in the history of the world.



FIG. 2.—PRIMITIVE FIRE-MAKING. THE STICK-AND-GROOVE METHOD.

The stick-and-groove method—as we may call it—of getting a flame was much better than guarding fire and carrying it from place to place; yet it was, nevertheless, a very clumsy method. The wood used had to be perfectly



FIG. 3.—THE FIRE DRILL.
(Simple Form.)

dry, and the rubbing required a vast amount of work and patience. Sometimes it would take hours to produce the spark. After a while—and doubtless it was a very long while—it was found that it was better to keep the end of the stick in one spot and twirl it (Fig. 3) than it was to plow to and fro with it. The



FIG. 4.—FIRE DRILL.
(Improved Form.)

twirling motion made a hole in which the heat produced by the friction was confined in a small space.

At first the drilling was done by twirling the stick between the palms of the hands, but this made the hands too hot for comfort, and the fire-makers learned to do the twirling with a cord or thong³ wrapped around the stick (Fig. 4). You see, the upper end of the stick which serves as a drill turns in a cavity in a mouthpiece which the operator holds between his teeth. If you should undertake to use a fire-drill of this kind, it is likely that your jaws would be painfully jarred.

By both the methods described above, the fire was obtained by rubbing or *friction*. The friction method seems to have been used by all primitive peoples, and it is still in use among savages in various parts of the world.



FIG. 5.—
STRIKING FIRE.

The second step in fire-making was taken when it was discovered that a spark can be made by striking together a stone and a piece of iron ore. Strike a piece of flint against a piece of iron ore known as pyrites, or fire-stone, and you will make sparks fly. (Fig. 5.) Let these sparks fall into small pieces of dried moss or powdered charcoal, and the *tinder*, as the moss or the charcoal is called, will catch fire. It will glow, but it will not blaze. Now hold a dry splinter in the glowing tinder, and fan or blow with the breath and the splinter will burst into a

flame. If you will tip your splinter with sulphur before you place it in the burning tinder, you will get a flame at once. This was the strike-a-light, or *percussion*, method of making a fire. It followed the friction method, and was a great improvement upon it because it took less work and a shorter

time to get a blaze. The regular outfit for fire-making with the strike-a-light consisted of a tinder-box, a piece of steel, a piece of flint, and some splinters tipped with sulphur (Fig. 6). The flint and steel were struck together, and the sparks thus made fell into the tinder and made it glow. A splinter was applied as quickly as possible to the tinder, and when a flame was produced the candle which rested in the socket on the tinder-box was lighted. As soon as the splinter was lighted the cover was replaced on the tinder-box, so as to smother the glowing tinder and save it for another time.

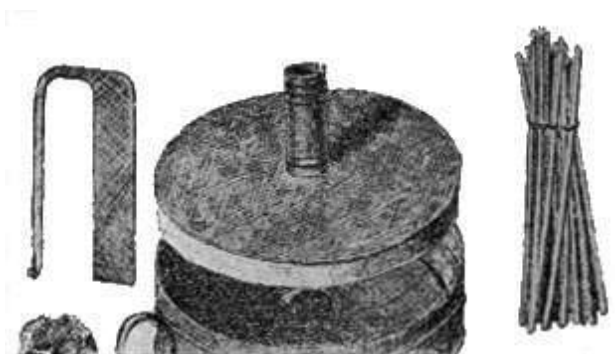


FIG. 6.—TINDER BOX, FLINT, STEEL, AND SULPHUR-TIPPED SPLINTERS.

The strike-a-light method was discovered many thousands of years ago, and it has been used by nearly all the civilized nations of the world.⁴ And it has not been so very long since this method was laid aside. There are many people now living who remember when the flint and steel and tinder-box were in use in almost every household.

About three hundred years ago a third method of producing fire was discovered. If you should drop a small quantity of sulphuric acid into a mixture of chlorate of potash and sugar, you would produce a bright flame. Here was a hint for a new way of making a fire; and a thoughtful man in Vienna, in the seventeenth century, profited by the hint. He took one of the sulphur-tipped splinters which he was accustomed to use with his tinder-box, and dipped it into sulphuric acid, and then applied it to a mixture of chlorate of potash and sugar. The splinter caught fire and burned with a blaze. Here was neither friction nor percussion. The chemical substances were simply brought together, and they caught fire of themselves; that is to say, they caught fire by *chemical* action.

The discovery made by the Vienna man led to a new kind of match—the chemical match. A practical outfit for fire-making now consisted of a bottle of sulphuric acid (vitriol) and a bundle of splints tipped with sulphur, chlorate of potash, and sugar. Matches of this kind were very expensive, costing as much as five dollars a hundred; besides, they were very unsatisfactory. Often when the match was dipped into the acid it would not catch fire, but would smolder and sputter and throw the acid about and spoil both the clothes and the temper. These dip-splint matches were used in the eighteenth century by those who liked them and could afford to buy them. They did not, however, drive out the old strike-a-light and tinder-box.

In the nineteenth century—the century in which so many wonderful things were done—the fourth step in the development of the match was taken. In 1827, John Walker, a druggist in a small English town, tipped a splint with sulphur, chlorate of potash, and sulphid of antimony, and rubbed it on sandpaper, and it burst into flame. The druggist had discovered the first *friction-chemical* match, the kind we use to-day. It is called friction-chemical because it is made by mixing certain chemicals together and rubbing them. Although Walker's match did not require the bottle of acid, nevertheless it was not a good one. It could be lighted only by hard rubbing, and it sputtered and threw fire in all directions. In a few years, however, phosphorus was substituted on the tip for antimony, and the change worked wonders. The match could now be lighted with very little rubbing, and it was no longer necessary to have sandpaper upon which to rub it. It would ignite when rubbed on any dry surface, and there was no longer any sputtering. This was the *phosphorus* match, the match with which we are so familiar.

After the invention of the easily-lighted phosphorus match there was no longer use for the dip-splint or the strike-a-light. The old methods of getting a blaze were gradually laid aside and forgotten. The first phosphorus matches were sold at twenty-five cents a block—a block (Fig. 7) containing a hundred and forty-four matches. They were used by few. Now a hundred matches can be bought for a cent. It is said that in the United States we use about 150,000,000,000 matches a year. This,



FIG. 7.—A "BLOCK" OF

on an average, is about five matches a day for each person.

MATCHES.

There is one thing against the phosphorus match: it ignites too easily. If one is left on the floor, it may be ignited by stepping upon it, or by something falling upon it. We may step on a phosphorus match unawares, light it, leave it burning, and thus set the house on fire. Mice often have caused fires by gnawing the phosphorus matches and igniting them. In one city thirty destructive fires were caused in one year by mice lighting matches.

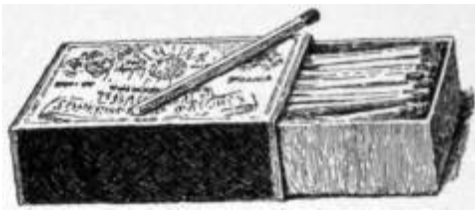


FIG. 8.—A BOX OF MODERN SAFETY MATCHES.

To avoid accident by matches, the *safety match* (Fig. 8) has recently been invented. The safety match does not contain phosphorus. The phosphorus is mixed with fine sand and glued to the side of the box in which the matches are sold. The safety match, therefore, cannot be lighted unless it is rubbed on the phosphorus on the outside of the box. It is so much better than the old kind of phosphorus match that it is driving the latter out of the market. Indeed, in some places it is forbidden by law to sell any kind of match but the safety match.

The invention of the safety match is the last step in the long history of fire-making. The first match was lighted by rubbing, and the match of our own time is lighted by rubbing; yet what a difference there is between the two! With the plowing-stick or fire-drill it took strength and time and skill to get a blaze; with the safety match an awkward little child can kindle a fire in a second.

And how long it has taken to make the match as good as it is! The steam-engine, the telegraph, the telephone, and the electric light were all in use before the simple little safety match.

THE STOVE

From the story of the match you have learned how man through long ages of experience gradually mastered the art of making a fire easily and quickly. In this chapter, and in several which are to follow, we shall have the history of those inventions which have enabled man to make the best use of fire. Since the first and greatest use of fire is to cook food and keep the body warm, our account of the inventions connected with the use of fire may best begin with the story of the stove.

The most important uses of fire were taught by fire itself. As the primitive man stood near the flames of the burning tree and felt their pleasant glow, he learned that fire may add to bodily comfort; and when the flames swept through a forest and overtook a deer and baked it, he learned that fire might be used to improve the quality of his food. The hint was not lost. He took a burning torch to his cave or hut and kindled a fire on his floor of earth. His dwelling filled with smoke, but he could endure the discomfort for the sake of the fire's warmth, and for the sake of the toothsome of the cooked meats. After a time a hole was made in the roof of the hut, and through this hole the smoke passed out. Here was the first stove. The primitive stove was the entire house; the floor was the fireplace and the hole in the roof was the chimney (Fig. 1). The word "stove" originally meant "a heated room." So that if we should say that at first people lived in their stoves, we should say that which is literally true.



FIG. 1.—THE PRIMITIVE STOVE.

Early inventions in cooking consisted in simple devices for applying flame directly to the thing which was to be cooked. The first roasting was doubtless done by fastening the flesh to a pole placed in a horizontal position above the fire and supported as is shown in Figure 2.⁵ The



FIG. 2.—PRIMITIVE COOKING.

horizontal bar called a spit was originally of wood, but after man had learned to work in metals an iron bar was used. When one side of the flesh was roasted the spit was turned and the other side was exposed to the flames. The spit of the primitive age was the parent of the modern grill and broiler.

Food was first boiled in a hole in the ground. A hole was filled with water into which heated stones were thrown. The stones, by giving off their heat, caused the water to boil in a very short time. After the art of making vessels of clay was learned, food was boiled in earthen pots suspended above the fire.

The methods of warming the house and cooking the food which have just been described were certainly crude and inconvenient, but it was thousands of years before better methods were invented. The long periods of savagery and barbarism passed and the period of civilization was ushered in, but civilization did not at once bring better stoves. Neither the ancient Egyptians nor the ancient Greeks knew how to heat a house comfortably and conveniently. All of them used the primitive stove—a fire on the floor and a hole in the roof. In the house of an ancient Greek there was usually one room which could be heated when there was need, and this was called the "black-room" (*atrium*)—black from the soot and smoke which escaped from the fire on the floor.

But we must not speak harshly of the ancients because they were slow in improving their methods of heating, for in truth the modern world has not done as well in this direction as might have been expected. In a book of travels written only sixty years ago may be found the following passage: "In Normandy, where the cold is severe and fire expensive, the lace-makers, to keep themselves warm and to save fuel, agree with some farmer who has cows in winter quarters to be allowed to carry on their work in the society of the cattle. The cows would be tethered in a long row on one side of the apartment, and the lace-makers sit on the ground on the other side with their feet buried in the straw." Thus the lace-makers kept themselves warm by the heat which came from the bodies of the cattle; the cows, in other words,

served as stoves. This barbarous method of heating was practised in some parts of France less than sixty years ago.

The ancient peoples around the Mediterranean may be excused for not making great progress in the art of heating, for their climate was so mild that they seldom had use for fire in the house. Nevertheless there was in use among these people an invention which has in the course of centuries developed into the stove of to-day. This was the *brazier*, or warming-pan (Fig. 3). The brazier was filled with burning charcoal and was carried from room to room as it was needed. The unpleasant gases which escaped from the charcoal were made less offensive, but not less unhealthy, by burning perfumes with the fuel. The brazier has never been entirely laid aside. It is still used in Spain and in other warm countries where the necessity for fire is rarely felt.



FIG. 3.—A ROMAN BRAZIER.



FIG. 4.—A ROMAN HYPOCAUST.

The brazier satisfied the wants of Greece, but the colder climate of Rome required something better; and in their efforts to invent something better, the ancient Romans made real progress in the art of warming their houses. They built a fire-room—called a *hypocaust*—in the cellar, and, by means of pipes made of baked clay, they connected the hypocaust with different parts of the house (Fig. 4). Heat and smoke passed up together through these pipes. The poor ancients, it seems, were forever persecuted by smoke. However, after the wood in the hypocaust was once well charred, the smoke was not so troublesome. The celebrated baths (club-rooms) of ancient Rome were heated by means of hypocausts with excellent results. Indeed, the hypocaust had many of the features and many of the merits of our modern furnace. Its weak feature was that it had no separate pipe to carry away the smoke. But as there were no chimneys yet in the world, it is no wonder there was no such pipe.

The Romans made quite as much progress in the art of cooking as they did in the art of heating. Perhaps the world has never seen more skilful cooks than those who served in the mansions of the rich during the period of the Roman Empire (27 B.C.-476 A.D.). In this period the great men at Rome abandoned their plain way of living and became gourmands. One of them wished for the neck of a crane, that he might enjoy for a longer time his food as it descended. This demand for tempting viands developed a race of cooks who were artists in their way. Upon one occasion a king called for a certain kind of fish. The fish could not be had, but the cook was equal to the emergency. "He cut a large turnip to the perfect imitation of the fish desired, and this he fried and seasoned so skilfully that his majesty's taste was exquisitely deceived, and he praised the root to his guests as an excellent fish." Such excellent cooking could not be done on a primitive stove, and along with the improvements in the art of cooking, there was a corresponding improvement at Rome in the art of stove-making.

When Rome fell (476 A.D.), many of the best features of her civilization perished with her. Among the things that were lost to the world were the Roman methods of cooking and heating. When the barbarians came in at the front door, the cooks fled from the kitchen. The hardy northerners had no taste for dainty cooking. Hypocausts ceased to be used, and were no longer built. For several hundred years, in all the countries of Europe, the fireplace was located, as of old, on the floor in the center of the room, while the smoke was allowed to pass out through a hole in the roof.

The eleventh century brought a great improvement in the art of heating, and the improvement came from England. About the time of the Conquest (1066) a great deal of fighting was done on the roofs of English fortresses, and the smoke coming up through the hole in the center of the roof proved to be troublesome to the soldiers. So the fire was moved from the center of the floor to a spot near an outside wall, and an opening was made in the wall just above the fire, so that the smoke could pass out. Here was the origin of the *chimney*. Projecting from the wall above the fire was a hood, which served to direct the smoke to the opening. At first the opening for the smoke extended but a few feet from the fire, but it was soon found

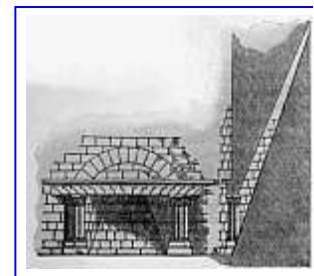


FIG. 5.—A CHIMNEY AND FIREPLACE IN AN OLD ENGLISH CASTLE.

that the further up the wall the opening extended the better was the draft. So the chimney was made to run diagonally up the wall as far as possible. The next and last step in the development of the chimney was to make a recess in the wall as a fireplace, and to build a separate structure of masonry—the chimney—for the smoke. By the middle of the fourteenth century chimneys were usually built in this way (Fig. 5). As the fireplace and chimney cleared the house of soot and smoke, they grew in favor rapidly. By the end of the fifteenth century they were found in the homes of nearly all civilized people.



FIG. 6.—A STOVE OF THE MIDDLE AGES.

The open fireplace was always cheerful, and it was comfortable when you were close to it; but it did not heat all parts of the room equally. That part next to the fireplace might be too warm for comfort, while in another part of the room it might be freezing. About the end of the fifteenth century efforts were made to distribute heat throughout the room more evenly. These efforts led to the invention of the modern stove. We have learned that the origin of the stove is to be sought in the ancient brazier. In the middle ages the brazier in France took on a new form. Here was a fire-box (Fig. 6) with openings at the bottom for drafts of air and arrangements at the top for cooking things. This French warming-pan (*réchaud*) was the connecting-link between the ancient brazier and the modern stove. All it lacked of being a stove was a pipe to carry off the smoke, and this was added by a Frenchman named Savot, about two hundred years ago. We owe the invention of the chimney to England, but for the stove we are indebted to France. The Frenchman built an iron fire-box, with openings for drafts, and connected the box with the chimney by means of an iron flue or pipe. Here was a *stove* which could be placed in the middle of the room, or in any part of the room where it was desirable, and which would send out its heat evenly in all directions.

The first stoves were, of course, clumsy and unsatisfactory; but inventors kept working at them, making them better both for cooking and for heating. By the middle of the nineteenth century the stove was practically what it is to-day (Fig. 7). Stoves proved to be so much better than fireplaces, that the latter were gradually replaced in large part by the former.



FIG. 7.—THE MODERN STOVE.

Our affection, however, for a blazing fire is strong, and it is not likely that the old-fashioned fireplace (Fig. 8) will ever entirely disappear.



FIG. 8.—AN OLD-FASHIONED FIREPLACE AND OVEN.

The French stove just described is intended to heat only one room. If a house with a dozen rooms is to be heated, a dozen stoves are necessary. About one hundred years ago there began to appear an invention by which a house of many rooms could be heated by means of one stove. This invention was the *furnace*. Place in the cellar a large stove, and run pipes from the stove to the different rooms of the house, and you have a furnace (Fig. 9). Doubtless we got our idea of the furnace from the Roman hypocaust, although the Roman invention had no special pipe for the smoke. The first furnaces sent out only hot air, but in recent years steam or hot water is sent out through the pipes to *radiators*, which are simply secondary stoves set up in convenient places and at a distance from the source of the heat, the furnace in the cellar. Furnaces were invented for the purpose of heating large buildings, but they are now used in ordinary dwellings.



FIG. 9.—A MODERN

FURNACE.

In its last and most highly developed form, the stove appears not only without dust and smoke, but also without even a fire in the cellar. The modern *electric* stove, of course, is meant. Pass a slight current of electricity through a piece of platinum wire, and the platinum becomes hot. You have made a diminutive electric stove. Increase the strength of your current and pass it through something which offers greater resistance than the platinum, and you get more heat. The electric stove is a new invention, and at present it is too expensive for general use, although the number of houses in which it is used is rapidly increasing, and in time it may drive out all other kinds of stoves. It will certainly drive all of them out if the cost of electricity shall be sufficiently reduced; for it is the cleanest, the healthiest, the most convenient, and the most easily controlled of stoves.

THE LAMP

Next to its usefulness for heating and cooking, the greatest use of fire is to furnish light to drive away darkness. Man is not content, like birds and brutes, to go to sleep at the setting of the sun. He takes a part of the night-time and uses it for work or for travel or for social pleasures, or for the improvement of his mind, and in this way adds several years to life. He could not do this if he were compelled to grope in darkness. When the great source of daylight disappears he must make light for himself, for the sources of night-light—the moon and stars and aurora borealis and lightning—are not sufficient to satisfy his wants. In this chapter we shall follow man in his efforts to conquer darkness, and we shall have the story of the lamp.

We may begin the story with an odd but interesting kind of lamp. The firefly or lightning-bug which we see so often in the summer nights was in the earliest time brought into service and made to shed its light for man. Fireflies were imprisoned in a rude box—in the shell of a cocoanut, perhaps, or in a gourd—and the light of their bodies was allowed to shoot out through the numerous holes made in the box. We must not despise the light given out by these tiny creatures. "In the mountains of Tijuca," says a traveler, "I have read the finest print by the light of one of these natural lamps (fireflies) placed under a common glass tumbler (Fig. 1), and with distinctness I could tell the hour of the night and discern the very small figures which marked the seconds of a little Swiss watch."

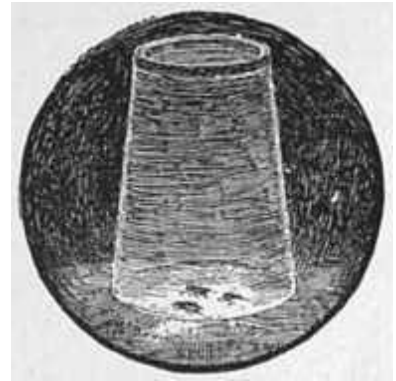


FIG. 1.—A FIREFLY LAMP.

Although fireflies have been used here and there by primitive folk, they could hardly have been the first lamp. Man's battle with darkness really began with the *torch*, which was lighted at the fire in the cave or in the wigwam and kept burning for purposes of illumination. A burning stick was the first lamp (Fig. 2). The first improvement in the torch was made when



FIG. 2.—A
BURNING
STICK WAS
THE FIRST
LAMP.

slivers or splinters of resinous or oily wood were tied together and burned. We may regard this as a lamp which is all wick. This invention resulted in a fuller and clearer light, and one that would burn longer than the single stick. A further improvement came when a long piece of wax or fatty substance was wrapped about with leaves. This was something like a candle, only the wick (the leaves) was outside, and the oily substance which fed the wick was in the center.

In the course of time it was discovered that it was better to smear the grease on the *outside* of the stick, or on the outside of whatever was to be burned; that is, that it was better to have the wick *inside*. Torches were then made of rope coated with resin or fat, or of sticks or splinters smeared with grease; here the stick resembled the wick of the candle as we know it to-day, and the coating of fat corresponded to the tallow or paraffin. Rude candles made of oiled rope or of sticks smeared with fat were invented in primitive times, and they continued to be used for thousands of years after men were civilized. In the dark ages—and they were dark in more senses than one—torch-makers began to wrap the central stick first with flax or hemp and then place around this a thick layer of fat. This torch gave a very good light, but about the time of Alfred the Great (900 A.D.) another step was taken: the central stick was left out altogether, and the thick layer of fat or wax was placed directly around the wick of twisted cotton. All that was left of the original torch—the stick of wood—was gone. The torch had developed into the *candle* (Fig. 3). The candles of to-day are made of better material than those of the olden time, and they are much cheaper; yet in principle they do not differ from the candles of a thousand years ago.



FIG. 3.—THE
CANDLE.

I have given the development of the candle first because its forerunner, the torch, was first used for lighting. But it must not be forgotten that along with the torch there was used, almost from the beginning, another kind of lamp. Almost as soon as men discovered that the melted fat of animals



FIG. 4.—A SHELL FILLED WITH OIL AND USED AS A LAMP.

would burn easily—and that was certainly very long ago—they invented in a rude form the *lamp* from which the lamp of to-day has been evolved. The cavity of a shell (Fig. 4) or of a stone, or of the skull of an animal, was filled with melted fat or oil, and a wick of flax or other fibrous material was laid upon the edge of the vessel. The oil or grease passed up the wick by capillary action,⁶

and when the end of the wick was lighted it continued to burn as long as there were both oil and wick. This was the earliest lamp. As man became more civilized, instead of a hollow stone or a skull, an earthen saucer or bowl was used. Around the edge of the bowl a gutter or spout was made for holding the wick. In the lamp of the ancient Greeks and Romans the reservoir which held the oil was closed, although in the center there was a hole through which the oil might be poured. Sometimes one of these lamps would have several spouts or nozzles. The more wicks a lamp had, of course, the more light it would give. There is in the museum at Cortona, in Italy, an ancient lamp which has sixteen nozzles. This interesting relic (Fig. 5) was used in a pagan temple in Etruria more than twenty-five hundred years ago.



FIG. 5.—AN ETRUSCAN LAMP 2500 YEARS OLD.

Lamps such as have just been described were used among the civilized peoples of the ancient world, and continued to be used through the Middle Ages far into modern times. They were sometimes very costly and beautiful

(Fig. 6), but they never gave a good light. They sent out an unpleasant odor, and they were so smoky that they covered the walls and furniture with soot. The candle was in every way better than the ancient lamp, and after the invention of wax tapers—candles made of wax—in the thirteenth century, lamps were no longer used by those who could afford to buy tapers. For ordinary purposes and ordinary people, however, the lamp continued to do service, but it was not improved. The eighteenth century had nearly passed, and the lamp was still the unsatisfactory, disagreeable thing it had always been.



FIG. 6.—AN ANCIENT LAMP.

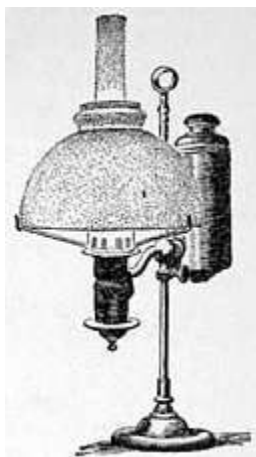


FIG. 7.—AN ARGAND LAMP.

Late in the eighteenth century the improvement came. In 1783 a man named Argand, a Swiss physician residing in London, invented a lamp that was far better than any that had ever been made before. What did Argand do for the lamp? Examine an ordinary lamp in which coal-oil is burned. The *chimney* protects the flame from sudden gusts of wind and also creates a draft of air,² just as the fire-chimney creates a draft. Argand's lamp (Fig. 7) was the first to have a chimney. Look below the chimney and you will see open passages through which air may pass upward and find its way to the wick. Notice further that as this draft of air passes upward it is so directed that, when the lamp is burning, an extra quantity of air plays directly upon the wick. Before Argand, the wick received no supply of air. Now notice—and this is very important—that the wick of our modern lamp is flat or circular, but *thin*. The air in abundance plays upon both sides of the thin wick, and burns it without making smoke. Smoke is simply half-burned particles (soot) of a burning substance. The particles pass off half-burned because enough air has not been supplied. Now Argand, by making the wick thin and by causing plenty of air to rush into the flame, caused all the wick to be burned and thereby caused it to burn with a white flame.

After the invention of Argand, the art of lamp-making improved by leaps and by bounds. More progress was made in twenty years after 1783 than had been made in twenty centuries before. New burners were invented,

new and better oils were used, and better wicks made. But all the new kinds of lamps were patterned after the Argand. The lamp you use at home may not be a real Argand, but it is doubtless made according to the principles of the lamp invented by the Swiss physician in 1783.

Soon after Argand invented his lamp, William Murdock, a Scottish inventor, showed the world a new way of lighting a house. It had long been known that fat or coal, when heated, gives off a vapor or gas which burns with a bright light. Indeed, it is *always* a gas that burns, and not a hard substance. In the candle or in the lamp the flame heats the oil which comes up to it through the wick and thus causes the oil to give off a gas. It is this gas that burns and gives the light. Now Murdock, in 1797, put this principle to a good use. He heated coal in a large vessel, and allowed the gas which was driven off to pass through mains and tubes to different parts of his house. Wherever he wanted a light he let the gas escape at the end of the tube (Fig. 8) in a small jet and lighted it. Here was a lamp without a wick. Murdock soon extended his gas-pipes to his factories, and lighted them with gas. As soon as it was learned how to make gas cheaply, and conduct it safely from house to house, whole cities were rescued from darkness by the new illuminant. A considerable part of London was lighted by gas in 1815. Baltimore was the first city in the United States to be lighted by gas. This was in 1821.

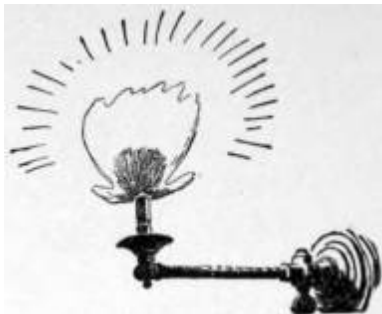


FIG. 8.—THE GAS JET.

The gas-light proved to be so much better than even the best of lamps, that in towns and cities almost everybody who could afford to do so laid aside the old wick-lamp and burned gas. About 1876, however, a new kind of light began to appear. This was the *electric* light. The powerful *arc light* (Fig. 9), made by the passage of a current of electricity between two carbon points, was the first to be invented. This gave as much light as a hundred gas-jets or several hundred lamps. Such a light was excellent for lighting streets, but its painful glare and its sputtering rendered it unfit for use within doors. It was

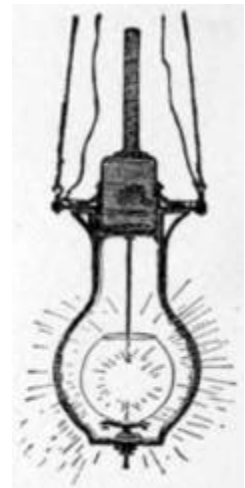


FIG. 9.—AN
EARLY ARC
LAMP.

not long, however, before an electric light was invented which could be used anywhere. This was the famous Edison's *incandescent* or glow lamp (Fig. 10), which we see on every hand. Edison's invention is only a few years old, yet there are already more than thirty million incandescent lamps in use in the United States alone.

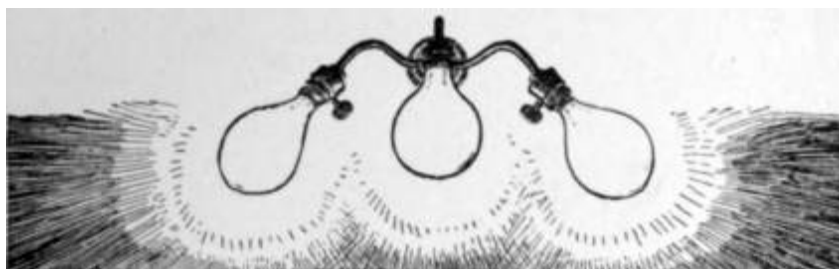


FIG. 10.—AN INCANDESCENT ELECTRIC LIGHT.

The torch, the candle, the lamp, the gas-light, the electric light,—these are the steps of the development of the lamp. And how marvelous a growth it is! How great the triumph over darkness! In the beginning a piece of wood burns with a dull flame, and fills the dingy wigwam or cave with soot and smoke; now, at the pressure of a button, the house is filled with a light that rivals the light of day, with not a particle of smoke or soot or harmful gas. Are there to be further triumphs in the art of lighting? Are we to have a light that shall drive out the electric light? Only time can tell.

THE FORGE

After men had learned how to use fire for cooking and heating and lighting they slowly learned how to use it when working with metals. In the earliest times metals were not used. For long ages stone was the only material that man could fashion and shape to his use. During this period, sometimes called the "stone age," weapons were made of stone; dishes and cooking utensils were made of stone; and even the poor, rude tools of the age were made of stone (Fig. 1).



FIG. 1.—IMPLEMENTS OF THE STONE AGE.

In the course of time man learned how to make his implements and weapons of metals as well as of stone. It is generally thought that bronze was the first metal to be used and that the "stone age" was followed directly by the "bronze age," a period when all utensils, weapons, and tools

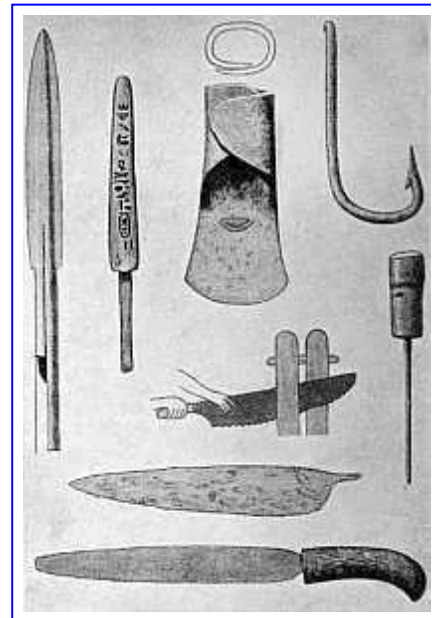


FIG. 2.—IMPLEMENTS OF THE BRONZE AGE.

were made of bronze (Fig. 2). It is easy to believe that bronze was used before iron, for bronze is made of a mixture of tin and copper and these two metals are often found in their pure or natural state. Whenever primitive man, therefore, found pieces of pure copper and tin, he could take the two metals and by melting them could easily mix them and make bronze of them. This bronze he could fashion to his use. There is no doubt that he did

this at a very early age. In nearly all parts of the world there are proofs that in primitive times, many articles were made of bronze.

If primitive man were slow to learn the use of iron it was not because this metal was scarce, for iron is everywhere. "Wherever, as we go up and down, we see a red-colored surface, or a reddish tint upon the solid substances of the earth, we see iron—the bank of red clay, the red brick, the red paint upon the house wall, the complexion of rosy youth, or my lady's ribbon. Even the rosy apple derives its tint from iron which it contains."⁸ But although iron is so abundant it is seldom found in its pure or natural state. It is nearly always mixed with other substances, the mixture being known as iron ore. Primitive man could find copper and tin in their pure state but the only pure iron he could find was the little which fell from heaven in the form of meteors, and even this was not perfectly pure for meteoric iron is also mixed slightly with other metals.

The iron which lay about primitive man in such abundance was buried and locked tightly in an *ore*. To separate the iron from the other substances of the ore was by no means an easy thing to do. Iron can best be extracted from the ore by putting the ore in a fire and melting out the iron. Place some iron ore in a fire and if the fire is hot enough—and it must be very hot indeed—the iron will leave the ore and will gather into a lump at the bottom of the fire. To separate the iron from its ore in this way is to make iron. When and where man first learned the secret of making iron is of course unknown. A camp-fire in some part of the world may have shown to man the first lump of iron, or a forest fire sweeping along and melting ores in its path may have given the first hint for the manufacture of iron.

Iron making at first doubtless consisted in simply melting the ore in an open heap of burning wood or charcoal, for charcoal is an excellent fuel for smelting (melting) ores. But this open-fire method was wasteful and tedious and at a very early date the smelting of the ore was done in a rude sort of a furnace. A hole ten or twelve feet deep was dug in the side of a hill. In the hole were placed charcoal and iron ore, first a layer of charcoal, then a layer of the ore. At the top of the mass there was an opening and at the bottom there were several openings. When the mass was set on fire the openings produced a good strong draft, the charcoal was consumed, and the ore was smelted. The product was a lump of *wrought iron*, known as the *bloom*.



FIG. 3.—THE
PRIMITIVE FORGE.

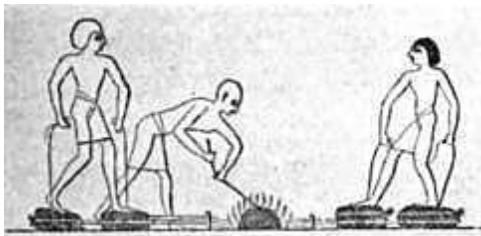


FIG. 4.—BELLOWS WORKED BY
THE FEET.

The hillside furnace worked well enough when the wind was favorable, but when the wind was unfavorable there was no draft and no iron could be



FIG. 5.—THE
WOODEN BELLOWS.

made. So ironmakers found a way by which the air could be driven into the furnace by artificial means. They invented the *bellows*, a blowing apparatus (Fig. 3) which was usually made of goat skins sewed together and which was operated either by the hands or by the feet (Fig. 4). Sometimes the bellows consisted of a hollow log in which a piston was worked up and down (Fig. 5). After the invention of the bellows, ironmakers could make their iron whenever and wherever they pleased, for they could force air into their furnaces at any time and at any place. This rude bellows forcing a draft of air into a half-closed furnace filled with a burning mass of charcoal and iron ore was the first form of the forge, one of the greatest of all inventions.

With the invention of the forge the stone age gradually passed away and the iron age was ushered in. Tools and weapons could now be made of iron. And great was the difference between iron tools and stone tools. To cut down a tree with a flint hatchet required the labor of a man for a month, while to clear a forest with such an implement was an impossible task. But the forge gave to man iron for the sharp cutting tools, for the ax and knife and chisel and saw. With these he became the master of wood and he could

now easily cut down trees and build houses and make furniture and wagons and boats.

As time went on and man advanced in civilization, iron was found to be the most useful of metals. Iron can be shaped into many forms. It can be drawn into wire of any desired length or fineness, it may be bent in any direction, it may be sharpened, or hardened, or softened, at pleasure. "Iron accommodates itself to all our wants and desires and even to our caprices. It is equally serviceable to the arts, the sciences, to agriculture and war; the same ore furnishes the sword, the plowshare, the scythe, the pruning-hook, the needle, the spring of a watch or of a carriage, the chisel, the chain, the anchor, the compass and the bomb. It is a medicine of much virtue and the only metal friendly to the human frame."⁹

A metal that was so useful was needed in large quantities, yet the primitive forge could turn out only small quantities of iron. A day's labor at the bellows would produce a lump weighing only fifteen or twenty pounds. As a result of this slowness in manufacture there was always in primitive and ancient times a scarcity of iron. Indeed in some countries iron was a precious metal, almost as precious as silver or gold. In many countries, it is true, there were thousands of forges at work, but in no country was the supply of iron equal to the demand. The old forge could not supply the demand, yet centuries passed before any great improvement was made in the progress of iron making.

Near the close of the Middle Ages improvements upon the primitive forge began to be made. In the sixteenth century ironmakers in Germany began to smelt ore in closed furnaces and to build their furnaces higher and to make them larger (Fig. 6). Sometimes they built their furnaces to a height of twenty or thirty feet. About this time also a better and a stronger blast was invented. Water-power instead of hand-power began to be used for operating the bellows. In some cases wooden bellows—great wooden pistons working in tubs—were substituted for the old bellows of leather. By the end of the sixteenth century so many improvements had been made upon the primitive forge that it no longer resembled the forge of ancient times. So the new forge received a new name and was called a *blast furnace*.¹⁰ You should observe, however, that the blast furnace was simply the old forge built with a large closed furnace and provided with a more powerful blast.



FIG. 6.—A BLAST FURNACE OF THE MIDDLE AGES.

The invention of the blast furnace marked the beginning of a new era in the history of iron making. In the first place there was produced in the blast furnace a kind of iron that was entirely different from that which was produced in the primitive forge. In the primitive forge there was made a lump of practically pure unmelted iron, known as wrought iron. In the blast furnace there was produced a somewhat impure grade of melted iron, known as *cast* iron, or *pig*¹¹ iron. In the second place, the blast furnace produced iron in quantities vastly greater than it was ever produced by the old forge. In the blast furnace more iron could be made in a day than could be made by the forge in a month. In some of the early blast furnaces a thousand pounds of iron could be made at one melting and we read of one early furnace that produced 150 tons of iron in a year.

But even with the blast furnace it was still difficult to make enough iron to supply the ever-increasing demands of the industrial world. In the sixteenth and seventeenth centuries machinery was brought into use more than ever before and of course more iron was needed for the construction of the machines. There was ore enough for all the iron that was needed but it was difficult to get fuel enough to smelt the ore. Charcoal was still used as the fuel for smelting (Fig. 7), and in order to get wood for the charcoal great inroads were made upon the forests. In England in the early part of the eighteenth century Parliament had to put a check upon the manufacture of iron in certain counties in order to save the forests of those counties from utter destruction. It then became plain that if iron making were to be continued on a large scale a new kind of fuel would have to be used in the furnaces. So men set their wits to work to find a new kind of fuel. As far back as 1619 Dud Dudley in the county of Warwick, England, undertook to use ordinary soft coal in his furnaces but his experiment was not very successful or very profitable. More than a century after this an English ironmaker named Abraham Darby began (in 1735) to



FIG. 7.—MAKING CHARCOAL.

use *charred coal* in his blast furnaces, and his experiments were successful. Here was the new fuel which was so badly needed. Charred coal is simply *coke* and coke could be had in abundance. So the new fuel was soon used in all parts of England and by the end of the eighteenth century coke was driving charcoal out of blast furnaces (Fig. 8).



FIG. 8.—A PITTSBURGH COKE OVEN.

About the time the use of coke for smelting became general, an Englishman named Neilson brought about another great change in the process of iron making. Before Neilson's time the blast driven into the furnace had always been one of cold air. Neilson learned that if the air before entering the furnace were heated to a temperature of 600 degrees it would melt twice the amount of ore and thus produce twice the amount of iron without any increase in the amount of fuel. So he invented (in 1828) a *hot blast* for the blast furnace (Fig. 9). With the use of coke and with the hot blast the production of iron increased enormously. But there was need for all the iron that could be made. Indeed it seems that the world can never get too much iron. About the time the hot blast was invented iron chains instead of ropes began to be used for holding anchors, iron plows began to be made in great numbers (p. 83), iron pipes instead of hollow wooden logs began to be used as water-mains in cities, and iron rails began to be used on railroads. To supply iron for all these purposes kept ironmakers busy enough, even though they burned coke in their furnaces and made use of the hot air blast.



FIG. 9.—A MODERN BLAST FURNACE.

But ironmakers were soon to become busier than ever before. About the middle of the nineteenth century Sir Henry Bessemer invented a new process of making steel. Steel is only iron mixed with a small amount of carbon. Ironmakers have known how to make steel—and good steel, too—for thousands of years, but before the days of Bessemer the process had always been slow and tedious, and the cost of steel had always been very great. Bessemer undertook to make steel in large quantities and at low prices. In his experiments amid showers of molten metal he often risked his life, but his perseverance and courage were rewarded. By 1858 he had invented a process by which tons of molten iron could be run into a furnace and in a few minutes be converted into a fine quality of steel. This invention of Bessemer was the last great step in the history of the forge.



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Underwood, N. Y.

FIG. 10.—GREAT STEEL RAIL PASSING
THROUGH ROLLER STEEL MILL.

Now that steel could be made in great quantities and at a low cost it was put to uses never dreamed of in former times. Soon the railroad rail was made of steel (Fig. 10), bridges were made of steel, ships of war were plated with steel. Then ocean grayhounds and battleships were made of steel, still later steel freight cars and steel passenger coaches were introduced, while in our own time we see vast quantities of steel used in the building of houses.

So while the invention of Bessemer marked the last step in the history of the forge it also marked the ending of the Age of Iron and the beginning of the wonderful age in which we live—the Age of Steel.



THE STEAM-ENGINE

We have now traced the steps by which man mastered the art of kindling a fire quickly and easily and have followed the progress that has been made in the most common uses of fire. But the story of a most important use of fire remains to be told, the story of its use in doing man's *work*. How important this use is, how much of the world's work is done through the agency of fire, a little reflection will make plain. Fire makes steam and what does steam do? Its services are so many you could hardly name all of them. The great and many services of steam are made possible by the fire-engine, or *steam-engine*, and the story of this wonderful invention will now be told.

That steam has the power to move things must have been learned almost as soon as fire was used to boil water. Heat water until it boils and the steam that is formed is bound to move something unless it is allowed to escape freely. It will burst the vessel if an outlet is not provided. That is why a spout has been placed on the tea-kettle. Where there is cooking, steam is abundant and the first experiments in steam were doubtless made in the kitchen (Fig. 1). It has been said that the idea of the steam-engine first occurred to Adam as he watched his wife's kettle boil.

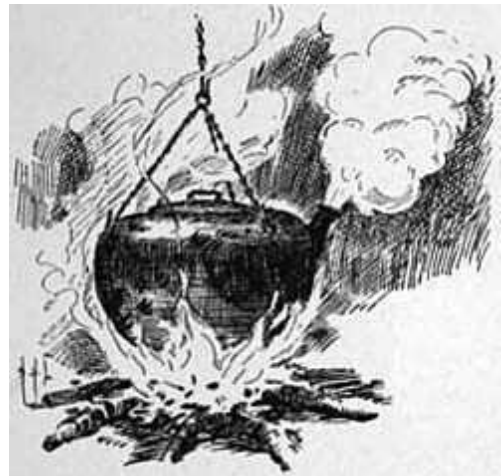


FIG. 1.—FIRST EXPERIMENTS
WITH STEAM.

Whatever may have happened in ancient kitchens, we are certain that there were no steam-engines until many centuries after Adam. The beginnings of this invention are not shrouded in so much mystery as are those of the match and the lamp and the forge. In giving an account of the steam-engine we can mention names and give dates from the very beginning



FIG. 2.—HERO'S
ENGINE, 120 B. C.

of the story. We know what the first steam-engine was like and we know who made it and when and where it was made. It was made 120 B. C. by Hero, a philosopher of Alexandria in Egypt. It was like the one shown in Figure 2. The boy applies the fire to the steam-tight vessel *p* and when steam is formed it passes up through the tube *o* and enters the globe which turns easily on the pivots. The steam, when it has filled the globe, rushes out of the short tubes *w* and *z* projecting from opposite sides of the globe and bent at the end in opposite directions. As it rushes out of the tubes the steam strikes against the air and the reaction causes the globe to revolve, just as in yards we sometimes see jets of water causing bent tubes to revolve. This was Hero's engine, the first steam-engine ever made.

Hero's engine was used only as a toy and it seems to represent all the ancients knew about the power of steam and all they did with it. It is not strange that they did not know more for there is no general rule by which discoveries are made. Sometimes even enlightened peoples have for centuries remained blind to the simplest principles of nature. The Greeks and Romans with all their culture and wisdom were ignorant of some of the plainest facts of science. It is a little strange, however, that after Hero's discovery was made known, men did not profit by it. It would seem that eager and persistent attempts would have been made at once to have steam do useful work, as well as furnish amusement. But such was not the case. Hero's countrymen paid but little attention to his invention and the steam-engine passed almost completely out of men's minds and did not again attract attention for nearly seventeen hundred years.

About the end of the fifteenth century Europe began to awaken from a long slumber and by the end of the sixteenth century its eyes were wide open. Everywhere men were now trying to learn all they could. The study of steam was taken up in earnest about the middle of the sixteenth century and by the middle of the next century quite a little had been learned of its nature and power. In 1629 an Italian, Branca by name, described in a book a steam-engine which would furnish power for pounding drugs in a mortar. There was no more need for such a machine then than there is now and of course

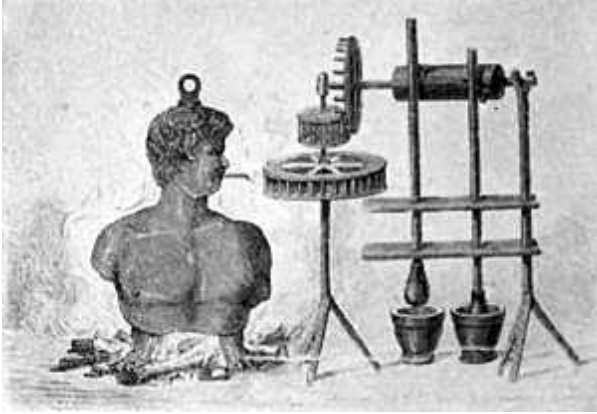


FIG. 3.—BRANCA'S ENGINE, 1629.

the inventor aroused no interest in his engine. You can easily understand how Branca's engine (Fig. 3) works. The steam causes the wheels and the cylinder to revolve. As the cylinder revolves, a cleat on it catches a cleat on the pestle and lifts the pestle a short distance and then lets it fall. Here the pestle instead of being raised by a human hand is raised by the force of steam. This engine would be

more interesting if an engine had actually been made, but there is no reason to believe that Branca ever made the engine he described. We owe much to him, nevertheless, for suggesting how steam might be put to doing useful work.

It was not very long before an Englishman put into practice what the Italian had only suggested. Edward Somerset, the Second Marquis of Worcester, in 1663 built a steam-engine that raised to the height of forty feet four large buckets of water in four minutes of time. This was the first useful work ever done by steam. Figure 4 shows the construction of Worcester's engine.

In this engine there was one improvement over former engines which was of the greatest importance: there was one vessel in which the steam was generated and another in which the steam did its work. The steam-engine now consisted of two great divisions, the boiler and the engine proper.

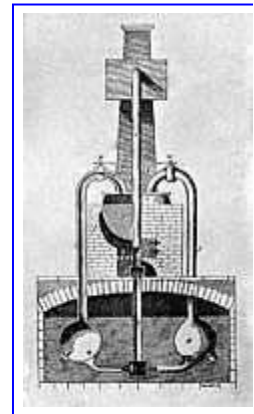


FIG. 4.—
WORCESTER'S
ENGINE, 1663.

Worcester spent a large part of his fortune in trying to improve the steam-engine, yet he received neither profit nor honor as a reward. He died poor and his name was soon forgotten. His service to the world was nevertheless very great. In his time the mines of England had been sunk very deep into the earth; and the deeper they were sunk the greater was the difficulty of lifting the water out of them and keeping them dry. The water

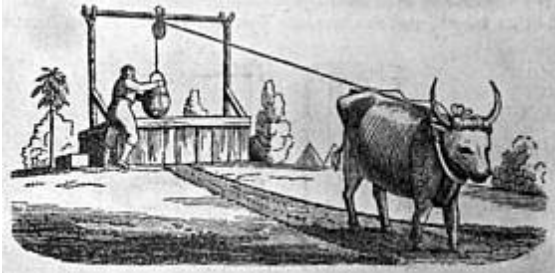


FIG. 5.—AN ANCIENT METHOD OF DRAWING WATER.

was lifted up from the mines by means of buckets drawn by horses or oxen (Fig. 5). Sometimes it took several hundred horses to keep the water out of a single mine. It was Worcester's object to construct an engine that would do the work of the horses. The engine he built could not do this, yet it furnished the idea—and the idea is

often the most important thing. It was not long before engines built upon Worcester's plan were doing useful work at the mines. At the opening of the eighteenth century the steam-engine had been put to work and was serving man in England and throughout the continent of Europe.

The first engines were not safe. Often the steam pressed too heavily upon the sides of the vessel in which it was compressed and there were explosions. About 1680 Denis Papin, a Frenchman, invented the *safety valve*, that is a valve that opens of its own accord and lets out steam when there is more in the vessel than ought to be there. About ten years later Papin gave the world another most valuable idea. In Worcester's engine the steam in the steam chest pressed directly on the water that was to be forced up. Papin showed a better way. He invented the engine shown in Figure 6. In this engine a small quantity of water was placed in the bottom of the cylinder *A*. Fitting closely in the cylinder was a *piston B* such as Papin had seen used in ordinary pumps. We will suppose that the piston is near the bottom of the cylinder and that a fire is built underneath. The bottom being made of very thin metal the water is rapidly converted into steam and thus drives the piston up to the top as shown in the figure. Here a latch *E* catches the piston-rod *H* and holds the piston up until it is time for it to descend. Now the fire is removed and the steam, becoming cold, is condensed and a vacuum is formed below the piston. The latch *E* now releases the rod *H* and the piston is driven down by the air above it, pulling with it the rope *L* which passes over the pulleys *TT*. As the rope descends it lifts a weight *W* or does other useful work. As the

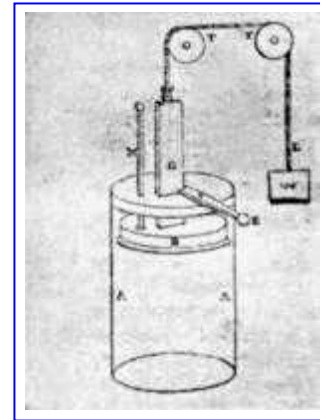


FIG. 6.—PAPIN'S ENGINE, 1690.

inventor of the piston Papin ranks among the greatest of those whose names are connected with the development of the steam-engine.

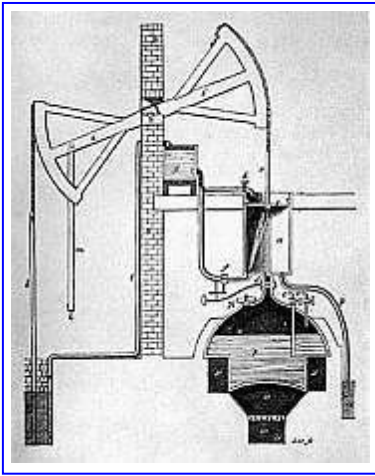


FIG. 7.—NEWCOMEN'S
ENGINE, 1705.

Our story has now brought us to the early part of the eighteenth century. Everywhere men were now trying to make the most of the ideas of Worcester and Papin. The mines were growing very deep. As the water in them was getting beyond control something extraordinary had to be done. Now it seems that whenever the world is in need of an extraordinary service someone is found to render that service. The man who built the engine that was needed was a humble blacksmith of Dartmouth, England, Thomas Newcomen. This master mechanic in 1705 constructed the best steam-engine the world had yet seen. We must study Newcomen's engine (Fig. 7) very carefully. The large beam *ii* moved freely up and down on the pivot *v*. One end of the beam was connected with the heavy pump-rod *k* by means of a rope or chain working in a groove and the other end was connected with the rod *r* in the same way. When steam from the boiler *b* passed through the valve *d* into the cylinder (steam-chest) *a* it raised the piston *s* and with it the piston-rod *r* thus slackening the rope and allowing the opposite end of the beam to be pulled down by the weight of the pump-rod *k*. As soon as the piston *s* reached the top of the cylinder the steam was shut off by means of the valve *d* and the valve *f* was turned and a jet of cold water from the tank *g* was injected into the cylinder *a* with the steam. The jet of cold water condensed the steam rapidly—steam is always condensed rapidly when anything cold comes in contact with it—and the water formed by the condensation escaped through the pipe *p* into the tank *o*. As soon as the steam in *a* is condensed, a vacuum was formed in the cylinder and the atmosphere above forced the piston down and at the same time pulled the pump-rod *k* up and lifted water from the well or mine. When the piston reached the bottom of the cylinder the valve *d* was opened and the piston again ascended. Thus the beam is made to go up and down and the pumping goes on. Notice that steam pushes the piston one way and the atmosphere pushes it back.

In Newcomen's engine the valves (*f* and *d*) at first were opened and shut (at each stroke of the piston) by an attendant, usually a boy. In 1713 a boy named Humphrey Potter, in order to get some time for play, by means of strings and latches, caused the beam in its motion to open and shut the valves without human aid. We must not despise Humphrey because his purpose was to gain time for play. The purpose of almost all inventions is to save human labor so that men may have more time for amusement and rest. Humphrey Potter ought to be remembered not as a lazy boy but as a great inventor. His strings and latches improved the engine wonderfully (Fig. 8). Before his invention the piston made only six or eight strokes a minute; after the valves were made to open and shut by the motion of the beam, it made fifteen or sixteen strokes a minute and the engine did more than twice as much work.



FIG. 8.—HUMPHREY POTTER'S LATCHES AND STRINGS.

Newcomen's engine as improved by Potter and others grew rapidly into favor. It was used most commonly to pump water out of the mines but it was put to other uses. In and about London it was used to supply water to large houses and in 1752 a flour mill near Bristol was driven by a steam-engine. In Holland Newcomen's engines were used to assist the wind-mills in draining lakes.



FIG. 9.—JAMES WATT STRIVING TO IMPROVE NEWCOMEN'S ENGINE.

For nearly seventy-five years engines were everywhere built after the Newcomen pattern. Improvements in a small way were added now and then but no very important change was made until the latter part of the eighteenth century, when the steam-engine was made by James Watt practically what it is to-day. This great inventor spent years in making improvements upon Newcomen's engine (Fig. 9) and when his labors were finished he had done more for the steam-engine than any man who ever lived. We must try to learn *what* he did. We can learn what Watt did by studying Figure 10. Here P is a piston working in a cylinder A *closed at both ends*. By the side of the cylinder is a *valve-chest* C into which steam passes from the pipe T. Connecting C with the cylinder there are *two* openings, one at the top of the cylinder and the other at the bottom. The valve-chest is provided with valves which are worked by means of the rod F, which moves up and down with the beam B, thanks to Humphrey Potter for the hint. The valves are so arranged that when steam enters the opening at the top of the cylinder it is shut off from the opening at the bottom, and when it enters the opening at the bottom it is shut off from the opening at the top. When the opening at the bottom is closed the steam will rush in at the upper opening and push the piston downward; when the piston has nearly reached the bottom of the cylinder the upper opening will be closed and steam will rush in at the bottom of the steam chest and push the piston upwards. Here was *one* of the things done by Watt for the engine: he contrived to make the steam push the piston down as well as up. You have observed that in Newcomen's engine steam was used only to push the piston *up*, the atmosphere being relied upon to push it down. Thus we may say that Watt's engine was the first *real steam-engine*, for it was the first that was worked entirely by steam. All engines before it had been worked partly by steam and partly by air.

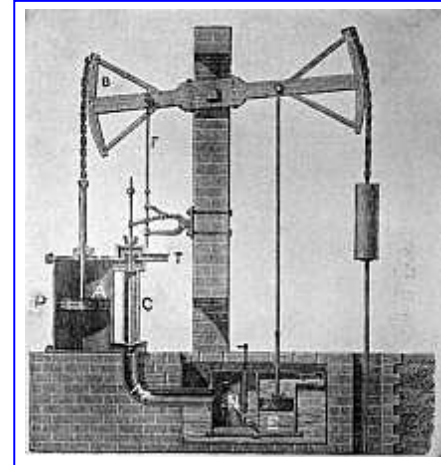


FIG. 10.—WATT'S ENGINE.

Watt's greatest improvement upon the steam-engine is yet to be mentioned. In Newcomen's engine when the cold water was injected into the cylinder it cooled the piston and when steam was let into the cylinder again a part of it, striking the cold piston, was condensed before it had time to do any work and the power of this part of the steam was lost. Watt did not allow

the piston to get cold, for he did not inject any cold water into the cylinder. In his engine as soon as the steam did its work it was carried off through the pipe *M* to the vessel *N* and there condensed by means of a jet of water which was injected into *N* (called the *condenser*) by means of a pump *E* worked by the motion of the beam, thanks again to Humphrey Potter for the idea. This condensation of the steam outside of the cylinder and at a distance from it prevented the piston (and cylinder) from getting cold. In other words, in the Watt engine when steam entered the cylinder it went straight to work pushing the piston. No steam was lost and no power was lost and the cost of running the engine was greatly reduced.

It cannot be said that Watt invented the steam-engine—no one can claim that honor—yet he did so much to make it better that he well deserves the epitaph which is inscribed on his monument in Westminster Abbey. This inscription is as follows:

NOT TO PERPETUATE A NAME
WHICH MUST ENDURE WHILE THE PEACEFUL ARTS
FLOURISH
BUT TO SHEW
THAT MANKIND HAVE LEARNT TO HONOR THOSE
WHO BEST DESERVE THEIR GRATITUDE
THE KING
HIS MINISTERS AND MANY OF THE NOBLES
AND COMMONERS OF THE REALM
RAISED THIS MONUMENT TO
JAMES WATT
WHO DIRECTING THE FORCE OF AN ORIGINAL
GENIUS
EARLY EXERCISED IN PHILOSOPHIC RESEARCH
TO THE IMPROVEMENT OF
THE STEAM ENGINE
ENLARGED THE RESOURCES OF HIS COUNTRY
INCREASED THE POWER OF MAN
AND ROSE TO AN EMINENT PLACE
AMONG THE MOST ILLUSTRIOUS FOLLOWERS OF

SCIENCE
AND THE REAL BENEFACTORS OF THE WORLD
BORN AT GREENOCH MDCCXXXVI
DIED AT HEATHFIELD IN STAFFORDSHIRE
MDCCCXIX

But the story of the steam-engine does not end with Watt. It will be remembered that in the engines of Nero and of Branca the steam did its work by reaction or by impulse. Now soon after the time of Watt, inventors turned their thoughts to the old engines of Nero and Branca and began to experiment with engines that would do their work by a direct impact of steam. After nearly a century of experimenting and after many failures there was at last developed an engine known as the *steam-turbine*. In this engine the steam does its work by impinging or pushing directly upon blades (Fig. 11) which are connected with the shaft which is to be turned, and it does this in much the same manner that we saw the steam do its work in Branca's engine. One of the greatest names connected with the steam turbine is that of Charles Algernon Parsons of England. In 1884 this great inventor patented a steam-turbine which proved to be a commercial success and since that date the steam-turbine has been constantly growing in favor. So great has been its success on land and on sea that there are those who believe that the engine invented by Watt will in time be cast aside and that its place will be taken by an engine which is the most ancient as well as the most modern of steam motors.

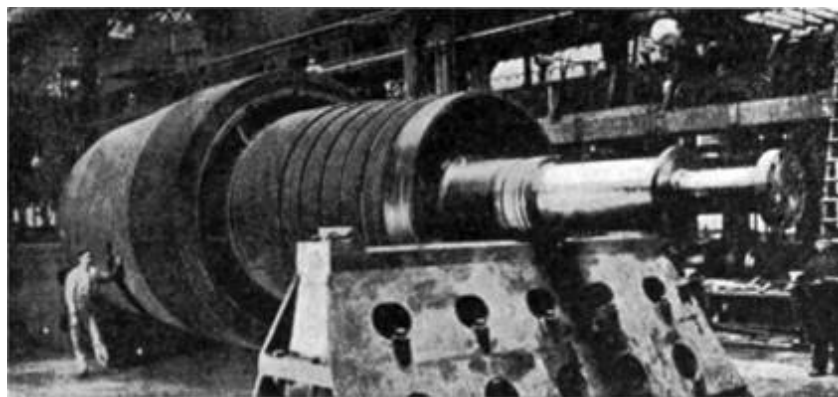


FIG. 11.—SHAFT OF A LARGE MARINE TURBINE.

Within the cylinder are thousands of blades upon which the steam acts directly in the turning of the shaft. In the largest turbines there are as many as 50,000 blades.



THE PLOW

You have now learned the history of those inventions which enabled man to gain a mastery over fire and to use it for his comfort and convenience. We shall next learn the history of an invention which gave man the mastery of the soil and enabled him to take from the earth priceless treasures of fruit and grain. This invention was the plow.

In his earliest state man had no use for the plow because he did not look to the soil as a place from which he was to get his food. The first men were hunters and they relied upon the chase for their food. They roamed from place to place in pursuit of their prey—the birds and beasts of the forest and the fishes of the stream. They did not remain long enough in one spot to sow seed and to reap the harvest. Still in their wanderings they found wheat and barley growing wild and they ate of the seeds of these plants and learned that the little grains were good for food. They learned, too, that if the seeds were planted in a soil that was well stirred the plants would grow better than they would if the seeds were planted in hard ground. So by the time men had grown tired of wandering about and were ready to settle down and live in one spot they had learned two important facts: they knew they could add to their food supply by tilling the soil, and they knew that they could grow better crops if they would stir the soil before planting the seed.

For the stirring of the soil the primitive farmer doubtless first used a sharpened stick such as wandering tribes carry for the purpose of digging up eatable roots, knocking fruits down from trees, and breaking the heads of enemies. Such a stick known as the *Katta* (Fig. 1) is carried by certain tribes in Australia, and we are told by travelers that the Kurubars of Southern India use a sharp stick when digging up the ground. The digging stick is used by savages in many parts of the world and we may regard it as the oldest of implements used for tilling the soil.



FIG. 1.—THE KATTA OR
DIGGING STICK.



FIG. 2.—THE FIRST PLOW.

The first plow was a forked stick or a limb of a tree with a projecting point (Fig. 2). With this implement the ground was broken not by digging but by dragging the fork or projecting point of the stick through the ground and forming a continuous furrow. In this forked stick we see two of the principal parts of the modern plow. The fork of the stick is the *share*, or cutting part of the plow, while the main part of the stick is the *beam*.

An improvement upon the simple forked stick is seen in Figure 3, which is copied from an ancient monument in Syria (in Asia Minor). The old Syrian plow consists almost wholly of the natural crooks of a branch of a tree, the only artificial piece being the brace *e* which connects the share and the beam and holds them firm. In this crooked stick we have three of the main parts of the modern plow, the beam (*a*), the share (*c-b*) and the handle (*d*). The plow in this form requires the services of two persons—one to draw the plow and one to guide it and keep it in the ground. It is said that it was with a plow of this kind that the servants of Job were plowing when they were driven from their fields by the Sabeans.

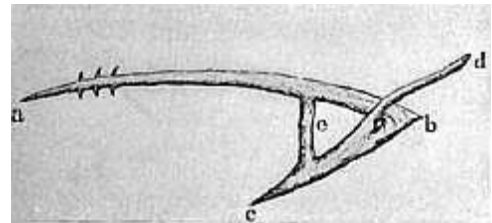


FIG. 3.—THE SYRIAN PLOW KNOWN AS JOB'S PLOW.



FIG. 4.—PLOW DRAWN BY HUMAN LABOR.

The first plows were drawn by the strength of the human body (Fig. 4). Upon a very old monument of ancient Egypt, the

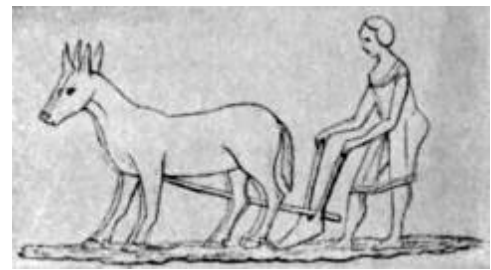


FIG. 5.—THE EGYPTIAN PLOW.

country which seems to have been the first home of the plow, we have a plowing scene which shows a number of men dragging a plow by means of a rope. But primitive man was not at all fond of labor and in the course of time he tamed wild bulls and horses and made them draw the plows. So upon another Egyptian monument of a later date we

have a picture of a plowing scene in which animals are drawing the plow (Fig. 5). In this Egyptian plow we see improvements upon the crooked stick of the Syrians. The Egyptian plow, you observe, has a broader share. It will, therefore, make a wider furrow and will plow more ground. Moreover, it has two handles instead of one. Taking it altogether, the Egyptian plow was a fairly good implement.

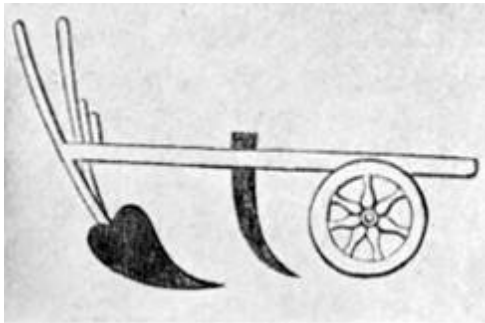


FIG. 6.—PLINY'S PLOW, 70 A. D.

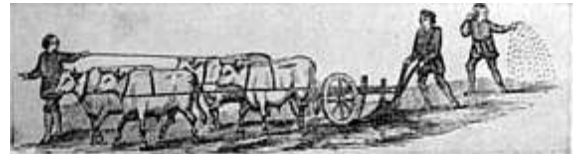


FIG. 7.—AN OLD SAXON PLOW, 1000 A. D.

Many centuries passed before any real improvement was made upon the old Egyptian plow. If there were any improvement anywhere it was among the Romans. We read in Pliny—a Roman writer of the first century—of a plow that had wheels to regulate the depth of the plow and also a *coulter*, that is, a knife fixed in front of the share to make the first cut of the sod (Fig. 6). But such a plow was not in general use in Pliny's time. A thousand years later, however, the plow with wheels and coulter was doubtless in common use. In a picture taken from an old Saxon print we see (Fig. 7) a plow which was used in the time of William the Conqueror (1066). Here the plow has a coulter inserted in the beam and there are two wheels to regulate the depth to which the plow may go. This Saxon plow is drawn by four fine oxen and it is plainly a great improvement upon the old Egyptian plow.

But improvements in the plow during the dark ages came very slowly. At the time of the discovery of America the plow was still the clumsy wooden thing it was five hundred years before. In the sixteenth and seventeenth centuries, however, when improvements were being made in so many things, it was natural that men should begin to think of trying to improve the plow. In an old book published in 1652 we read of a double plow—one which would plow two furrows at one time. A picture (Fig. 8) of the double plow is given in the book but there is no proof that such a plow was ever made or ever used. The world did not as yet need a double plow, although the time was to come when it would need one.

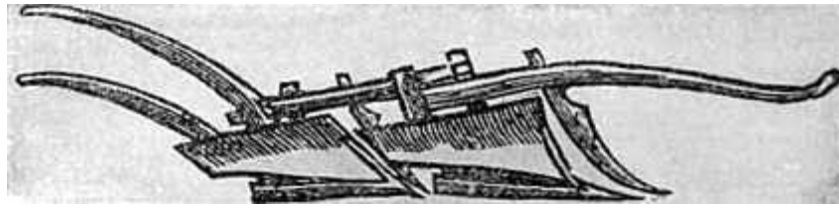


FIG. 8.—A DOUBLE PLOW OF THE SEVENTEENTH CENTURY.
(This plow was proposed but was never made.)

In the early part of the eighteenth century we begin to see real improvements in plow making. About this time Dutch plowmakers began to put *mold-boards* on their plows. The purpose of the mold-board is to lift up and turn over the slice of sod cut by the share. Without the mold-board the plow simply runs through the ground and stirs it up. With the mold-board of the Dutch plow (Fig. 9) the sod was turned completely over and the weeds and grass were covered up. This was the kind of plow that was needed, for if the weeds and grass are not covered up the best effects of plowing are lost. So the mold-board was a great improvement and its invention marks a great event in the history of the plow.

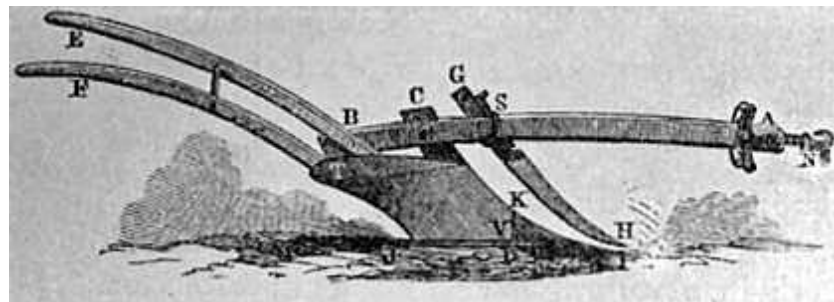


FIG. 9.—THE DUTCH PLOW SHOWING THE MOLD-BOARD.

The Dutch plow was taken as a model for English plows and, in fact, for the plows of all nations. The mold-board grew rapidly into favor and by the end of the eighteenth century it was found on plows in all civilized nations. But the plow was still made mostly of wood (Fig. 10) and it was still an awkward and a poorly constructed affair. The method of making plows about the year 1800 has been described as follows: "A mold-board was hewed from a tree with the grain of the timber

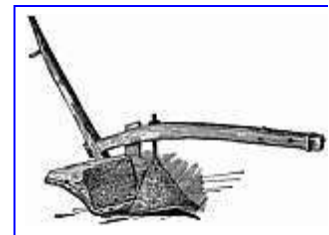


FIG. 10.—A
COLONIAL PLOW.

running as nearly along its shape as it could well be obtained. On to this mold-board, to prevent its wearing out too rapidly, were nailed the blade of an old hoe, thin strips of iron, or worn out horseshoes (Fig. 10). The land side was of wood, its base and sides shod with thin plates of iron. The share was of iron with a hardened steel point. The coulter was tolerably well made of iron. The beam was usually a straight stick. The handles, like the mold-board, were split from the crooked trunk of a tree or as often cut from its branches. The beam was set at any pitch that fancy might dictate, with the handles fastened on almost at right angles with it, thus leaving the plowman little control over his implement, which did its work in a very slow and most imperfect manner."

But about the end of the eighteenth century the world was beginning to need a plow that would do its work rapidly and well. Population was everywhere increasing and it was necessary to till more ground than had ever been tilled in former times. Especially was a good plow needed in the United States where there were vast areas of new ground to be broken. And it was in the United States that the first great improvements in the plow were made. Foremost among those who helped to make the plow a better implement was the statesman, Thomas Jefferson. This great man while traveling in France in 1788 was struck by the clumsiness of the plows used in that country. In his diary he wrote: "The awkward figure of their mold-board leads one to consider what should be its form." So Jefferson turned his attention to mold-boards. He saw that the mold-board ought to be so shaped that it would move through the ground and turn the sod with the least possible resistance and he planned for a mold-board of this kind. By 1793 he had determined what the proper form of a mold-board should be and had in actual use on his estate in Virginia several plows which had mold-boards of least resistance. Mr. Jefferson's patterns of the mold-board have, of course, been improved upon, but he has the honor of having invented the first mold-board that was constructed according to scientific and mathematical principles.¹²

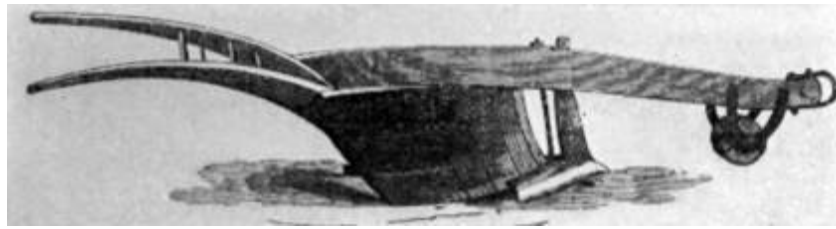


FIG. 11.—DANIEL WEBSTER'S PLOW.

About the time Jefferson was working upon the mold-board, Charles Newbold, a farmer of Burlington, New Jersey, was also doing great things for the improvement of the plow. We have seen that the plow of this time was a patch work of wood and iron. Newbold thought the plow ought to be made wholly of iron and about 1796 he made one of cast iron, the point, share, and mold-board all being cast in one piece. But the New Jersey farmers did not take kindly to the iron plow. They said that iron poisoned the crops and caused weeds to grow faster than ever. So Newbold could not sell his plows and he was compelled to give up the business in despair.

But soon the iron plow was to have its day. In 1819 Jethro Wood of Scipio, New York, took out a patent for a plow which was made of cast iron and which combined the best features of the plow as planned by Jefferson and by Newbold. In Wood's plow (Fig. 12) the several parts—the point, share and mold-board—were so fastened together that when one piece wore out it could easily be replaced by a new piece. In Newbold's plow when one part wore out the whole plow was rendered useless. Wood's plow became very popular and by 1825 it was rapidly driving out the half-wooden, half-iron plows of the olden time. Great improvements of course have been made upon the plow since 1819, but in the main features the best plows of to-day closely resemble the implement invented by Jethro Wood. Since our greatness as a nation is due largely to the plow all honor should be given to the memory of this inventor. "No citizen of the United States," said William H. Seward, "has conferred greater benefits on his country than Jethro Wood."

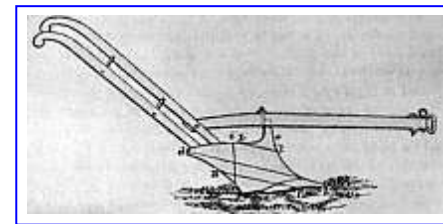


FIG. 12.—JETHRO WOOD'S PLOW, 1819.



FIG. 13.—THE GANG PLOW DRAWN BY HORSES.

But the plow of Jethro Wood, as excellent as it was, did not fully meet the needs of the western farmer. The sod of the vast prairies could not be broken fast enough with a plow of a single share. So about the middle of the nineteenth century the *gang plow*, a hint for which had been given long before (p. 78) was invented, and as this new plow moved along three or four or five furrows were turned at once. At first the gang plow was drawn by horses (Fig. 13) but later it was drawn by steam (Fig. 14).

The great gang plow drawn by steam marked the last step in the development of the plow. The forked stick drawn by human hands and making its feeble scratch on the ground had grown until it had become a mighty machine drawn across the field by an unseen force and leaving in its wake a broad belt of deeply-plowed and well-broken soil.



FIG. 14.—PLOWING BY STEAM.

The plow is drawn across the field by means of cables. Sometimes a traction engine moves along with the plow.

THE REAPER

After man had invented his rude plow and had learned how to till the soil and raise the grain, it became necessary for him to learn how to harvest his crop, how to gather the growing grain from the fields. The invention of the plow, therefore, must have soon been followed by the invention of the *reaper*.

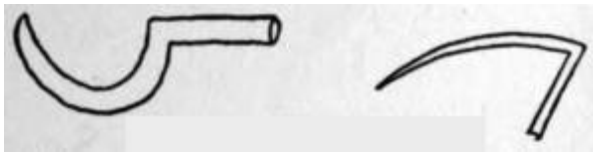


FIG. 1.—PRIMITIVE SICKLES.

The first grain was doubtless cut with

the rude straight knives used by primitive man. In time it was found that if the knife were bent it would cut the grain better. So the first form of the reaper was a curved or bent knife known as the sickle or reaping hook (Fig. 1). The knife was fastened at one end to a stick which served as a handle. When using the sickle the harvester held the grain in one hand and cut it with the other. (Fig. 2).



FIG. 2.—REAPING WITH THE SICKLE.

When the sickle first began to be used is of course unknown. Among the remains of the "stone age" (p. 39) are implements of flint which resemble the sickle, while among the remains of the so-called "bronze age" many primitive sickles made of bronze have been found. Nor do we know *where* the sickle was first used, although Egypt seems to have been the first home of the sickle just as it was the first home of the plow. Upon the wall of a building of ancient Thebes is a picture of an Egyptian harvest scene. Two men with sickles are cutting the wheat. A man following the reapers seems to be gleaning, that is, picking up the wheat that the reapers have cut. Other harvesters are carrying the grain to the threshing place where it is tramped out by the slow feet of oxen. A primitive sickle such as was used by the

Egyptians was used by all civilized nations in ancient times, by the Hebrews, by the Greeks, and by the Romans.



FIG. 3.—AN
EARLY
SCYTHE.

The first improvement upon the primitive sickle was made by the Romans. About the year 100 A. D. the Roman farmers, who were at the time the best farmers in the world, began to use a kind of scythe for cutting grass. The Roman scythe was simply an improved form of the sickle; it was a broad, heavy blade fastened on a long straight handle, resembling the pruning hook of to-day (Fig. 3). The scythe was swung with both hands and it was used chiefly for cutting grass.

For more than a thousand years after the appearance of the Roman scythe agriculture in Europe was everywhere neglected and little or no improvement was made in farming implements. About the end of the Middle Ages, however, improvements in the form of the scythe began to appear. In Flanders farmers began to use an implement known as the Hainault scythe (Fig. 4). This scythe had a fine broad blade and a curved handle. When reaping with this scythe the reaper with his left hand brought the stalks of grain together with a hook and with his right hand he swung the scythe and cut the grain. This scythe was an improvement upon the sickle but it was still a very awkward implement.

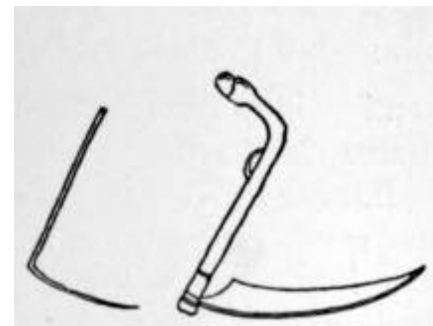


FIG. 4.—THE HAINAULT OR
FLEMISH SCYTHE, WITH
HOOK.

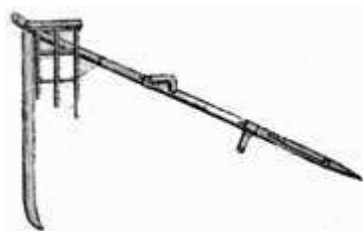


FIG. 5.—EARLY FORM OF
THE CRADLE SCYTHE.

The Hainault or Flemish scythe was followed by the *cradle scythe*. On this scythe (Fig. 5) there were wooden fingers running parallel to the blade. These fingers, called the cradle, caught the grain as it was cut and helped to leave it in a bunch. In the early cradle-scythe the fingers were few in number and they ran along the blade for only a part of its length, but in America during the colonial period the cradle was improved by lengthening the fingers and increasing their number. At the time of the Revolution the improved

American cradle was coming into use and by the end of the eighteenth century it was driving out the sickle.

But even the excellent American cradle-scythe could not meet the needs of the American farmer. The cast iron plow which was brought into use in the early part of the nineteenth century (p. 82) made it possible to raise fields of wheat vastly larger than had ever been raised before. But it was of no use to raise great fields of grain unless the crop could be properly harvested. Wheat must be cut just when it is ripe and the harvest season lasts only a few days. If the broad American fields were to be plowed and planted there would have to be a reaping machine that would cut the grain faster than human hands could cut it with the scythe (Fig. 6).



FIG. 6.—THE IMPROVED CRADLE SCYTHE.



FIG. 7.—THE FIRST REAPING MACHINE, 70 A. D.

So about the year 1800 inventors in Europe and in America took up the task of inventing a new kind of reaper. The first attempts were made in England where population was increasing very fast and where large quantities of grain were needed to feed the people. The first hints for a reaper were from a machine which was used in Gaul nearly 2,000 years ago. Pliny, who described for us a wonderful plow used in his time (p. 77), also describes this ancient reaper of the Gauls. It consisted of a large hollow frame mounted on two wheels (Fig. 7). At the front of the frame there was a set of teeth which caught the heads of grain and tore them off. The heads were raked into the box by an attendant. The machine was pushed along by an ox. This kind of machine was doubtless used in Europe for a while but it was not a success. It passed out of use and for many centuries it was entirely forgotten. Still, the first English reaping machines were made after the plan of this interesting old reaper of ancient Gaul.

The most remarkable of the early reapers was one invented by Henry Ogle, a schoolmaster of Remington, England. In 1822 Ogle constructed a model for a reaper which was quite different from any that had appeared before and which bore a close resemblance to the improved reapers of a later date. In Ogle's reaper (Fig. 8) the horse walked ahead beside the standing grain, just as it does now, and the cutting apparatus was at the right, just as it is now. The cutter consisted of a frame at the front of which was a bar of iron armed with a row of teeth projecting forward. Directly under the teeth lay a long straight edged knife which was moved to and fro by means of a crank and which cut the grain as it came between the teeth. A reel pushed the grain toward the knife and there was a platform upon which the grain when cut might fall. Ogle's machine did not meet with much success yet it holds a very high place in the history of reaping machines, for it had nearly all the parts of a modern reaper.

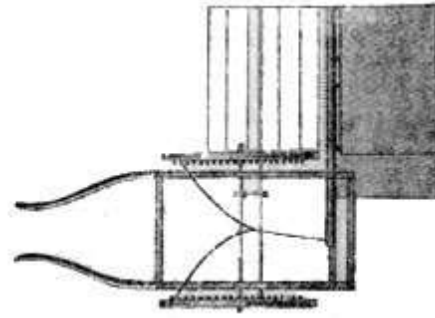


FIG. 8.—OGLE'S REAPER, 1822.



FIG. 9.—THE FIRST MCCORMICK REAPER.

English inventors did much to prepare the way for a good reaping machine but the first really successful reaper, the first reaper that actually reaped, was made in the United States. In the summer of 1831, Cyrus McCormick, a young blacksmith living in the Shenandoah Valley in Virginia, made a trial of a reaper which he and his father had invented—how much they had learned from Ogle we do not know—and the trial was successful (Fig. 9). With two horses he cut six acres of oats in an afternoon. "Such a thing," says Mr. Casson in his life of McCormick, "at the time was incredible. It was equal to the work of six laborers with scythes or twenty-four peasants with sickles. It was as marvelous as though a man had walked down the street carrying a dray horse on his back."

Although McCormick had his reaper in successful operation by 1831 he did not take out a patent for the machine until 1834. One year before this (in

1833) Obed Hussey, a sailor living in Baltimore, took out a patent for a reaper that was successful and that was in many respects as famous a machine as McCormick's. So while McCormick was the first in the field with his invention, Hussey was the first to secure a patent. The machines of McCormick and Hussey were very much alike: both had the platform, the iron bar armed with guards and the long knife moving to and fro. The most remarkable feature of

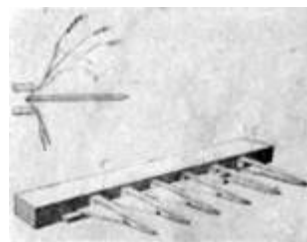


FIG. 10.—THE KNIFE
BLADE OF HUSSEY'S
REAPER.

Hussey's machine was the knife which consisted of thin triangular plates of steel sharpened on two edges and riveted side by side upon a flat bar (Fig. 10). The saw-like teeth of Hussey's knife caught the wheat between the guards and cut it better than any knife that had as yet appeared. Both the McCormick reapers and the Hussey reapers were practical and successful and each of these inventors performed a noble part in giving the world the reaper it needed.

The McCormick and the Hussey reapers gave new life to farming in the United States. Especially was the reaper a blessing to the Western farmers. In 1844 McCormick took a trip through the West, passing through Ohio, Michigan, Illinois, and Iowa. As he passed through Illinois he saw how badly the reaper was needed. He saw great fields of ripe wheat thrown open to be devoured by hogs and cattle because there were not enough laborers to harvest the crops. The farmers had worked day and night and their wives and children had worked but they could not harvest the grain; they had raised more than the scythe and sickle could cut. McCormick saw that the West was the natural home for the reaper and in 1847 he moved to Chicago, built a factory, and began to make reapers. In less than a year he had orders for 500 machines and before ten years had passed he had sold nearly 25,000 reapers. It was these reapers that caused the frontier line to move westward at the rate of thirty miles a year.

Improvements upon the machines of Hussey and McCormick came thick and fast. One of the first improvements was to remove the grain from the platform in a better way. With the first machines a man followed the reaper (Fig. 9) and removed the grain with a rake. Then a seat was provided and the man sat (Fig. 11) on the reaper and raked off the grain. Finally the *self-raking* reaper was invented. In this machine, as it appeared in its

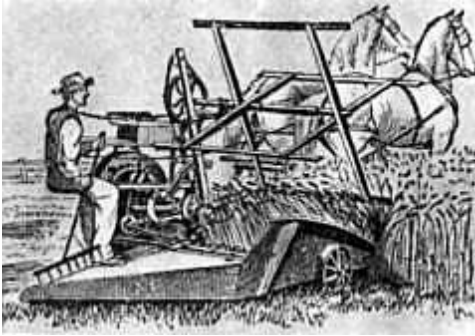


FIG. 11.—REAPER PROVIDED WITH SEAT FOR THE RAKER.

completed form about 1865, the reel and rake were combined.

The reel consisted of a number of revolving

arms each of which carried a rake (Fig. 12).

As the arms revolved they not only moved the standing grain toward the knife, but they also swept the platform and raked off the wheat in neat bunches ready to be bound into sheaves. So the self-raking reaper saved the labor of the man who raked the wheat from the platform.

Because it saved the labor of one man the self-raking reaper was for a time the king of reaping machines. But it did not remain king long, for soon there came into the harvest fields a reaper that saved the labor of several men. This was the *self-binder*. With the older machines, as the grain was raked off the platform it was gathered and bound into sheaves by men who followed the reaper, one reaper requiring the services of three or four or five human binders. With the self-binder (Fig. 13) the grain was gathered into sheaves and neatly tied without the aid of human hands. At first, wire was used in binding the sheaves but by 1880 most self-binders were using twine. So the self-binder saved the labor not only of the man who raked the grain from the platform but it saved the labor of all the binders as well.

The last step in the development of the reaper was taken when the *complete harvester* was invented. This machine cuts the standing grain, threshes it, winnows¹³ it, and places it in sacks (Fig. 14). As this giant reaper travels over the field one sees on one side the cutting bar 15 to 25 feet in length slicing its way through the wheat, while on the other side of the machine streams of grain run into sacks which, as fast as they are filled, are



FIG. 12.—SELF-RAKING REAPER.

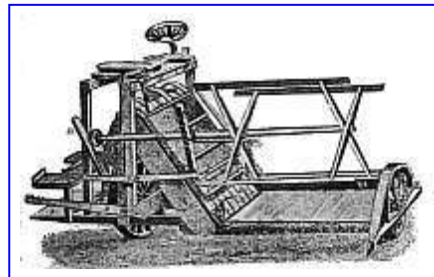


FIG. 13.—A SELF-BINDING REAPER.

hauled to the barn or to the nearest railway station. The complete harvester is either drawn by horses—30 or 40 in number—or by a powerful engine. It cuts and threshes 100 acres of wheat in a day and the cost is less than 50 cents an acre. It does as much work in a day as could have been done by a hundred men before the days of McCormick. Of all the wonderful machines used by farmers the most wonderful is the complete harvester, the latest and the greatest of reapers.

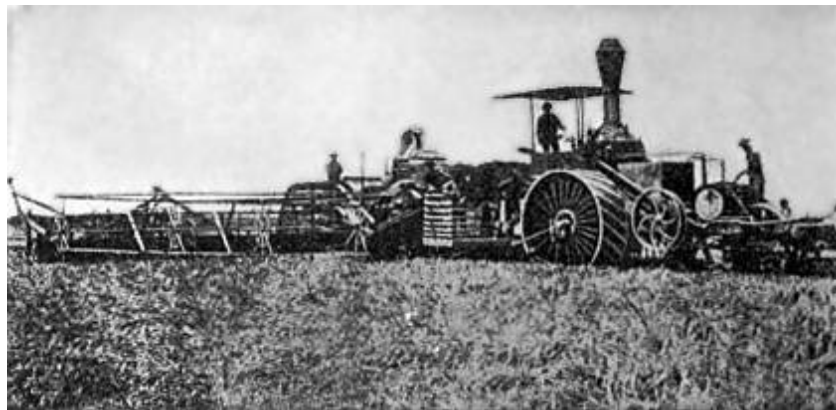


FIG. 14.—A COMBINED HARVESTER AND THRESHER.

THE MILL

The first mill was a hole made in a stationary rock (Fig. 1). The grain was placed in the hole and crushed with a stone held in the hand. On Centre street in Trenton, New Jersey, not many years ago one of these primitive mills could still be seen and there are evidences that such mills once existed in all parts of the world. In those places where the earth did not supply the stationary rock, stones were brought from afar and hollowed out into cup-like form and in these the grinding was done.



FIG. 1.—THE FIRST MILL.

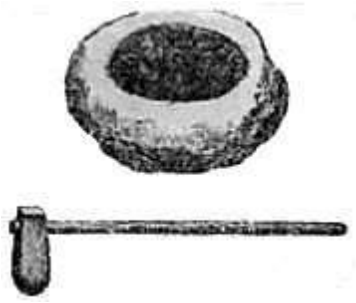


FIG. 2.—THE KNOCKING-STANE.

The mill which consisted of a hole in a rock and a stone in the hands was followed by the "knocking-stane" and mallet (Fig. 2). The "knocking-stane" was a mortar, or cup-shaped vessel made of stone; the mallet was usually made of wood. The grain was placed in the mortar and struck repeatedly with the mallet, the beating being kept up until a coarse flour was produced. This is an exceedingly rude method of crushing grain, yet this is the way the people in some parts of Scotland grind their barley at the present time.

At a very early date the "knocking-stane" was laid aside for the mortar and pestle (Fig. 3) almost everywhere. In this mill the grain instead of being struck with a hammer was pounded with a pestle. The bottom of the pestle was frequently covered with iron in which grooves were cut. As the man pounded he found that when he gave the pestle a twirling or rotary motion as it fell it ground the grain much faster. We may be sure that after this was learned the twirling motion was always given.



FIG. 3.—MORTAR AND PESTLE MILL.



FIG. 4.—THE SLAB-MILL.

The mortar and pestle were followed by the slab-mill (Fig. 4). Here the grain was ground by being rubbed between two stones. Dr. Livingstone, the great African explorer, gives the following description of a slab-mill which he saw in operation in South Africa. "The operator kneeling grasps the upper millstone with both hands and works it backwards and forwards in the hollow of the lower millstone, in the same way

that a baker works his dough. The weight of the person is brought to bear on the movable stone and while it is pressed and pushed forward and backward one hand supplies every now and then a little grain to be bruised and ground."

As we have seen, the primitive miller gradually learned that the pestle did better work when it fell with a twirling motion. This little bit of experience led to important results in the development of the mill. If the grinding were done better with a twirling motion, why not have as much of the twirling motion as possible? Why not make the upper stone go round and round? This was what was done. The upper stone was caused to turn round and round. The wheel-mill, the mill of the upper and nether millstone (Fig. 5), was invented. When and where it was invented we cannot tell for it was in use among all civilized peoples before history began to be written. There were many kinds of wheel-mills among the nations of antiquity and in principle they were all alike in construction. How they worked may be learned by studying Figure 5 which represents a mill used in ancient India. The upper stone is placed upon the pivot projecting from the center of the lower (nether) stone, and caused to revolve by means of the handle. The grain when placed in the hollow at the center of the upper stone (Fig. 5) works its way down between the stones and comes out at the circumference ground, bran and flour together. The mill was fed with grain by the operator. The first hopper was a human hand.

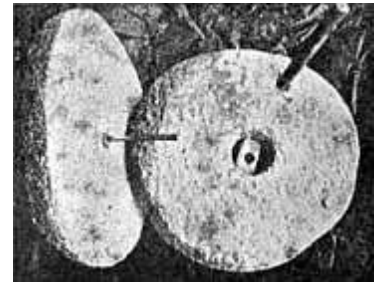


FIG. 5.—THE UPPER AND NETHER MILLSTONE.



FIG. 6.—AN ANCIENT JEWISH MILL.

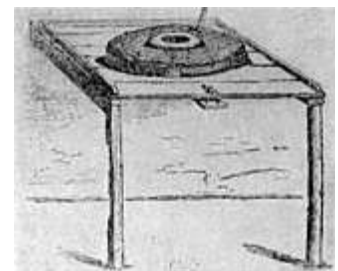


FIG. 8.—A SCOTTISH QUERN.



FIG. 7.—AN OLD ROMAN

MILL.



FIG. 9.—
POMPEIAN
FLOUR MILL, 79
A. D.

We have here several pictures of ancient mills. Figure 6 is an ancient Jewish mill. As we look at it we may recall the words, "Two women shall be grinding at a mill, the one shall be taken, and the other left."¹⁴ Figure 7 is an old Roman mill bearing a strong resemblance to the coffee mill that is used in our kitchens. Figure 8 is a Scottish quern, a mill that may still be found in use, it is said, in some parts of Scotland. Figure 9 is an old flour mill dug from the ruins of the city of Pompeii which was destroyed by an eruption in the year 79 A. D. Figure 10 shows the construction of this interesting mill. The upper (outer) stone is shaped like an hour-glass, the upper half of which serves as a hopper; the lower half turns upon the cone-shaped lower stone and does the grinding. The mill was operated by the projecting handles, the operators walking round and round the mill. Sometimes it was turned by human power, sometimes by horses or oxen.

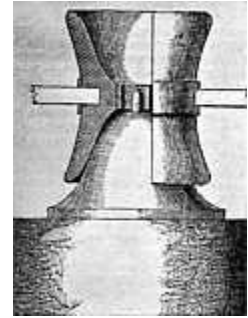


FIG. 10.—
SHOWING THE
INTERIOR OF
POMPEIAN
MILL.



FIG. 11.—THE FIRST
WATER-MILL, 50 B. C.

The Pompeian mill shows that as early as the first century the Romans ground their grain by animal power. Indeed about this time a still greater change was made in the method of grinding grain. When Julius Cæsar flourished (50 B. C.) men began to harness the power of running water and make it turn their mills (Fig. 11). From Figure 12 we may easily learn how this was done. The running water turns the wheel and in doing so turns the upper millstone. A hopper is suspended from the roof by ropes. Through this

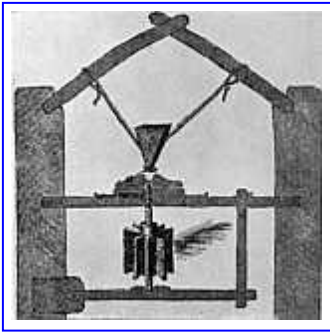


FIG. 12.—SHOWING
THE INTERIOR OF
THE FIRST WATER-
MILL.

the grain passes into the mill. Here was a great saving in human labor and a great advancement in mill making. A Roman writer of Cæsar's time appreciating how great a blessing was the invention of the water-mill exclaimed:

Ye maids who toiled so faithful at the mill
Now cease from work and from these toils be still;
Sleep now till dawn and let the birds with glee
Sing to the ruddy morn, on bush and tree;
For what your hands performed so long, so true,
Ceres¹⁵ has charged the water-nymphs to do;
They come, the limpid sisters, to her call,
And on the wheel with dashing fury fall;
Impel the axle with a whirling sound
And make the massive millstone reel around
And bring the floury heap luxuriant to the ground.

Nothing can be simpler than the water-mill described above; it was the old mill of the upper and nether millstones, the old hand mill turned by water. That was all. Yet, as simple as it was, many centuries passed after its invention before a new principle in flour making was discovered. There were inventions for lowering and raising the stone so as to grind finer or coarser as might be desired, and there were improvements in the kind of water wheels employed, and better methods of sifting the flour from the bran were discovered from time to time, but the water-mill invented in the time of Julius Cæsar remained practically unchanged until the early part of the nineteenth century, when the last step in the development of the mill was taken.¹⁶

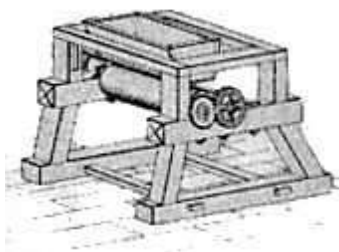


FIG. 13.—AN EARLY

About 1810 millers in Austria, more particularly those in Vienna, began to grind their grain by passing it between two horizontal rollers (Fig. 13). The rollers were spirally grooved and turned toward each other. There was a wide difference between this process and the one to which the world was accustomed, yet the new method was found to be better than the old one.

FLOUR ROLLER-MILL.

Austrian flour and Austrian bread became famous.

The delicious Vienna bread on our tables of course has never seen Vienna. It is called "Vienna bread" because it is made out of a kind of flour which was first ground in the Austrian capital. The Austrian way of grinding grew rapidly into favor among millers everywhere. In the United States where there was so much wheat to be ground the roller process was taken up eagerly and improved upon as only Americans know how to improve upon an idea. In the flour mills of the West the grain was soon passing through a series of rollers. By the first pair of rollers the grain was simply cracked into pieces somewhat coarse. Then after being bolted (sifted) it was passed between a second pair of rollers and reduced to a greater fineness. Then it was bolted again and passed between a third pair of rollers. The rolling and sifting continued until a practically pure flour was obtained. A pure flour is the modern miller's ideal. He wants a branless flour and a flourless bran. The old stone mill could not grind this kind of flour. Before the roller mill appeared there was always bran in the flour and flour in the bran.

The invention of the flour roller-mill (Fig. 14) is the last step in the development of the mill. The roller process has almost entirely driven out all other processes. Now and then we see by the roadside an old fashioned mill with the upper and nether stone, but we seldom see one that is prosperous and thriving. Millers, like everybody else in these days, do business on a large scale and to make flour on a large scale they must use the roller-mill. Thus the hole in the rock in which a handful of grain was laboriously crushed has, through long ages of growth, become the great factory in which thousands of barrels of flour are made in a day.

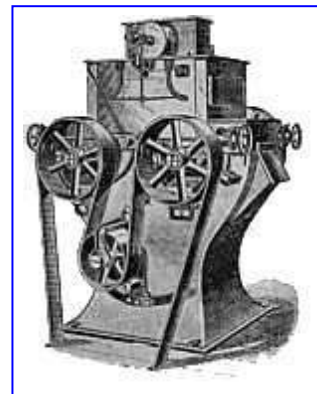


FIG. 14.—A MODERN
FLOUR ROLLER-
MILL.



A. H. C. 1850

THE LOOM

Have you ever seen a loom? It would not be a wonder if you have not. In these days the average person seldom sees one. Everyone knows in a vague sort of way that clothes and carpets are made of wool or silk or cotton, as the case may be, and that they are woven upon an instrument called a loom. This is about as much as we usually know about the clothes we wear or the carpets we walk upon. We buy these things from the store and that is all there is to it. In the olden times, and not so very long ago either, everybody knew something about weaving, at least every girl and woman knew something of the art, and a loom was as familiar an object in the household then as a sewing machine is now.

Matrons and maidens sat in snow-white caps and in kirtles
Scarlet and blue and green, with distaff spinning the golden
Flax for the gossiping loom, whose noisy shuttle within doors
Mingled their sounds with the whir of the wheels and the songs of the
maidens.

This picture of home life in Acadia two hundred years ago would have served as a picture of home life almost everywhere in the civilized world. From the beginning of history until modern times most of the weaving was done by the women in the home.

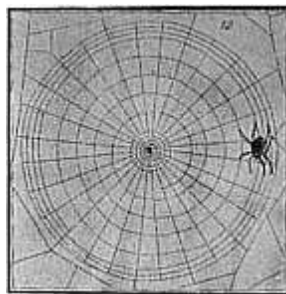


FIG. 1.—THE FIRST
LESSON IN
WEAVING.

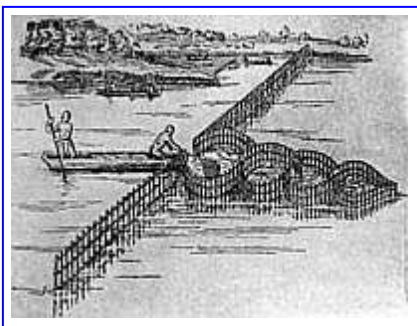


FIG. 2.—A WIER TRAP OF THE VIRGINIA INDIANS.

The earliest practical weaver on record is the spider and it may be that man learned his first lesson in weaving from this skilled little workman (Fig. 1); or the beautiful nest of the weaver-bird may have given to human beings the first hints in the weaving



FIG. 3.—PRIMITIVE BASKET MAKING.

art. Whoever may have been his teacher, it is certain that man learned how to weave in the earliest stages of existence. It is thought that his first effort in this direction consisted in making cages for animals and wiers (traps) for catching fish (Fig. 2) by interlacing vines or canes or slender boughs. The next step was taken when women began to make baskets and cradles and mats by interlacing long slender strips of wood (Fig. 3).

Basket weaving led to cloth weaving, and this led to the loom. In Figure 4 we see the simplest and oldest form of the loom. It consisted of a single stick (yarn beam) of wood about four feet long. This was the first form of the loom—just a straight stick of wood and nothing more. From the stick the threads which run lengthwise in the cloth were suspended. These threads are known as the *warp*. The threads which run breadthwise in the cloth are known as the *weft*, or *woof*. As the woman's deft fingers pass along with the weft she carries the thread over the first warp thread, under the second, over the third, under the fourth, and so on. Here we have not only the simplest form of the loom but the simplest kind of cloth.



FIG. 4.—THE PRIMITIVE LOOM.

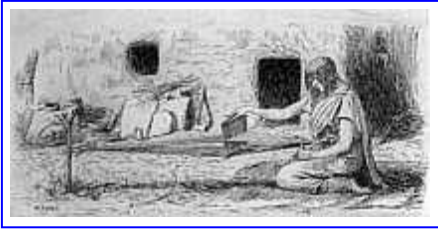


FIG. 5.—THE PUEBLO LOOM.

In the loom worked by the Pueblo woman (Fig. 5) a new piece appears. This is the frame through which the threads of the warp pass and which the woman is

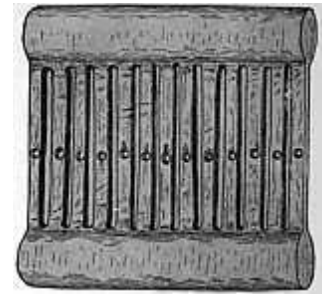


FIG. 6.—THE HEDDLE.

holding in her right hand. The frame is called a heald, or *heddle* (Fig. 6). The heddle is of the greatest importance in the construction of the loom and it is well worth while to understand what it does. In the loom operated by the Chilcoot woman (Fig. 4) you noticed that the weaver passed the weft thread above and below the alternate threads of the warp. This required a separate movement for every thread of the warp; if there were a hundred threads a hundred movements were required to pass the weft across once. Now the heddle used by the Pueblo woman separated the fifty warp threads that were to pass above the weft thread from the fifty that were to pass below it, making an opening called, a *shed*. When the shed was made the weft thread could be passed across at one movement. One movement instead of a hundred! How was this accomplished? Fifty alternate warp threads were passed through the holes in the bars of the heddle frame, one thread through each hole; the other fifty alternate threads passed between the bars of the heddle frame. Now suppose the entire warp of a hundred threads is stretched tight and firm between the woman's body and the yarn beam. With her right hand she *raises* the heddle and thus lifts the fifty threads which pass through the holes in the bars, while the other fifty threads remain unmoved. This movement makes the passage or shed through which she passes the weft with the left hand. After beating the weft thread close to the cloth either with the fingers or with a sword-like stick, she lowers the heddle with its fifty threads, the other fifty still remain fixed and unmoved. Another shed is formed and the weft is passed through again. Thus with the raising and lowering of the heddle the weft is passed backward and forward and the weaving goes on quite rapidly. If you care to do so you can make a Pueblo loom and can weave a belt on it.

In the old African loom represented in Fig. 7 we find several improvements upon the loom of the Pueblo woman. In the first place, it has two heddles instead of one. These are operated by the feet, leaving the hands



FIG. 7.—AN OLD AFRICAN LOOM.

free to do other work. In the second place, the wooden frame which the weaver holds in his right hand is not to be seen in the Pueblo loom. This frame called the *batten*, or *lathe*, contains the *reed*, which is a series of slats or bars between which the threads of the warp pass after they leave the heddle. When the weaver has thrown the weft through the shed he brings the batten down hard and the reed drives the last weft thread close to the woven part of the cloth. The reed takes the place of the sword-like stick used by the Pueblo woman. Last and most important: in the African's left hand is the *shuttle*, or little car—weaver's ship, the Germans call it—which carries the weft across (Fig. 8).

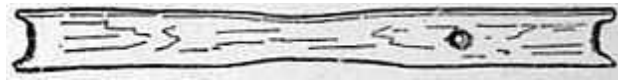


FIG. 8.—A PRIMITIVE SHUTTLE.

The loom described above seems to be clumsy and rude when compared with a loom of the present day, yet it is really the kind of loom which was used by nearly all civilized people from the dawn of their civilization to the middle of the eighteenth century. It is the loom of history and poetry and song. Upon a loom of this kind was woven Joseph's coat with its many colors and the garment which the fair Penelope made when she deceived her suitors. Of course as the centuries passed the parts of the loom were better made and weavers became more skilful. In Figure 9 we have the loom as it appeared in the sixteenth century. If we inspect it closely we shall find it to be merely the old African loom mounted on stout upright timbers instead of being mounted on a tripod made of poles. With her feet the weaver works the heddle, with her right hand she throws the shuttle, with her left she draws toward her the swinging batten and drives the weft home with the reed.



FIG. 9.—A LOOM OF THE SIXTEENTH CENTURY.

The year 1733 is a most important date in the development of the loom for in that year John Kay, a practical loommaker of Lancashire, England,

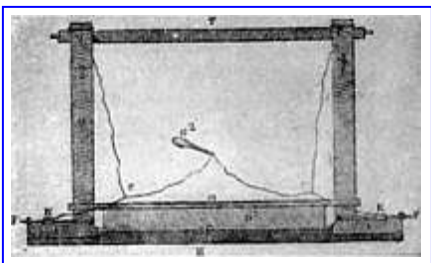


FIG. 10.—KAY'S FLYING SHUTTLE.

invented the flying shuttle and thus did more for the loom than any man whom we can distinguish by name. To appreciate the great service of Kay we must recall how the shuttle was operated before his time. You remember it was thrown through the shed by one of the weaver's hands and caught and returned by the other hand. Sometimes it was caught and returned by a boy. This was at best a slow process and unless the weaver had an assistant to return the shuttle only narrow pieces could be woven. The common width of cloth, three-fourths of a yard, had its origin in necessity. The weaver's arms were not long enough to weave a wider piece. "The essence of Kay's invention was that the shuttle was thrown from side to side by a mechanical device instead of being passed from hand to hand. One hand only was required for the shuttle while the other was left free to beat up the cloth (with the batten) after each throw, and the shuttle would fly across wide cloth as well as narrow." You will be able to understand Kay's invention by studying Figure 10 which shows how the flying shuttle worked. *G* is a groove (shuttle-race) on which the shuttle runs as it crosses through the shed leaving its thread behind it. *I* and *I* are boxes which the shuttle (Fig. 11) enters at the end of the journey. In each box is a driver *K* sliding freely on the polished rod *F*. The weaver with his right hand pulls the handle *H* and *K* drives the shuttle to the opposite side. With his left hand he works the reed, with his feet he works the heddle.

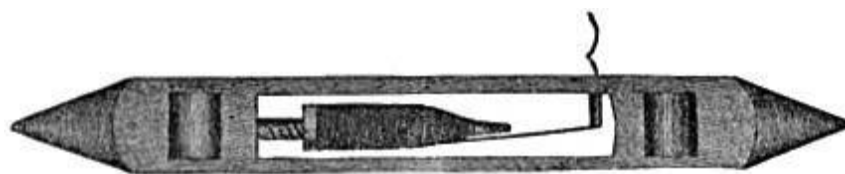


FIG. 11.—A MODERN SHUTTLE.

The profits of Kay's invention were stolen, his house was destroyed by a mob and he himself was driven to a foreign country where he died in poverty. Yet he deserves high rank among the benefactors of mankind, for the flying shuttle doubled the power of the loom and improved the quality of the cloth woven. Kay's invention was the first step in a great industrial revolution. The increased power of the loom called for more yarn than the

old spinning wheel could supply. Hargreaves and Arkwright set their wits to work and made their wonderful spinning machine, and the demands of the loom were supplied. So great was the supply of yarn that the hand loom was behind with its work. Then in order to keep up with the spinning machine the *power-loom* was invented. Heddle and batten and shuttle were now driven by a force of nature and all the weaver had to do was to keep the shuttle filled with thread and see that his loom worked properly. At first the water-wheel was used to drive the power-loom but later the steam-engine was made to do this work. All this was changing the face of the civilized world. Hitherto weavers and spinners had worked for themselves in their homes or in their own shops; now they were gathered in large factories where they worked as wage earners for an employer. Hitherto industry had been carried on in small villages; the great factories drew the people to large industrial centers and the era of crowded cities began.

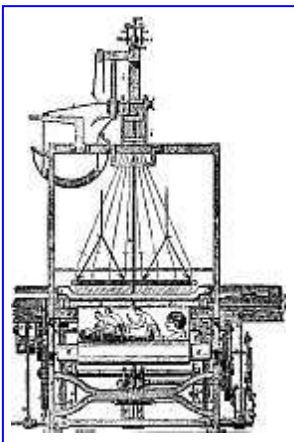


FIG. 12.—THE JACQUARD LOOM.

Following the invention of the power-loom in the latter half of the eighteenth century came the invention of Joseph Jacquard of Lyons, France. This very ingenious man in 1801 invented a substitute for the heddle. We cannot readily understand the workings of Jacquard's wonderful "attachment," as his substitute for the heddle is called, but we ought to know what the great Frenchman did for the loom. In Figure 12 you see that the cloth which is exposed shows that beautiful designs have been woven into it. This is what Jacquard did for the loom. He made it weave into the cloth whatever design, color or tint one might desire. He made the loom a mechanical artist rivaling in excellence the work of a human artist. The Jacquard loom has brought about a revolution in man's, and especially in woman's dress. With the old loom, colors and designs could be woven into cloth but only very slowly, and goods with fancy patterns were made at a cost that was so great that only the rich could afford to buy. In the olden times, therefore, almost everybody wore plain clothes. With Jacquard's attachment the most beautiful figures can be cheaply woven into the commonest fabrics. As far as weaving is concerned, it costs no more to have beautiful figures in cotton goods than it does to have them in silk. As a result the poor as well as the rich can dress as their taste and fancy may suggest.

The last century brought improvements in the weaving art as every century before it brought improvements, but the changes made since Jacquard's time need not concern us. The story of the loom ends with the Jacquard "attachment." Perhaps no other of man's inventions has a more interesting development than the loom. We can see it grow, piece by piece. First a simple stick from which dangle the threads of the warp; then the heddle, then the shuttle, then the reed, then the shuttle-race and the swiftly flying shuttle, and last the Frenchman's wonderful device for weaving in colors and fancy figures.

THE HOUSE

Man has always been a builder. Like squirrels and beavers and birds he provides himself a home as by instinct. The kind of house erected by a people in the beginning depended upon the surroundings, upon the enemies that prowled about, upon the climate, upon the building materials close at hand. In a hilly, rocky region primitive folk built one kind of house, in a forest they built another kind, in a low marshy district they built still another kind. In all cases they took the materials that were the easiest to get and erected the kind of dwelling place that would afford the greatest safety and comfort.

If one could have traveled over the earth during the first days of man's history one would doubtless have found that dwellings were made of wood, for in those days the greater part of the earth was covered with forests. To build a home in the forest was the simplest of tasks. All that was necessary was to fasten together the tops of several saplings, interlace the saplings with boughs (Fig. 1) and cover the frame with skins of animals or thatch it with leaves and grass. A cone-shaped structure of this pattern, a tent, or hut, or wigwam, was the first house of all primitive people who lived where there was plenty of wood.



FIG. 1.—BUILDING A HOUSE WITH WOOD.



FIG. 2.—A CAVE-DWELLING.

In many regions, especially in parts of northwestern Europe, the wigwam or hut was not always the most suitable dwelling place for early man. In hilly and mountainous districts and along streams where shores were overhung by rocks or pierced by caverns the first inhabitants found that a hollow in the earth was the best kind of house. Sometimes the house of

the cave-dwellers was made by Nature (Fig. 2); sometimes it was an artificial living-place dug in the side of a hill or mountain. The cave was truly a rude and gloomy home, yet there was a time when large numbers of the human race lived in caves. The Zuni Indians of Arizona in seeking a refuge from their enemies built their homes far up in steep cliffs where it was almost impossible for a stranger to go.

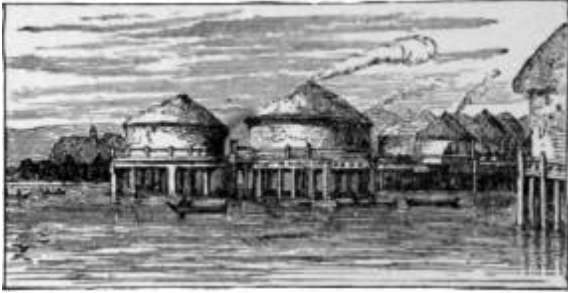


FIG. 3.—LAKE-DWELLINGS, RESTORED.
(From Troyon.)

Coming down from the highlands to the lowlands where there were swamps and marshes or where inland lakes were numerous, we find that the first houses were built upon piles driven in the water or in the mud (Fig. 3). These lake-dwellings, as houses of this kind were called, were generally connected with the mainland by gangways of wooden piers, although sometimes they could be approached only by boat. In the floors of some of these curious dwellings were trapdoors through which baskets could be lowered for catching fish in the lake below. The children of the lake-dwellers were tethered by the feet to keep them from falling into the water. The beautiful city of Venice in its infancy was a community of lake-dwellers. The rough canoe of the lake-dwelling time has developed into the graceful gondola, and the rude wooden pier has grown to be the magnificent Rialto.

In many regions the most convenient building material is stone and all over the earth there are proofs to show that building with stone began at a very early date. The stones in the earliest stone structures were rough and unhewn and were laid without mortar or cement (Fig. 4) yet they were sometimes fitted together with such nicety that a thin knife blade could not be passed between them. Remains of stone houses built many thousands of years ago may be seen in Peru, Mexico, Italy, and Greece. These primitive dwellings were humble and simple, but they were made of good material and they



FIG. 4.—A PRIMITIVE STONE HOUSE.

were well built. They have weathered the storms of ages and they have remained standing while later and more pretentious buildings have crumbled and disappeared.

The illustrations of early building which have been given will make plain the truth that the people of a particular country have taken the materials nearest at hand and have constructed their homes according to their particular needs. Now since the beginnings of house building have been different in different parts of the earth, the story of the house will not be the same in all countries. In China and Japan, where the light bamboo has always flourished and has always been used in building, the house has had one development; in countries where granite and marble and heavy timber abound it has had another and an entirely different development. What then is the story of the house as we see it in our country? Can this story be told? As one passes through an American city looking at the public buildings and churches and stores and dwellings can one go back to the beginning and trace step by step the growth of the house and tell how these came to be what they are? Let us see if this cannot be done.

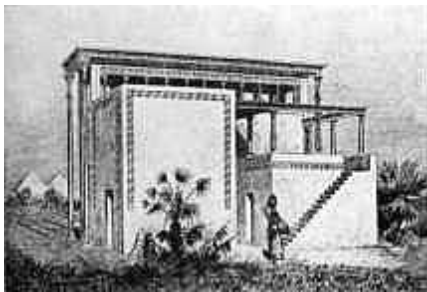


FIG. 5.—AN EGYPTIAN HOUSE.



FIG. 6.—AN ANCIENT HEBREW DWELLING.

Our story takes us back many thousands of years to Egypt, the cradle of civilization. From Egypt it will take us to Greece, thence to Rome, thence to the countries of Northern Europe, thence to America. What kind of houses did the Egyptians first build? They built as simple a structure as can be imagined; they erected four walls and over these they placed a flat roof (Fig. 5). The roof was made flat because in Egypt there is scarcely any rain and there was no need for a roof with a slant. In all those countries where rain seldom falls, or never falls, the flat roof is the natural roof (Fig. 6). Although their buildings were simple in construction the Egyptians left behind them most remarkable specimens of the builder's art. Their pyramids and monuments and sphinxes and palaces have always been foremost among

the great wonders of the world. Figure 7 shows the interior of an ancient

Egyptian palace. This palace had only an awning for a roof. That was all that was necessary to keep out the rays of the sun. Notice the lofty pillars or columns of this building. You see they are adorned above or below with the figure of the lotus, the national flower of the Egyptians. The column, as we shall see, plays an important part in the history of the house and it was ancient Egypt that gave the world its first lessons in the art of making columns.



FIG. 8.—A GREEK DWELLING.

From Egypt we pass over "the sea" to Greece. The Greeks borrowed ideas wherever they could and in the matter of architecture they borrowed heavily from Egypt. But they did not borrow the flat roof of the Egyptians. In Greece there was some rainfall and this fact had to be taken into account when building a house; the roof had to slant so that the rain could run off. Now the Greeks taught the world the best way to make a slanting roof. They made the roof to slant in two directions from a central ridge (Fig. 8) instead of having the entire roof to slant in one direction like an ugly shed. The slant was gentle because there was no snow to be carried off. The roof of two slants formed a gable. The Greeks, then, were the inventors of the *gable*. The column they borrowed from Egypt. But whenever the Greeks borrowed an invention or an idea they nearly always improved upon it. Instead of slavishly imitating the Egyptian columns they tried to make better ones and they were so successful that they soon became the teachers of the world in column making.

The oldest and strongest of the Greek columns belong to what is known as the Doric order (Fig. 9), a name given to them because they were first made by the Dorians, the original Greek dwellers in Europe. Aside from the flutes or channels which ran throughout its length the Doric column was perfectly plain. In the older Doric columns even the flutes are absent. Its *capital* or top, was without ornament. Later the graceful and elegant Ionic pillar (Fig. 9) came into fashion. We can always distinguish an Ionic column



FIG. 7.—INTERIOR OF AN ANCIENT EGYPTIAN PALACE.

by the volute or scroll at its capital. The latest of the Greek columns was the Corinthian (Fig. 9), the lightest, the most slender and the most richly decorated of all. A cluster of acanthus leaves at its capital is the most prominent ornament of the Corinthian column. The Greeks carried the art of column making to such perfection that even to this day we imitate their patterns. A column in a modern building is almost certain to be a Greek column. It is worth one's while, therefore, to be able to tell one Greek column from another. One can do this by remembering (1) that the Doric column is perfectly plain and has no capital; (2) that the Ionic column has a scroll at the capital; (3) that the capital of the Corinthian column is adorned with a cluster of acanthus leaves.

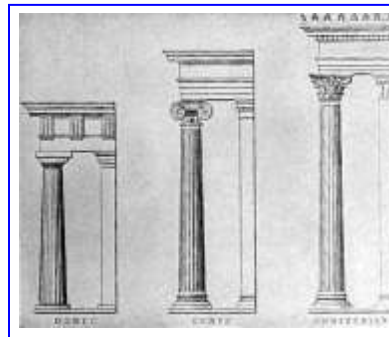


FIG. 9.—THE THREE ORDERS OF COLUMNS.



FIG. 10.—AN OLD ROMAN ARCH.

Our story now takes us to Italy. Greece fell before the power of Rome 146 B.C., but before she fell she had taught her conquerors a great deal about architecture. Indeed the Romans took up the art of building where the Greeks left it. They needed the

Greek gable for they had rains, and the Greek column recommended itself to them on account of its beauty. They used the best features of Grecian architecture and added a feature that was largely their own. This was the *arch*. The Greeks, like the Egyptians before them, bridged over the openings of doors and windows and the spaces between columns by means of straight wooden beams or long blocks of stone. The Romans bridged over these spaces with the arch (Fig. 10). If you will study the arch you will see that it is a curved structure which is supported by its own curve. You will also see that it is a structure of great strength. The greater the weight placed upon it, providing its bases are supported, the stronger it gets. In teaching the world how to make arches Rome added to the house an element of great strength and beauty. With the arch came the tall building. In Greece a house



FIG. 11.—INTERIOR OF A ROMAN CLUB HOUSE.

was never more than two stories high. In Rome arch rose upon arch (Fig. 11); the dome which is itself a kind of arch appeared and palaces were piled story upon story until they seemed to reach the skies.



FIG. 12.—A
DWELLING IN
NORTHERN EUROPE.

From Italy we pass to northern Europe. The power of Rome fell 476 A.D., but before that date the greater part of Europe had been Romanized, and the Roman way of building with column and arch and dome had been learned in France and Germany and England. But the climate of those countries was different from that of Italy and a slight change in the Roman way of building was necessary. In the northern countries there were heavy rains and snows and a roof with a gentle slope was not suitable for carrying off large quantities of water and snow. A gable (Fig. 12) with a sharp slant was necessary. Hence throughout northern Europe the roofs were built much steeper than they were in Italy and Greece, although in other respects the northern houses resembled more or less closely those of the older southern countries.

The pointed roof which was made necessary by the climate of the north prepared the way for a new style of building, the *pointed* or *Gothic* style. This style began to appear in the twelfth century and by the end of the thirteenth century—that remarkable century again—the buildings of all northern Europe were Gothic. The new style began with a change in the arch. The Roman arch was a semi-circle and was therefore described from one center. The Gothic arch was formed by describing it from two centers instead of one and was therefore a pointed arch. As the pointed arch grew in favor it became the fashion to shape other parts of the building into points wherever it was possible to do this. The rounding dome became a spire "pointing heavenward"; the windows and doors were pointed and so were the ornaments and decorations. For several centuries buildings fairly bristled with points (Fig. 13). The finest example of Gothic architecture is the glorious cathedral at Cologne.

During the thousand years of the Dark Ages (476-1453) the glories of the civilization of ancient Greece and Rome faded almost completely from human vision. Events of the sixteenth century brought those glories again



FIG. 14.—A
RENAISSANCE
DWELLING.

into view and Europe was dazzled by them. Men everywhere became dissatisfied with the things around them. They longed for ancient things. They read ancient authors, they imitated ancient artists, they imbibed the wisdom of ancient teachers. This was the period of the Renaissance, the time when the world was born anew—as it pleased men to think and say. The world of the present died and the old world of Greece and

Rome was brought to life. Of course in the new order of things architecture underwent a change. *It* was born again; *it* experienced a renaissance. The pointed style grew less pleasing to the builder's eye, and wherever he could he placed in his building something that was Greek or Roman, here an arched doorway, there a Greek column. There resulted from these changes a style that was neither Gothic, Grecian nor Roman, but a mixture of all these. This mixed style was named after the period in which it arose. When you see a building that strongly resembles the buildings of ancient Greece and Rome and at the same time has features which belong to other styles you may safely say that the building belongs to the renaissance style. (Fig. 14.) The most noble and beautiful examples of renaissance architecture are the church of St. Peter's at Rome and the church of St. Paul at London.



FIG. 13.—POINTED
STYLE.

Typical scheme of a fully developed French cathedral of the 13th century. (From Viollet-le-Duc's "Dict. de l'Architecture.")



FIG. 15.—A COLONIAL MANSION.

The Cliveden Chew Mansion, where the Battle of Germantown was fought.

We now pass over to America. About the time the old world was born anew the new world was found. The houses of the first settlers in America were of course rude and ugly but as the colonies grew in population and wealth more expensive and beautiful houses were built. As we should expect, the colonists built their best houses in the style that was then in fashion in the old world and that was the renaissance style. They did not, however, copy the old world architecture outright. They had different materials, a different climate and a different class of workmen and they had to build according to these changed conditions. The result was a style of building that has been called colonial (Fig. 15). Colonial architecture was simply American renaissance. And that is what it is to-day. To say that a house is in the colonial style is to say that it represents a certain architect's ideas as to what is best and most beautiful in all styles.

The story of the house really ends with the period of the renaissance. Since the sixteenth century nothing really new in architecture has been discovered and men have been wedded to no particular style. When we want to build a house we choose from all the styles and build according to our tastes. Our story of the house, however, will not be complete without a brief account of what has been called *elevator* architecture. The high price of land in large cities makes it necessary to run buildings up to a considerable height if they are to be profitable. Now if a building is more than five stories high it

must have an elevator, or lift, and if an elevator is to be put in, the building might as well be run up nine or ten stories. American business men learned this thirty or forty years ago and began to build high, and they have been building higher and higher ever since. There are tall buildings in other countries but the "sky-scraper" of twenty-five and thirty stories is found only in the United States (Fig. 16).



(Copyright 1911 by Underwood & Underwood, N. Y.)

FIG. 16.—ELEVATOR ARCHITECTURE.

The tower-like structure in the distance is a building more than forty stories in height.

Thus we may see in the house of to-day a long and unbroken story. Where the roof is flat it is Egyptian; where it slants gently in two directions it is Greek; where it is steep or sharply pointed it is Gothic. The columns are Greek, the rounded arches are Roman. The whole is the result of the thousands of years of effort which man has given to the task of providing for himself a safe, convenient and beautiful home.

THE CARRIAGE

We are very proud in our day of our means of transportation. If one wishes to send a present to a friend a thousand miles away a few cents spent in postage will take the article to its destination. If for the sake of higher prices a fruit grower wishes to sell his crops in a distant city, the railroad people will haul it for him at a very small cost. If you wish to visit a friend in town several blocks away, there is the electric car ready to take you for a nickel. If your friend is several hundred miles away, the steam car will take you in a few hours at a cost of not more than two or three cents a mile. I am living in the country sixteen miles from the city in which my work lies, and for nine cents I am carried to the place of my business in less than half-an-hour. What has been the history of the inventions which make transportation so comfortable, rapid and cheap? Our subject divides itself into two parts, transportation on land and transportation on water or the story of the Carriage and the story of the Boat. We will have the story of the carriage first.

Man's only carriage at first was of course his own feet. When he wanted to go to any place he had to take "Walker's hack," if a playful expression may be pardoned. As a traveler on foot, man soon surpassed all other animals. He could walk down the deer and wear out the horse. When it came to carrying things from place to place, in the beginning he had to rely upon his own limbs and muscles. It was not long, however, before he learned that there were good ways and bad ways of carrying things, and he soon set about finding the best way. We may believe that he began by making a snug bundle and carrying it on his shoulder. Then he found that he could carry a heavier burden upon his back, and he invented a pack or frame on which he could carry things on his back (Fig. 1) after the manner of one of our modern pack peddlers.



FIG. 1.—A
HUMAN
BURDEN
BEARER.
(From a
Model in



FIG. 2.—A SHIP OF
THE DESERT.

In the course of time man tamed one or more of the wild beasts which roamed near him. Then the burden was shifted from the back of a man to the back of a beast. The first beast of burden in South America was the llama; in India it was the elephant; in Arabia it was the camel (Fig. 2). In Europe and in parts of Asia and in Egypt the horse first became man's burden bearer and the nations which had the services of this swift and strong animal outstripped the other nations of the world. "Which is the most useful of animals?" asked one Egyptian god of another. "The horse," was the reply, "because the horse enables a man to overtake and slay his enemy."

National
Museum.)

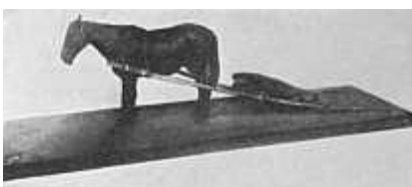


FIG. 3.—A CART WITHOUT
WHEELS.

(From a Model in the
National Museum.)

It is often easier to drag a thing along than it is to carry it. This fact led to the invention of what we may call the first and simplest form of carriage. This was the drag or travail (tra-vay'), a cart without wheels (Fig. 3). Two long saplings were fastened at the large end to the strap across the horse's breast and the small end upon which the burden was placed dragged upon the ground. Mr. Arthur Mitchell in his delightful book, "The Past in the Present," tells us that he saw carts of this kind in actual use in the highlands of Scotland as late as 1864! An improvement upon the travail was the sledge made of the forked limb of a tree (Fig. 4). This primitive sledge was really a travail consisting of one piece.



FIG. 4.—A PRIMITIVE SLEDGE.

(From a Model in
National Museum.)

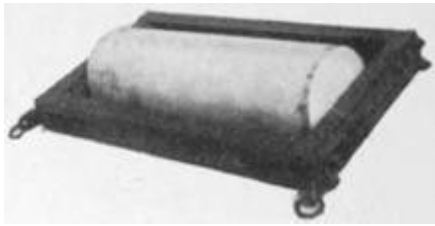


FIG. 5.—THE FIRST CART.

In many cases it is easier to roll a thing than it is to drag it. This fact led to another step in the development



FIG. 6.—HAULING
TOBACCO.

(From a Model in
National Museum.)

of the carriage; it led from the cart without wheels to a cart with a wheel—a most important step in the history of inventions. The first wheeled cart was simply a log from each end of which projected an axle (Fig. 5). The axle fitted in the holes of a frame upon which the body of the cart was placed and to which the horse or the ox was attached. As the cart moved along, wheel (log and axle) turned together. The very ancient method of moving a load by rolling it along was in use in the United States not so very long ago. As late as 1860 in some of the southern States hogsheads of tobacco (Fig. 6) were rolled over country roads in the manner just described and as late as 1880 the fishermen of Nantucket used as a fish cart a vehicle that had only a barrel for its wheel. (Fig. 7.) The common wheel-barrow and the one-wheeled carts which are still used in China and Japan had their origin in the rolling log.



FIG. 7.—A NANTUCKET FISH
CART.

(From a Model in the
National Museum.)

We are told by some writers that the rolling log (the one-wheeled cart) was followed by the two-wheeled cart, on which the wheels were the ends of a log and the axle was the middle portion of the log hewn down to a proper size (Fig. 8). Here wheels and axle turned together precisely like a modern

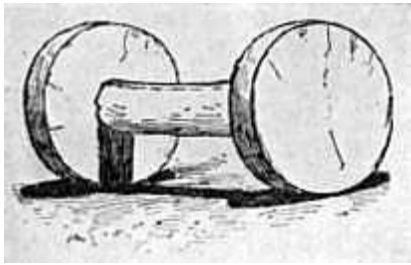


FIG. 8.—A CART WITH WHEELS AND AXLE IN ONE PIECE.

car wheel. This makes a very pretty story but I am afraid the solid two-wheeled affair represented in Figure 8 is only imaginary, and that in a true



FIG. 9.—CART WITH A SOLID WHEEL.

account of the development of the cart it has no place. The true beginning of the two-wheeled cart may be learned from Figure 9. Here the wheels are two *very short* logs through the center of which are holes in which the round ends (axles) of a piece of timber (the axle-tree) fit. When the cart moves, the wheels turn upon the axle. The one-wheeled cart had at first *one log* turning *with* the axle; the two-wheeled cart at first had as its wheels two very short logs turning *on* the axles.

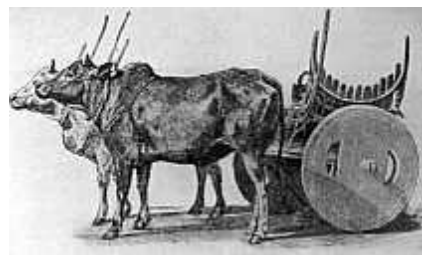


FIG. 10.—CART WITH WHEEL PARTLY SOLID.

(From a Model in the National Museum.)



FIG. 11.—WHEELS WITH SPOKES.
(From National Museum.)

The first two-wheeled carts were a great improvement upon the single rolling log, yet they were exceedingly heavy and clumsy. The trouble was with the wheel. This was very



FIG. 12.—AN ANCIENT EGYPTIAN CHARIOT SHOWING HUB, SPOKES, FELLY AND RIB.
(From National Museum.)

thick and with the exception of the hole in which the axle went it was entirely solid. Wheelwrights at a very early date saw that the problem was to make the wheel light and at the same time to keep it strong. Little by little this problem was solved. At first crescent-shaped holes were made in the wheel (Fig. 10). This made the wheel lighter, but did not weaken it. In its next form the wheel was even less solid than before. It now consisted of four curved pieces of wood (Fig. 11) held together by four spokes. In this wheel there was a hub, but the spokes were not inserted in it; they were fastened about it. In the Egyptian chariot (Fig. 12) we find the wheel in the last stage of its interesting and remarkable development. Here the spokes, six in number, are inserted in the hub from which they radiate to the six pieces of the felly or inner rim. Around the felly is the outer rim or tire made of wood and fastened to the felly with thongs. The wheel of to-day has more iron in it, and has more spokes and is lighter and stronger than the old Egyptian wheel, yet in its main features it is made like it.

A light running two-wheeled carriage was used by all the civilized nations of the ancient world. Three thousand years ago in the great and wicked city of Nineveh chariots raced up and down the paved streets "jostling against one another in the broad ways, with the crack of the whip, the rattle of the wheel and the prancing of horses." The chariot played an important part in the life of the Greeks and Romans, in their racing contests and in their wars, and throughout the Middle Ages it was the only vehicle in general use in Europe. As time passed it was of course made lighter and stronger and better. The doctor's gig so charmingly described by Holmes in his "Wonderful One Hoss Shay" may be taken as an illustration of the full development of the two-wheeled carriage (Fig. 13).



FIG. 13.—WONDERFUL ONE HOSS SHAY.
(From National Museum.)

Bring the hind part of one Egyptian chariot opposite to the hind part of another, lash the two chariots together, remove the tongue of one of the chariots and you have made a chariot of four wheels or a *coach*. The form of the most ancient of four-wheeled carriages leads to the belief that the coach was first made by joining together two two-wheeled chariots in the way just described. The ancient Egyptians had their four-wheeled chariots but only

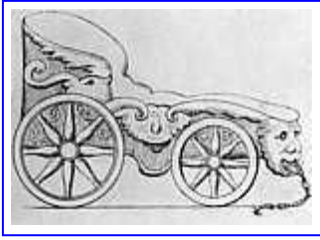


FIG. 14.—AN
ANCIENT ROMAN
CHARIOT.

their gods and their kings had the privilege of riding in them. For centuries none but the great and the powerful rode in coaches. The Roman chariot (Fig. 14), bad imitations of which we see nowadays in circus processions, was used only in the splendid triumphal processions which entered Rome after a great victory. In the Middle Ages we get a glimpse of a four-wheeled carriage now and then, but usually the king or a queen is lounging

in it (Fig. 15). The coach could not be generally used in Europe in medieval times because the roads were so bad. The excellent roads made by the Romans had not been kept in good condition. Traveling had to be done either on horseback or in the two-wheeled carriage. In 1550 there were but three coaches in Paris and in London there was but one. In 1564, however, we find Queen Elizabeth riding in a coach (Fig. 16)



FIG. 15.—A COACH OF
THE MIDDLE AGES.

on her way to see her lover, Lord Leicester. Insert more spokes and lighter ones in the wheels of this coach of the queen's, put on rubber tires and mount the body on elliptical springs¹⁷ and we will have the coach of to-day.



FIG. 16.—QUEEN ELIZABETH'S COACH.

THE CARRIAGE

Continued

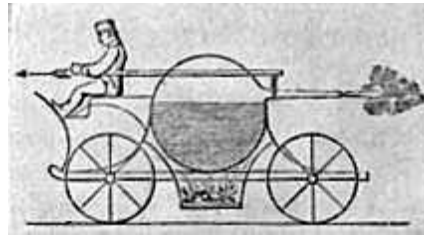


FIG. 1.—NEWTON'S STEAM CARRIAGE, 1680.

In the last chapter the story of the Carriage was brought up to the reign of Queen Elizabeth of England. In the century following Elizabeth's reign a new and most remarkable step in the development of the carriage was taken. You remember that in the seventeenth century there was a great deal of experimenting with steam (p. 58). Among other experiments was one made by Sir Isaac Newton. This great philosopher tried in 1680 to make a steam-carriage, or *locomotive*, as we call it. Figure 1 shows the principle upon which he tried to make his carriage work. The steam was to react against the air, as in the case of Hero's engine (p. 56) and thus push the carriage along. Newton's experiment was not satisfactory but the idea of a steam-carriage was now in men's heads and the hope of making one continued to be cherished. In 1769 Cugnot, a French army officer, invented a steam-carriage of three wheels (Fig. 2) but it was a very poor one. It traveled only three or four miles an hour, it could carry but three persons, and it had to stop every ten minutes to get up steam. Cugnot, however, deserves to be ranked among the great inventors for he showed that a steam-engine could be attached to a carriage and could push it along. In other words he showed that steam could



FIG. 2.—CUGNOT'S STEAM CARRIAGE, 1769.

be used for transportation as well as for working pumps and turning the wheels of factories. And that was just what was needed most in the latter part of the eighteenth century. Man needed assistance in traveling; he especially needed help in carrying things from place to place. The steam-engine was keeping the mines dry and making it possible to mine great quantities of coal and was turning the wheels of great factories where the spinning-jenny and the new power loom (p. 119) were consuming enormous quantities of cotton and wool. Now if the steam-engine could also be made to carry the coal and cotton and wool to the factory, and the manufactured products from the factory to the market, the industrial revolution would be complete indeed.

Inventors everywhere put their wits together to construct an engine that would draw a load. The great Watt tried to make one, but having failed, he came to the conclusion that the steam-engine could do good work only when standing still. Among those who entered the contest was Richard Trevithick, a Cornish miner, born in 1771. Trevithick when a lad at school was able to work six examples in arithmetic while his teacher worked one. He proved to be as quick in mechanics as he was in mathematics. He began his experiments with steam when a mere boy, and as early as 1796 he had built a steam-locomotive which would run on a table. By 1801 he had constructed a steam-carriage (Fig. 7). Three years later (1804) Trevithick exhibited a locomotive which carried ten tons of iron, seventy men, and five wagons a distance of nine and one-half miles at the rate of five miles an hour. This was the first steam carriage that actually performed useful work. The honor of inventing the first successful locomotive, therefore, belongs to Richard Trevithick, although he never received the honor that was due him.

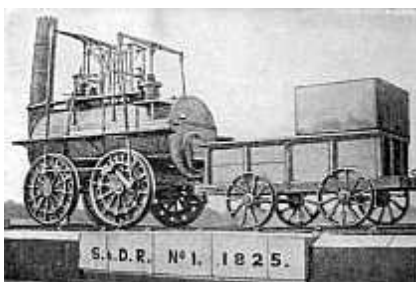


FIG. 3.—STEVENSON'S
LOCOMOTIVE, 1828.

The honor
went to George
Stephenson, of
Wylam, near
Newcastle,
England.

Stephenson's

parents were so

poor that they could not afford to send him to school long enough for him to learn to read and write. In his eighteenth year,



FIG. 4.—THE "BEST FRIEND."
THE FIRST LOCOMOTIVE
BUILT FOR ACTUAL
SERVICE IN THE UNITED
STATES.

however, he attended a night school and learned something of the common branches. In his childhood Stephenson lived among steam-engines. He began as an engine boy in a colliery and was soon promoted to the position of fireman. At an early age he was trying to build the locomotive that the world needed so badly, one that would do good work at a small cost. Trevithick's locomotive was too expensive. Stephenson wanted a locomotive that would pay its owner a profit. At the age of thirty-three he had solved his problem. In 1814 he exhibited a locomotive that would run ten or twelve miles an hour and carry passengers and freight cheaper than horses could carry them. Eleven years later he was operating a railroad between Stockton and Darlington, England. The steam carriage was now a success (Fig. 3). The iron horse was soon transporting passengers and freight in all the civilized countries of the world (Fig. 4). Observe that the first passenger car was simply the old coach joined to a locomotive.

The locomotive worked wonders in travel and in carrying loads, yet men were not satisfied with it. We never are satisfied with our means of transportation. No matter how comfortably or cheaply or fast we may travel we always want something better. In the latter part of the nineteenth century the great cities of the world were becoming over-crowded. The people could not be carried from one part of a city to another without great discomfort. The street cars drawn by horses could not carry the crowds and the elevated steam cars were not satisfactory. Wits were set to work to relieve the situation and about thirty years ago the *electric car* (Fig. 5) was invented. Without horse or locomotive this quick-moving car not only successfully handles the crowds which move about the city but it also relieves over-crowding by enabling thousands to reach conveniently and cheaply their suburban homes. It also does the work of the steam car and carries passengers long distances from city to city.



FIG. 5.—A TROLLEY CAR.



FIG. 6.—A HORSELESS
CARRIAGE OF THE
SIXTEENTH CENTURY.

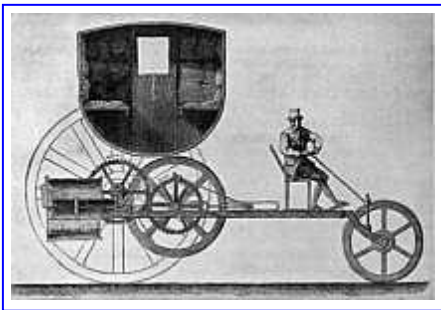


FIG. 7.—THE FIRST
AUTOMOBILE.



FIG. 8.—GOOD-BY TO THE
HORSE.

A late development in carriage making is seen in the automobile. As far back as the sixteenth century a horseless carriage was invented (Fig. 6) and was operated on the streets of a German city. But here the power was furnished by human muscle. The first real automobile (Fig. 7) was invented in 1801, by the man who invented the first successful locomotive. Trevithick's road locomotive—for that is what an automobile really is—did not work well because the roads upon which he tried it were in very bad condition. Inventors after Trevithick for a long time paid but little attention to the road locomotive; they bestowed their best thought upon the locomotive that was to be run upon rails—the railroad locomotive. In recent years, however, they have been working on the so-called automobile and they have already given us a horseless carriage that can run on a railless road at a rate as great as that of the fastest railroad locomotives. To what extent is this newest of carriages likely to be used? It is already driving out the horse. Will it also drive out the electric car and the railroad locomotive? Are we coming to the time when the railroad will be no more and when all travel and all hauling of freight will be done by carriages and wagons without horses on roads without rails? The answers to these questions can of course only be guessed.

The last and latest form of the carriage is seen in the *flying-machine*, the automobile of the air. In all ages men have watched with envy the movements of birds and have dreamed of flying-machines, but only in modern times has man dared to take wings and glide in bird-like fashion through the air. The first actual flying by a human being was done by a Frenchman named Bresnier, who, in 1675, constructed a machine similar to that shown in the right hand picture at the top of Figure 9. Bresnier worked his wings with his feet and hands. Once he jumped from a second story window and flew over the roof of a cottage. From the days of Bresnier on to the present time man has taxed his wits to the utmost to conquer the air, and in his efforts to do this he has invented almost every conceivable kind of machine. About the middle of the nineteenth century inventors began to apply steam to the flying-machine, and it is said that in 1842 a man named Philips was able, by the aid of revolving fans driven by steam, to elevate a machine to a considerable distance and fly across two fields. In 1896 Professor Langley, with a flying-machine driven by a small steam-engine, made three flights of about three-fourths of a mile each over the Potomac River, near Washington. This was the first time a flying-machine was propelled a long distance by its own power; it was the first aerial automobile. But the aerial steam carriage was never a success; the steam engine was too heavy. In the early years of the twentieth century inventors began to use the light gasoline engine to drive their flying-machines and then real progress in the art of flying began, and so great has been that progress that the automobiles of the air are becoming rivals of those on the land.

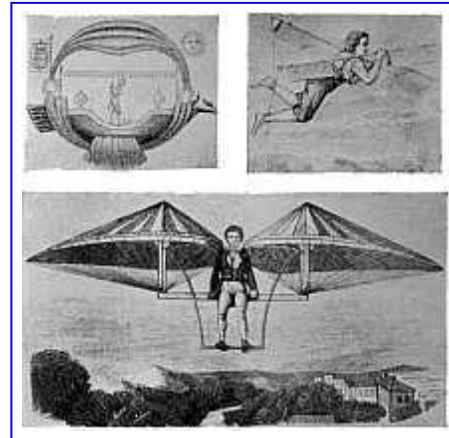


FIG. 9.—SOME UNSUCCESSFUL FLYING MACHINES OF A HUNDRED YEARS AGO.

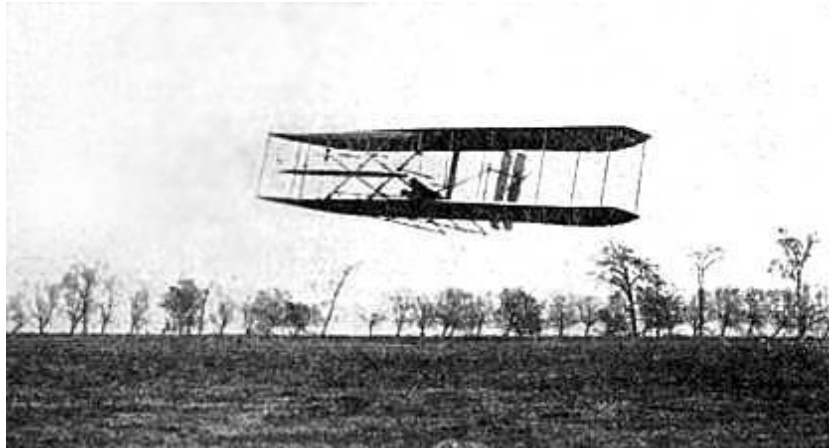


FIG. 10.—A SUCCESSFUL FLYING MACHINE OF TO-DAY.

THE BOAT



FIG. 1.—THE FIRST BOAT.

At first, when a man wanted to cross a deep stream, he was compelled to swim across. But man at his best is a poor swimmer, and it was not long before he invented a better method of traveling on water. A log drifting in a stream furnished the hint. By resting his body upon the log and plashing with his hands and feet he found he could move along faster and easier. Thus the log was the first boat and the human arm was the first oar. Experience soon taught our primitive boatman to get on top of the log and paddle along, using the limb of a tree for an oar (Fig. 1). But the round log would turn with the least provocation and its passenger suffered many unceremonious duckings. So the boatman made his log flat on top. It now floated better and did not turn over so easily. Then the log was made hollow, either by burning (Fig. 2), or by means of a cutting instrument. Thus the canoe was invented. Very often if the nature of the tree permitted it, the log was stripped of its bark, and this bark was used as a canoe.



FIG. 2.—THE INVENTION OF THE CANOE.

The canoe was one of the earliest of boats, but it is not in line with the later growth. The ancestry of the modern boat begins with the log and is traced through the raft rather than through the canoe. By lashing together



FIG. 3.—THE RAFT—
SHOWING ALSO
EARLY USE OF THE
SAIL.

several logs it was found that larger burdens could be carried. Therefore the boat of a single log grew into one of several logs—a raft (Fig. 3). By the time man had learned to make a raft he had learned something else: he had learned to row his

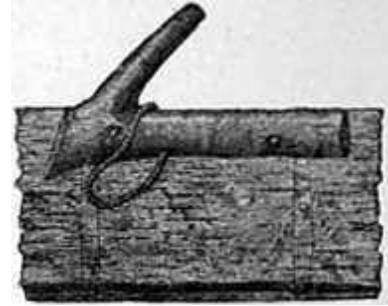


FIG. 4.—A PRIMITIVE
OARLOCK.

boat along by pulling at an oar instead of pushing it along with a paddle. But in order to row there must be something against which the oar may rest; so the oarlock (Fig. 4) was invented. Rafts were used by nearly all the nations of antiquity. Herodotus, the father of history, tells us that they were in use in ancient Chaldea. In Figure 3 we have a kind of raft that may still be seen on some of the rivers of South America. Here a most important step in boat-building has been taken. A *sail* has been hoisted and one of the forces of nature has been bidden to assist man in moving his boat along.

The raft was bound to develop into the large boat. The central log was used as a keel and about this was built a boat of the desired shape and size. Stout timbers, called ribs, slanted from the keel, and on the ribs were fastened planks running lengthwise with the vessel. To keep out the water the seams between the planks were filled with pitch or wax. Thus the raft grew into a large spoon-shaped vessel (Fig. 5). The early boat was usually propelled by oars, although a single sail sometimes invoked the assistance of the wind. It had no rudder and no deck, and if there was an anchor it was only a heavy stone.

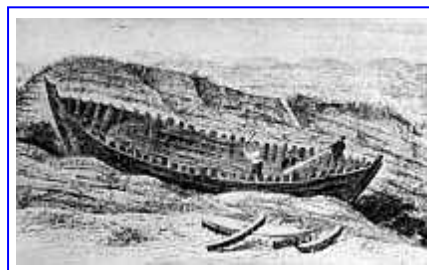


FIG. 5.—"THUS THE RAFT
GREW INTO A LARGE,
SPOON-SHAPED VESSEL."



FIG. 6.—THE PO-
SITION OF THE
RUDDER IN
ANCIENT
TIMES.

In the early history of the boat there was no such thing as a rudder. The oarsman had to steer his craft as best he could. With the appearance of larger boats, however, a steersman comes into view. He steers by means of a paddle held over the stern of the boat. Within historic times, probably about the time of Homer (1100 B. C.), the rudder appears as an oar with a broad blade protruding through a hole in the side of the boat well to the stern (Fig. 6). Throughout the whole period of ancient history boats were steered by rudders of this kind.

The anchor came later than the rudder. Of course even in primitive times there were methods of securing the vessel to the ground under water but they were very crude. Sometimes a sack of sand was used as an anchor, sometimes a log of wood covered with lead was thrown overboard to hold the boat in its place. In Homer's time the anchor was a bent rod with a single fluke. About 600 B. C. Anacharsis, one of the seven wise men of Greece, gave a practical turn to his wisdom and invented an anchor with two flukes (Fig. 7). The invention received the name of "anchor" from the name of the inventor.



FIG. 7.—ANCIENT
ANCHORS.



FIG. 8.—A ROMAN
GALLEY OF ONE TIER
OF OARS, INTRODUC-
ING THE RUDDER.

It was in the Mediterranean Sea that the boat had its most rapid development. As early as we can get a glimpse of that wonderful body of water it was alive with boats (called galleys) that had well-laid keels and lofty sides, and rudders, and sails. The greatest of the earlier navigators were the Phœnicians whose boats had traversed 5,000 years ago the whole course of the Mediterranean and had even ventured beyond the Straits of Gibraltar. The ancient Greeks also were a great sea-going people, and their merchantmen or trading boats visited every part of the known world. But it was the Romans who at last became

masters of the ancient seas. The Roman galley, therefore, may be taken as the representative boat of ancient times. What kind of a boat was the Roman galley? It was propelled chiefly by oars, just as nearly all the boats of antiquity were. Occasionally a sail was hoisted when the wind was favorable but the main reliance was the rower's arm. Men had not yet learned to use the sail to the best advantage. The older galleys had one row of oarsmen (Fig. 8), but as the struggle for the mastery of the sea became keener the boats were made larger and more rowers were necessary. Galleys with two and three, and even four rows of oarsmen were built by the Roman navy. When there was more than one row of oars the rowers sat on benches one above another. The oarsmen were slaves or prisoners captured in war, and their life was most wretched.¹⁸ They were chained to the benches on which they sat, and were compelled to row as long as a spark of life was left. Sometimes they dipped their oars to the music of the flute, but more often it was to the crack of the lash. Figure 9 shows us how the Roman galley looked when Rome was at the height of her power (100 A. D.). Here is a vessel about 400 feet long and about 50 feet across its *deck*, a part of the boat, by the by, which was not to be seen in the earlier galleys. The boat is a trireme, that is, it has openings for three tiers of oars, and it is propelled by several hundred oarsmen. For steering purposes it has four stout paddles, two on each side near the stern. Two masts instead of one carry the sail which, considering the size of the boat, would seem to be insufficient. This galley of the first century of our era represents the full development of the boat in ancient times.



FIG. 9.—A ROMAN GALLEY
WITH THREE BANKS OF
OARS.



After the downfall of Rome (476 A. D.) it was a long time before there was any real progress in boat-making. The glimpses we get now and then of vessels in the Middle Ages almost make us feel that boat-building was going backward rather than forward. But such was not the case. The ship in which William of Normandy sailed (Fig. 10) when he crossed over the Channel to give battle to Harold (1066 A. D.) was not so impressive as a Roman

FIG. 10.—THE SHIP IN WHICH WILLIAM THE CONQUEROR CROSSED THE CHANNEL IN 1066.

galley, yet it was, nevertheless, a better boat. In the first place William's boat was a better sailer; it relied more upon the force of the wind and less upon the oar. In the second place, it could be steered better, for the rudder had found its way to its proper place and was worked by a tiller. Finally, the shape of the Norman boat fitted it for fiercer battles with the waves.

If we should pass from the English Channel to the Adriatic we should find that boat-making had undergone the same changes. A Mediterranean galley of the fourteenth century (Fig. 11) shows fewer oars and more sails. Instead of three rows of oars and two sails as on the Roman galley, there are three sails and one row of oars. This was the tendency of the boat-builder in the Middle Ages; he crowded on the sail and took off the rowers. A war-boat of the sixteenth century (Fig. 12) shows that the last row of oarsmen has disappeared.

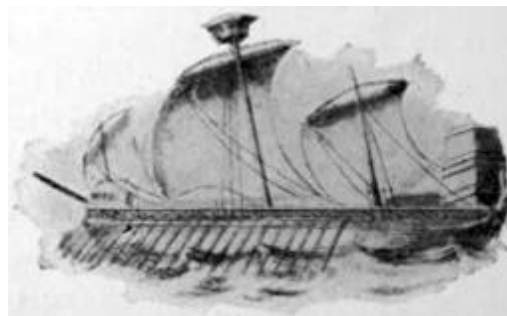


FIG. 11.—A MEDITERRANEAN GALLEY OF THE 14TH CENTURY.



FIG. 12.—A WAR-BOAT OF THE 16TH CENTURY, SHOWING THAT THE LAST ROW OF OARS HAD DISAPPEARED.

About the middle of the thirteenth century there began to appear on the decks of vessels almost everywhere in Europe, a little instrument that is of the greatest importance in the history of the boat. This was the *mariner's compass*. The use of the magnetic needle was known in China (Fig. 13) a thousand years before it was known to the Europeans, but in this, as in many other instances, the Chinese did not profit by their knowledge. Sailors have always sailed at night by the North star; but before the use of the compass was understood they could little more than guess their way when the night was dark and the stars could not be seen. With a mariner's needle on board they can tell the direction they are going no matter how dark the night. We can easily understand that sailors prized very highly the discovery of the compass. With the appearance of this faithful guide they became bolder and bolder and were soon venturing out upon the trackless expanse of the ocean. It was the compass that led to the discovery of the new world, for without it no sailor could have held his course due west long enough to reach the American coast.



FIG. 13.—A CHINESE COMPASS. AS THE CART MOVED THE HUMAN FIGURE IN FRONT ALWAYS POINTED NORTH.

After men had learned to carry their burdens on the broad back of the ocean, boat-building took on new life. All the great nations of Europe wanted a share in the new world that had just been found; but no nation could hope to profit greatly by the discovery of Columbus if its vessels were not swift and strong. So there arose a grim contest for the mastery of the Atlantic, just as in ancient times there had been a struggle for the mastery of the Mediterranean. Spain, France, Portugal, Holland and England all joined in the battle. When we see the kind of boats she sent out upon the oceans we are not surprised that England won. Compare the heavy, angular galley of the first century with the graceful ship of the sixteenth century and we see at once the progress the boat made in the Middle Ages (Fig. 14).

The log, the raft, the galley, the sailing-ship, these were the steps in the development of the boat up to the end of the seventeenth century. In the eighteenth century another step was taken. You remember that in that century inventors were everywhere trying to make a steam carriage. They were at the same time trying to make a steam boat. Their efforts to use steam



FIG. 14.—THE GREAT HARRY.

to drive boats were rewarded with success earlier than were their efforts to use it to draw carriages. This was to be expected. Boat-building has always moved along faster than carriage-building. Men were gliding about in well-built canoes before they had even the clumsiest of carts. The Londoners who gazed with admiration upon the *Great Harry* as it sailed on the Thames, had never seen as much as a lumbering coach. And so with the steamboat; it had crossed the

Atlantic before the locomotive could carry passengers from one town to the next.

France, England, Germany and America were all eager to have the first steamboat. In this race America won, although France and England came out with their colors flying. As far back as 1663 the Marquis of Worcester, of whom we have heard before (p. 59), described a vessel that could be moved by steam: "It roweth," he said, "it draweth, it driveth (if needs be) to pass London bridge against the stream at low water." It was one thing, however, to describe a steamboat, and quite another thing to make one. Worcester's steam-vessel existed only in the imagination of the inventor. Denys Papin, who did so much for the steam-engine, fitted out a boat with revolving paddles which were turned by horses. This was nothing new. The ancient Roman galley was sometimes propelled by paddle-wheels turned by horses or oxen. It is sometimes claimed that Papin turned the paddle-wheels of his boat by means of steam, but there are no grounds for the claim. If France wants the honor of having made the first steamboat she would do better to turn from Papin and look to Marquis of Jouffroy of Lyons, This nobleman, it is claimed, built a steamboat (Fig. 15) which made a successful trip on the river Soane, in the year 1783, before a multitude of witnesses. This claim may or may not be just. It may be as the French say: the boat after the trial trip may have been taken to pieces, the model may have been lost and the French Revolution may have swallowed up those who witnessed the trip.



FIG. 15.—THE MARQUIS OF JOUFFROY'S STEAMBOAT, 1783.

About the time the Frenchman is said to have been experimenting with his steamboat on the Soane similar experiments were being tried in many other places. In the latter part of the eighteenth century the idea of a steam-propelled boat seemed to be in the air. An English poet of the time was bold enough to prophesy:

Soon shall thy arm, Unconquered Steam, afar
Drag the slow barge and draw the rapid car,
Or on wide, waving wings, expanded bear
The flying chariot through the fields of air.

For the most part the prophesy has been fulfilled, although the steam flying-machine is not yet an accomplished fact. Among those who helped to make good the words of the poet was James Rumsey, of Sheppardtwn, Virginia. Rumsey in 1786 propelled, by means of steam, a boat on the Potomac River moving at the rate of five miles an hour. It is almost certain that this was the first boat ever drawn by steam. How did Rumsey drive his boat? A piston in a cylinder was worked by a steam-engine. When the piston was raised it brought water in and when it was pushed down it forced the water out behind and the reaction of the jet pushed the boat along. A remarkable revival of a very ancient idea! Just as Hero turned his globe by reaction, just as Newton pushed the first steam carriage along by reaction, so Rumsey pushed the first steamboat along by reaction.

If you will look on a map of the United States and observe the vast network of waterways which come to the different parts of the country you will understand how important a subject steam navigation must have been to the people of America in the latter part of the eighteenth century. Here was a tract of land containing millions upon millions of fertile acres, but it lacked good roads, and without roads it could not be developed. It was, however, traversed by thousands of miles of excellent water-roads and it was plain that if steamboats could be put upon these rivers the gain would be incalculable. The most pressing need of the time, therefore, was a steamboat. No one saw this more clearly than John Fitch. This talented but eccentric man served his country in the Revolution, and after the war was over roamed hither and thither for several years as a soldier of fortune. About 1785 he went to Philadelphia with a plan for a steamboat. He organized a company, and secured enough money to enable him to carry out his plans.

His boat was ready by August, 1787, and he made his trial trip in Philadelphia when the Constitutional Convention was in session. Many of the members of that distinguished body went down to the river to see how the new invention worked. It worked fairly well, but did not arouse much enthusiasm. Its speed was only three or four miles an hour and its movement was exceedingly awkward. It was pushed along by two sets of oars, one set entering into the water as the other came out. The steam rowboat of 1787 proved at least to be a failure, and was abandoned as worthless. Fitch afterward built another steamboat, but it also met with accident and came to naught. Heartbroken by his many failures the poor fellow at last ended his life with his own hand. He deserved a better fate, for his experiments taught the world a great deal about the steamboat.

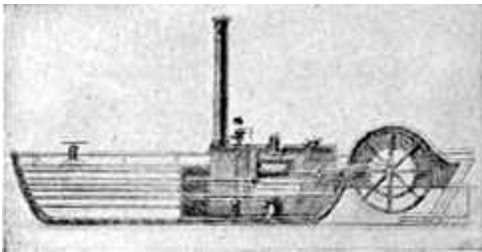


FIG. 16.—THE CHARLOTTE DUNDAS, 1802.

While Rumsey and Fitch were making their boats in America, European inventors were not idle. On the contrary they were so very active that they almost won the honor of making the first successful boat. One of these, William Symington, an Englishman, built a boat that may, with much justice, be called the first practical steamboat that was ever launched. This was the *Charlotte*

Dundas (Fig. 16) which made its trial trip on the Clyde and Firth Canal in 1802. On the *Charlotte* was a *paddle-wheel* instead of Fitch's two sets of paddles. The wheel was placed at the rear of the boat and was drawn by means of a crank which was turned by a rod attached to the piston-rod. Watt and his co-workers, a few years before, had shown how the steam-engine could be made to turn a wheel and Symington in the construction of his boat put this principle to good use. The *Charlotte* did so well that the Duke of Bridgewater ordered eight more boats like her to be built for use on the canal. Symington was elated for he thought he had at last made a successful steamboat, that is, a steamboat that would give to its owner a profit; but he was doomed to disappointment for the owners of the canal refused to allow steamboats to be employed upon it, and worse than this the duke soon died and the inventor's financial support was gone. The *Charlotte* was taken off the canal and laid in a creek where she fell to pieces. The really successful steamboat had not yet been built.

It was to be built first where it was needed most, and that was in America. It was built by a man who kept his eyes on Rumsey and Fitch and Symington, and made the best of what he saw. As all the world knows, this was Robert Fulton. In August of 1807 Fulton's steamboat the *Clermont*

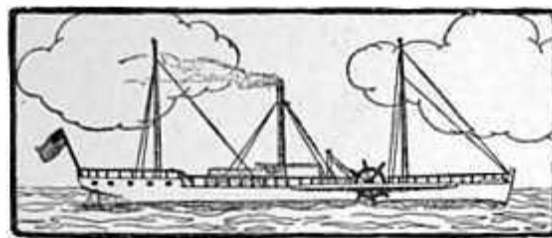


FIG. 17.—FULTON'S STEAMBOAT, CLERMONT.

(Fig. 17) made a trip on the Hudson River from New York to Albany, a distance of 150 miles, in thirty-two hours, and returned in thirty hours. Fulton advertised for passengers, and his boat was soon crowded. "The *Clermont*," says an English writer, "was the steamboat that commenced and continued to run for practical purposes, and for the remuneration of her owners." Here was the boat that was wanted—one that was financially profitable.

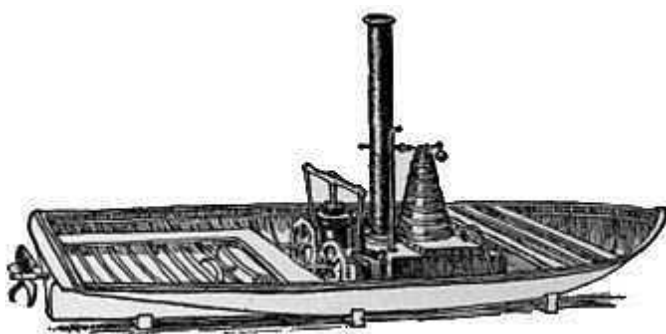
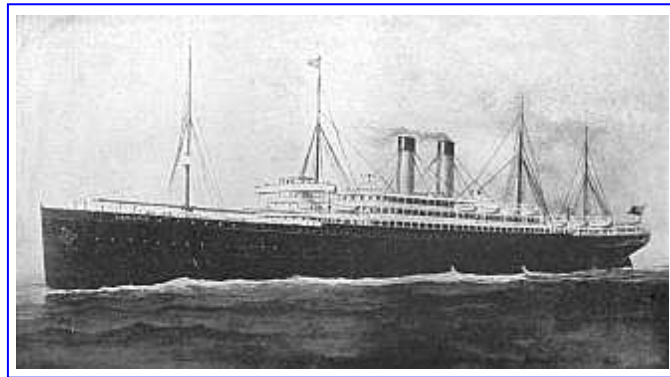


FIG. 18.—THE BOAT OF STEVENS.

The paddle-wheels of the *Clermont* were on the sides of the boat about midship. As the wheel turned, about half of it was in the water and about half was out. There were engineers, even in Fulton's day who did not believe the wheels ought to be on the sides of the boat. Look at waterfowl, they said,

look at the graceful swan; its feet do not work at its sides, half under the water and half out. Every animal that swims propels itself from behind, and its propellers are entirely under the water. So, thought these engineers, the paddle-wheel of a boat should be placed behind, and should be entirely covered by the water. John Stevens, an engineer of Hoboken, New Jersey, in 1805 built a steamboat according to this notion (Fig. 18). A close inspection of the wheel of the boat would show that it is spiral- or screw-like in shape. Stevens' boat made a trial trip on the Hudson and worked well; but after Fulton's great success the little steamer with its spiral-shaped wheel in the rear was soon forgotten. The idea of a screw-propeller, however, was not lost. It was taken up by John Ericsson, a Swedish engineer, who, in 1839, built, in an English shipyard for an American captain, the first screw-

propeller that crossed the Atlantic—the *Robert F. Stockton*. This was the last step in the development of the boat. Since 1839 there has been marvelous progress in ship-building, but the progress has consisted in improving upon the invention of Ericsson rather than in making new discoveries. With the screw-propeller in its present form we may close our story of the boat. The homely log propelled by rude paddles has become the magnificent floating palace.



THE ADRIATIC AT SEA.

THE CLOCK

"Tic-tac! tic-tac! go the wheels of time. We cannot stop them; they will not stop themselves." Time passing is life passing and the measurement of time is the measurement of life itself. How important then that our chronometers, or time measures, be accurate and faithful! It is said that a slight error in a general's watch caused the overthrow of Napoleon at Waterloo and thus changed the history of the world. Because of its great importance the measurement of time has always been a subject of deep human interest and the story of the clock begins with the history of primeval man.

The larger periods of time are measured by the motion of the heavenly bodies. The year and the four seasons are marked off by the motion of the earth in its long journey around the sun; the months and the weeks are told by the changing moon; sunrise and sunset announce the coming and the going of day. The year and the seasons and the day were measured for primeval man by the great clock in the heavens, but how were smaller periods of time to be measured? How was the passing of fractional parts of a day, an hour or a minute or a second to be noted? An egg was to be boiled; how could the cook tell when it had been in the water long enough? A man out hunting wished to get back to his family before dark: how was he to tell when it was time to start homeward?



FIG. 1.—A
PRIMITIVE
SUN-DIAL.

Plainly, the measurement of small portions of time was a very practical problem from the beginning. The first attempt to solve the problem consisted in observing shadows cast by the sun. The changing shadow of the human form was doubtless the first clock. As the shadow grew shorter the observer knew that noon was approaching; when he could reach out one foot and step on the shadow of his head he knew it was time for dinner; when his shadow began to lengthen he knew that evening was coming on. Observations of this kind led to the *shadow clock* or *sun-dial* (Fig. 1). You can make one

for yourself. On a perfectly level surface exposed all day to the sun, place in an upright position (Fig. 1) a stick about three feet long, and trace on the surface the shadows as they appear at different times of the day. A little study will enable you to use the shadows for telling the time. Sun-dials have been used from the beginning of time and they have not yet passed out of use. They may still be seen in a few public places (Fig. 2), but they are retained rather as curiosities than as real timekeepers.

For the sun-dial is not a good timekeeper for three reasons: (1) it will not tell the time at night; (2) it fails in the daytime when the sun is not shining; (3) it can never be used inside of a house.



FIG. 2.—A MODERN SUN-DIAL.

The sun-dial can hardly be called an invention; it is rather an observation. There were, however, inventions for measuring time in the earliest period of man's history. Among the oldest of these was the fire-clock, which measured time by the burning away of a stick or a candle. The Pacific islanders still use a clock of this kind. "On the midrib of the long palm-leaf they skewer a number of the oily nuts of the candle-nut-tree and light the upper one." As the nuts burn off, one after another, they mark the passage of equal portions of time. Here is a clock that can be used at night as well as in the daytime, in the house as well as out of doors. Mr. Walter Hough tells us that Chinese messengers who have but a short period to sleep place a lighted piece of joss-stick between their toes when they go to bed. The burning stick serves both as a timepiece and as an alarm-clock.

Fire-clocks of one kind or another have been used among primitive people in nearly all parts of the globe, and their use has continued far into civilized times. Alfred the Great (900 A. D.) is said to have measured time in the following way: "He procured as much wax as weighed seventy-two pennyweights, which he commanded to be made into six candles, each twelve inches in length with the divisions of inches distinctly marked upon it. These being lighted one after another, regularly burnt four hours each, at the rate of an inch for every twenty minutes. Thus the six candles lasted twenty-four hours."¹⁹

We all remember Irving's account of time-measurement in early New York: "The first settlers did not regulate their time by hours, but pipes, in the same manner as they measure distance in Holland at this very time; an admirably exact measurement, as the pipe in the mouth of a true-born Dutchman is never liable to those accidents and irregularities that are continually putting our clocks out of order." This, of course, is not serious, yet it is an account of a kind of fire-clock that has been widely used. Even to-day the Koreans reckon time by the number of pipes smoked.



FIG. 3.—AN EARLY FORM OF THE WATER-CLOCK.

If we could step on board a Malay proa we should see floating in a bucket of water a cocoanut shell having a small perforation through which the water by slow degrees finds its way into the interior. This orifice is so perforated that the shell will fill and sink in an hour, when the man on watch calls the time and sets it to float again. This sinking cocoanut shell, the first form of the water-clock, is the clock from which has been developed the timepiece of to-day. With it, therefore, the story of the clock really begins. In Northern India the cocoanut shell is replaced by a copper bowl (Fig. 3). At the moment the sinking occurs the attendant announces the hour by striking upon the bowl.

The second step in the development of the water-clock was made in China several thousand years ago. In the earlier Chinese clock the water, instead of finding its way into the vessel from the outside, was placed inside and allowed to trickle out through a hole in the bottom and fall into a vessel below. In the lower vessel was a float which rose with the water. To the float was attached an indicator which pointed out the hours as the water rose. By this arrangement, when the upper vessel was full, the water, by reason of greater pressure, ran out faster at first than at any other time. The indicator, therefore, at first rose faster than it ought, and after a while did not rise as fast as it ought to. After centuries of



FIG. 4.—CHINESE WATER-CLOCK AT CANTON.

experience with the two-vessel arrangement, a third vessel was brought upon the scene. This was placed above the upper vessel, which now became the middle vessel. As fast as water flowed from the middle vessel it was replaced by a stream flowing from the one above it. The depth of the water in the middle vessel did not change, and the water flowed into the lowest vessel at a uniform rate. Finally a fourth vessel was brought into use. The Chinese water-clock shown in (Fig. 4) has been running in the city of Canton for nearly six hundred years. Every afternoon at five, since 1321, the lowest jar has been emptied into the uppermost one and the clock thus wound up for another day.



FIG. 5.—
AN EARLY
GREEK
CLEPSYDR
A.

To follow the further development of the water-clock we must pass from China to Greece. In their early history the Greeks had nothing better than the sun-dial with which to measure time. About the middle of the fifth century B. C. there arose at Athens a need for a better timepiece. In the public assembly the orators were consuming too much time, and in the courts of law the speeches of the lawyers were too long. It was a common thing for a lawyer to harangue his audience for seven or eight hours. To save the city from being talked to death a time-check of some kind became necessary. The sun-dial would not answer, for the sun did not always shine, even in sunny Greece; so the idea of the water-clock was borrowed. A certain amount of water was placed in an amphora (urn), in the bottom of which was a small hole through which the water might slowly flow (Fig. 5). When the amphora was empty the speaker had to stop talking. The Greeks called the water-clock a *clepsydra*, which means "the water steals away." The orator whose time was limited by a certain amount of water would keep his eye on the clepsydra, just as a speaker in our time keeps his eye on the clock, and if he were interrupted he would shout to the attendant, "You there, stop the water," or would say to the one who interrupted him, "Remember, sir, you are in my water." The story goes that upon one occasion the speaker stopped every now and then to take a drink; the orator's speech, it seems, was as dry as his throat, and a bystander cried out: "Drink out of the clepsydra, and then you will give pleasure both to yourself and to your audience."

At first the Greeks used a simple form of the clepsydra, but they gradually adopted the improvements made by the Chinese, and finally added others. The great Plato is said to have turned his attention to commonplace things long enough to invent a clepsydra that would announce the hour by playing the flute. However this may have been, there was in use in the Greek world, about 300 B. C., a clepsydra something like the one shown in Fig. 6. This begins to look something like a clock. As the water drops into the cylinder *E* the float *F* rises and turns *G*, which carries the hour hand around. Inside of the funnel *A* is a cone *B* which can be raised or lowered by the bar *D*. In this way the dropping of the water is regulated. Water runs to the funnel through *H*, and when the funnel is full the superfluous water runs off through the pipe *I*, and thus the depth of the water in the funnel remains the same and the pressure does not change. Notice that when the hand in this old clock has indicated twelve hours it begins to count over again, just as it does on our clocks to-day. How easily it would have been to have continued the numbers on to twenty-four, as they do in Italy, and on the railroads in parts of Canada, to-day.

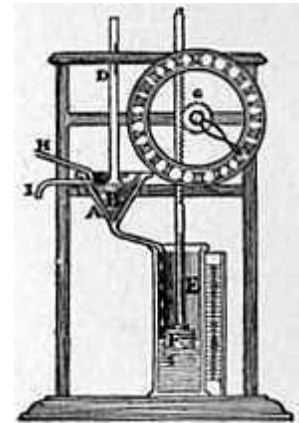


FIG. 6.—AN IMPROVED GREEK CLEPSYDRA.

If we pass from Greece to Rome, our usual route when we are tracing a feature of our civilization, we find that the Romans were slow to introduce new methods of timekeeping. The first public sun-dial in Rome was constructed about 200 B. C., an event which the poet Plautus bewailed:

Confound the man who first found out
How to distinguish hours! Confound them, too
Who in this place set up a sun-dial
To cut and hack my days so wretchedly
Into small portions! When I was a boy
My stomach was my sun-dial, one more sure,
Truer, and more exact than any of them,
This dial told me when 'twas the proper time
To go to dinner.

The water-clock was brought into Rome a little later than the sun-dial, and was used as a time-check upon speakers in the law courts, just as it had been in Athens. When the Romans first began to use the clepsydra it was already a very good clock. Whether it received any great improvements at their hands is not certain. Improvements must have been made somewhere, for early in the Middle Ages we find clepsydras in forms more highly developed than they were in ancient times. In the ninth century the Emperor Charlemagne received as gift from the King of Persia a most interesting timepiece which was worked by water. "The dial was composed of twelve small doors which represented the divisions of the hours; each door opened at the hour it was intended to represent, and out of it came the same number of little balls, which fell, one by one at equal distances of time, on a brass drum. It might be told by the eye what hour it was by the number of doors that were open; and by the ear by the number of balls that fell. When it was twelve o'clock, twelve horsemen in miniature issued forth at the same time, and, marching round the dial, shut all the doors." Less wonderful than the clock of the emperor, but more useful as an object of study, is the medieval clepsydra shown in Figure 7. This looks more than ever like the clock we are accustomed to see. It has weights as well as wheels. As the float *A* rises with the water it allows the weight *C* to descend and turns the spindle *B* on the end of which is the hand which marks the hours. Notice carefully that this is partly a water-clock and partly a *weight*-clock. The weight in its descent turns the spindle; the water regulates the rate at which the weight may descend.



FIG. 7.—A
MEDIEVAL
CLEPSYDRA.

The water-clock just described led easily and directly to the weight-clock. Clockmakers in the Middle Ages for centuries tried with more or less success to make clocks that would run by means of weights. In 1370, Henry De Vick, a German, succeeded in solving the problem. De Vick was brought to Paris to make a clock for the tower of the king's palace, and he made one that has become famous. In a somewhat improved form it can still be seen in Paris in the Palais de Justice. Let us remove the face of this celebrated timepiece and take a look at its works (Fig. 8). It had a striking part, and a timekeeping part, each distinct from the other. The figure shows only the

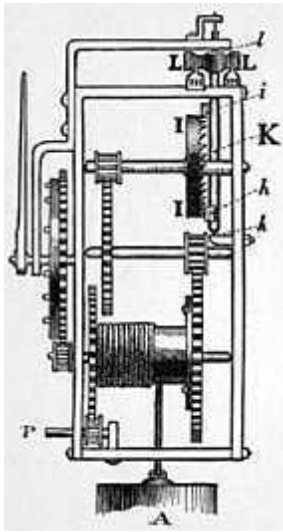


FIG. 8.—DE VICK'S CLOCK. THE FIRST WEIGHT CLOCK. (1370.)

timekeeping part. The weight (A), of 500 pounds, is wound up by a crank (the key) at P. O is the hour-hand. If A is allowed to descend, you can easily see how the whole system of wheels will be moved—and that

very rapidly. But if something does not prevent, A will descend faster and faster, the hour-hand will run faster and faster and the clock will run down at once. If the clock is to run at a uniform rate and for any length of time, the power of the weight must escape gradually. In the clepsydra (Fig. 1) the descent of the weight was controlled by the size of the stream

of flowing water. De Vick invented a substitute for the stream of flowing water. Fasten your attention upon the workings of the saw-toothed wheel II and the upright post K, which moves on the pivots l and k, and you may learn what he did. Fixed to the upper part of the post K is a beam or balance LL, at the ends of which are two small weights m and m, and projecting from the post in different directions are two pallets or lips i and h. Now, as the top of the wheel II turns toward you, one of its teeth catches the pallet i and turns the post K a part of the way round toward you. Just as the tooth escapes from i a tooth at the bottom of II (moving from you) catches the pallet h and checks the revolving post and turns it from you. Thus as II turns, it gives a to-and-fro motion to the post K and, consequently, a to-and-fro motion to the balance LL. II is called the *escapement* because the power of the descending weight gradually escapes from its teeth. In the clepsydra the trickling of water regulated the descent of the weight; in De Vick's clock the trickling of power or force from the escapement regulated the descent of the weight. The invention of this escapement is the greatest event in the history of the clock. The king was much pleased with De Vick's invention. He gave the clockmaker three shillings a day, and allowed him to sleep in the clock

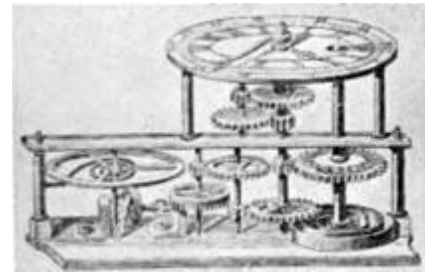


FIG. 9.—A CLOCK WITHOUT WEIGHTS.

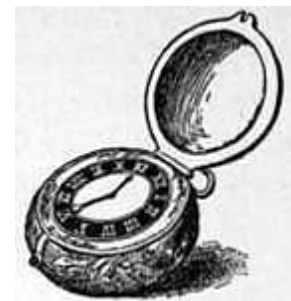


FIG. 10.—A WATCH OF THE 16TH CENTURY.

tower; a scanty reward indeed for one who had done so much for the world, for De Vick's invention led rapidly to the excellent timepieces of to-day, to both our watches and our clocks. After the appearance of the weight-clock, the water-clock gradually fell into disuse, and all the ingenuity of the clockmaker was bestowed upon weights and wheels and escapements and balances. A century of experimenting resulted in a clock without a weight (Fig. 9). In this timekeeper you recognize the beginnings of the modern watch. The uncoiling of a spring drove the machinery. Instead of the balancing beam with its weights as in De Vick's clock, a *balance wheel* is used. The escapement is the same as in the first weight-clock. The busy and delicately-hung little balance wheel in your watch is a growth from De Vick's clumsy balance beam. The spring-clock would run in any position. Because it could be carried about it led almost at once to the watch. Many places claim the distinction of having made the first watch, but it seems that the honor belongs to the city of Nürenburg. "Nürenburg eggs," as the first portable clocks were called, were made as early as 1470. The first watches were large, uncouth affairs, resembling small table clocks but by the end of the sixteenth century small watches with works of brass and cases of gold or silver were manufactured (Fig. 10).

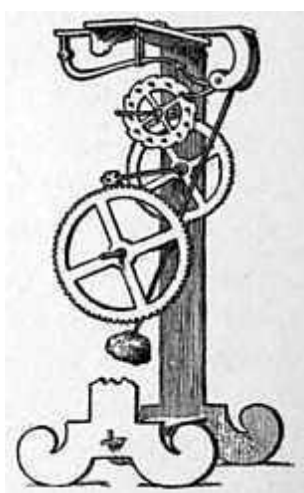


FIG. 11.—GALILEO'S PENDULUM. (1650.)

The last important step in the development of the clock was taken when the *pendulum* was brought into use. The history of the pendulum will always include a story told by Galileo. This great astronomer, the story runs, while worshipping in the cathedral at Pisa one day, found the service dull, and began to observe the swinging of the lamps which were suspended from the ceiling. Using his pulse as a timekeeper he learned that where the chains were

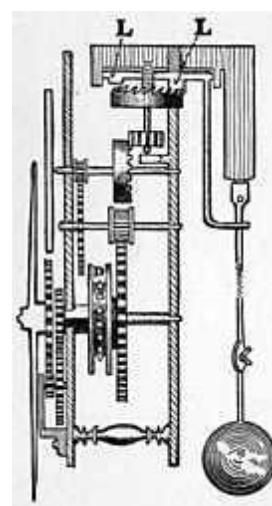


FIG. 12.—THE FIRST PENDULUM CLOCK. (1656.)

of the same length the lamp swayed to and fro in equal length of time, whether they traveled through a short space or a long space. This observation set the philosopher to experimenting with pendulums of different lengths. Among the many things he learned one of the most

important was this: a pendulum thirty-nine inches in length will make one vibration in just one second of time. Now, if the pendulum could only be kept swinging and its vibrations counted it would serve as a clock. Galileo, of course, saw this, and he caused to be made a machine for keeping the pendulum in motion (Fig. 11), but he did not make a clock; he did not connect his pendulum with the works of a clock. This, however, was done about the middle of the seventeenth century, although it is somewhat difficult to tell who was the first to do it. The honor is claimed by an Englishman, a Frenchman, and a Dutchman. The truth is, clockmakers throughout Europe were trying at the same time to make the best of the discoveries of Galileo, and several of them about the same time constructed clocks with pendulums. The one who seems to have succeeded first was Christian Huygens, a Dutch astronomer, who, in 1656, constructed a clock, the motions of which were regulated by the swinging of a pendulum (Fig. 12). The weight was attached to a cord passing over a pulley and gave motion to all the wheels, as in De Vick's clock. Like De Vick's clock also Huygens's clock had its escapement wheel acting upon two pallets. In the Dutchman's clock, however, the escapement, instead of turning a balance beam to and fro, acted upon the pendulum, giving it enough motion to keep it from stopping.

We need not carry our story further than the invention of Huygens. Timepieces are cheaper and better made and more accurate than they were two hundred years ago, but no really important discovery has been made since the pendulum was introduced.

THE BOOK

What is a book? It is an invention by means of which *thought* is recorded, and carried about in the world, and handed down from one age to another. Almost as soon as men began to think they began to make books and they will probably continue to make them as long as they continue to think. The story of the Book, therefore, takes us back to the very beginning of human existence.



FIG. 1.—TRADITION.
A Mural Decoration in the
Library of Congress.

At first thought was recorded and preserved by *tradition*. An account of a nation's deeds, its laws, the precepts of its religion were stamped, printed, on the memory of persons specially trained to memorize these things and hand them down by word of mouth from generation to generation (Fig. 1). These persons were usually priests, who underwent long years of daily and hourly training in memorizing what was to be handed down. The Sanskrit Vedas, the sacred scripture of the Hindoos, were for many centuries transmitted by tradition, and it is said it took forty years to memorize them. It is a wonder it did not take longer, for the Vedas make a volume as large as our Bible. It is believed that primitive people everywhere first adopted the method of tradition to record and preserve the thought which they did not wish to perish. We may say, then, that the first book was written on the tablet of the human memory.



FIG. 2.—THE QUIPU
OF THE PERUVIANS.

The first step in the growth of the book was taken when *memory aids* were invented. Sometimes we tie a knot in a handkerchief to help us to remember something. Now, it was just by tying knots that primitive man first lent assistance to the memory.

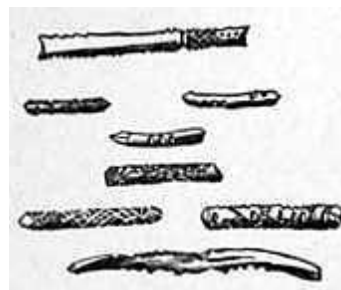


FIG. 3.—MESSAGE-
STICKS.

The first material book was doubtless a series of *knots* well represented by the *quipu* (Fig. 2) of the ancient Peruvians. This curious-looking book was written (tied) by one known as the officer of the knots. It contains an account of the strength of the Peruvian army, although it is confessed that its exact meaning cannot be made out. It was not intended to be read by any one who was not a keeper of the knots. Books made of knots were used by nearly all the ancient peoples of South America and by some of those of Asia. Akin to the knotted cord is the *notched stick*, which is still used in Australia by the savages to assist the memory of one who has a message to carry. Figure 3 shows a variety of such message-sticks. The lowest one—a crooked branch of a tree—contains an invitation to a dancing party. The notches are read by the messenger. The notched stick as an aid to memory is not confined to savage races. Many a highly civilized baker has kept his accounts by making notches in sticks and so has many a modern dairyman, as he has delivered milk from door to door.

Memory aids were followed by *picture-writing*. To express thought by means of pictures is an instinct shared alike by the lowest savage and the most enlightened people. All over the earth we find examples of early picture-writing. A beloved chief had died, a fierce battle had been fought, an exciting chase had occurred: promptly the event was pictured on a stone or on the skin of some animal. Pages might be filled with illustrations of these primitive picture-books, but we must be content with a single specimen (Fig. 4). This was found painted on a rock in California: "*We selected this as a camping place, but we have found nothing*," say the human figures *f, g, h, i*. The upturned palms say plainly, "*nothing, nothing*." "*One of our comrades (d) has died of starvation*," say the three lank figures at *c* pointing to their own lean bodies. "*We deeply mourn his loss*," says the sorrow-stricken *a*.

"We have gone northward," says *j*, his distinguished arm extended to the north.



FIG. 4.—PICTURE WRITING.

Practice in picture-making was bound to lead to shorter methods of expressing ideas. It was soon found that reduced pictures, or *picture-signs*, would suffice to express ideas. Thus, if the idea of sorrow was to be expressed it was not necessary to draw an elaborate picture of a sorrowful looking man like *a* in Figure 4; a weeping eye would express the idea just as well. Instead of numerous figures (*e*, *f*, *g*, *h*, *i*) weeping and saying, "nothing here," a single pair of empty palms would say the same thing just as clearly. In this way a pair of clasped hands came to mean "friendship"; two trees meant "a forest"; a calf running toward water meant "thirst." These picture-signs, of course, assumed the form in which they could be most easily and rapidly drawn. The weeping eye became ; the pair of extended palms ; the forest ; thirst . A simple picture of this kind became a fixed conventional sign for certain ideas; it was always drawn in the same way and it always stood for the same idea.

Picture-signs (ideographs) followed picture-writing in almost every country where the people were progressive. China was writing its books with picture-signs many thousands of years ago, and it is writing them in the same clumsy way still. Even in highly civilized countries picture-signs have not been entirely abandoned. Examine the advertising page of a newspaper or observe the business signs on the street and you will find picture-signs—pictures that are always made in the same way and that always stand for the same thing.

Each of the great nations of antiquity had its own peculiar system of writing, but the system that should interest us most is that of ancient Egypt, for it is to ancient Egypt that you must look for the origin of the book that is in your hands. The book in Egypt passed through the stages of tradition, memory aids, picture-writing and picture-signs (ideographs); then it passed into the *alphabetic* stage. Since the alphabet is certainly the most wonderful and perhaps the most useful of all inventions, and since it is an Egyptian invention, it is well worth your while to learn how the Egyptian picture-signs—hieroglyphics they are called—grew into letters, but if you wish to understand the change you will have to give the subject very close attention.

Well, here was the Egyptian system of picture-signs consisting of several thousand pictures of birds, beasts, reptiles, insects, trees, flowers, and objects of almost every description. Now suppose you were employed in writing *English* by means of several thousand picture-signs and in the course of an hour would have to write the words *manage*, *mansion*, *mantle*, *mandate*, might it not occur to you that it would be a good thing if that sound *man* could be represented by the picture-sign for man ()? And if you had to write *treacle*, *treason*, *treaty*, might you not feel like beginning these words with a tree ()? At some time in the remote past Egyptian scribes—priests they usually were—noticing that syllables identical in sound were constantly recurring in the different words, began to represent these *syllable-sounds* that occurred most frequently by *picture-signs*.²⁰ The picture-sign substituted for a syllable-sound was placed in the word not because it stood for an *idea*, but because it stood for a *sound*, just as in the case supposed above you would use the [symbol: man] or the [symbol: tree] not because it represented a thought, but because it had a certain sound. So certain Egyptian picture-signs began to be used to represent the sound of certain syllables. The picture-signs thus chosen were called *phonograms*.

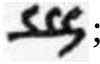
The phonogram led to the alphabet. The scribes in seeking a way to shorten their work found that the syllable itself could be broken up into separate sounds. For example, when they came to the syllable whose sound is spelled by our three letters *pad*, they found that it had three distinct sounds, namely: (1) one a lip sound which could be represented by the first

sound of the picture-sign  (a door); (2) one an open-throat or vowel sound which could be represented by the first sound of the picture-sign  (an eagle); (3) one a dental sound which could be represented by the first sound in the picture-sign  (a hand). So the scribes wrote the

syllable (p-a-d) with the three characters   . And so

with all the other sounds in the Egyptian language; each was represented by one of the picture-signs already used. Since there were only about twenty-five distinct elementary sounds in the Egyptian language, twenty-five picture-signs were sufficient to represent any sound or any word in the language. These twenty-five picture-sounds were the letters of the Egyptian alphabet. Twenty-five characters instead of thousands! Now the Egyptian youth could learn to read in three or four years, whereas under the old system it took fifteen or twenty years, just as it takes fifteen or twenty years for the Chinese youth to learn to read well.

Now that its origin has been explained, the story of the alphabet may be rapidly told. Indeed, its whole history can be learned from Figure 5. In column (a) are the three Egyptian picture-signs referred to above. Column (b) shows how the rapid writing of the priests reduced the old hieroglyphics

to script;  became ;  became  and  became

. The Phœnicians, who were great travelers, visited Egypt at a very early date and borrowed not only the idea of the alphabet, but also the forms of the Egyptian letters, as column *c* shows. Column *d* confirms the words of Herodotus, who tells us that the Greeks borrowed their alphabet from the Phœnicians. Column *e* shows that the Greeks handed the alphabet on to the Romans, who handed it on to us. Thus the three letters p, a, d come straight from the Egyptians and were originally a *door*, an *eagle*, and a *hand*, respectively. As it is with these three letters, so it is with nearly all the letters of our alphabet. If the letters on the page before you could be suddenly changed to their original form, you would behold a motley collection of birds, serpents, animals, tools, and articles of household use.

(a) Egyptian Picture Signs	(b) Hieroglyphics.	(c) Phonetician.	(d) Greek.	(e) Roman.
				
				
				

FIG. 5.—SHOWING THE DEVELOPMENT OF THE THREE LETTERS, P, A, AND D.

We must look to Egypt for the origin of the material form of our book as well as for the origin of our alphabetical characters. Before history had dawned the Egyptians had covered over with their writing nearly all the available surface on their pyramids and in their temples. At a time too far back for a date necessity seems to have compelled them to seek a substitute for stone. This they found in the *papyrus* plant, which grew in great luxuriance in the valley of the Nile. They placed side by side strips of the pith of the papyrus, and across these at right angles they placed another layer of strips. The two layers were then glued together and pressed until a smooth surface was formed. This made one sheet. To make a book a number of sheets were fastened together end to end. When in book form the papyrus was wound around a stick and kept in the form of a roll, a *volume* (Fig. 6). The roll was usually eight or ten inches wide, but its length might be upward of a hundred feet. This papyrus roll was the parent of our modern paper book, as the word papyrus is the original of our word paper. The pen used in writing upon papyrus was a split reed (*calamus*), and the ink a mixture of soot and gum.

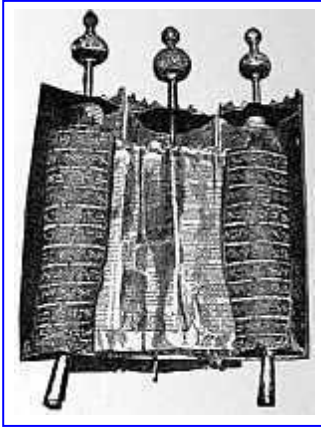


FIG. 6.—AN ANCIENT VOLUME.



FIG. 7.—THE OLDEST BOOK IN THE WORLD. WRITTEN NEARLY 5,000 YEARS AGO.

The most ancient volume in the world is an Egyptian papyrus (Fig. 7) now in the National Library of France. It was written nearly 5,000 years ago by an aged sage and contains precepts of right living. In this oldest of volumes we find this priceless gem:

"If thou art become great, if after being in poverty thou hast amassed riches and art become the first in the city, if thou art known for thy wealth and art become a great lord, let not thy heart become proud, for it is God who is the author of them for thee."

In Assyria and in other ancient countries of Central Asia letters were engraved on cylinders and these were rolled upon slabs of soft clay, making an impression of the raised letters, just as we make an impression with the seal of a ring. In the ruins of the cities of Assyria these old clay books may be found by the cart-load. The Assyrian cylinder was really the first printing press. In ancient Greece and Rome wooden tablets within which was spread a thin layer of wax were used as a writing surface in schools and in the business world. The writing on the wax was done with a sharp-pointed instrument of bone or iron called the *stylus*. But next to papyrus the most important writing material of antiquity was *parchment*, or the prepared skin of young calves and kids. The invention of parchment is said to have been due to the literary ambitions of two kings, the king of Persia and the king of Egypt. The king of Pergamus (250 B.C.) wishing to have the finest and largest library in the world was consuming enormous quantities of papyrus. The king of Egypt, who also wished to have the finest library in the world,

in order to cripple the plans of his literary rival, issued a command forbidding the exportation of papyrus from Egypt. The king of Pergamus, being unable to get papyrus except from Egypt, caused the skins of sheep to be prepared, and on these skins books for his library continued to be written. The prepared skins received the name of *pergamena*, because they were made in Pergamus, and from *pergamena* we get the word parchment. This is the story that has come down to us to explain the origin of parchment, but it cannot be accepted as wholly true. We know very well that the Old Testament was written in gold on a roll of skins long before there was a king of Pergamus. Indeed, writing was done on skins as far back as the picture-writing period.

After the invention of the alphabet and of paper (papyrus) books multiplied as never before. "Of making many books there is no end," exclaimed Solomon a thousand years before the Christian era. Greece in her early day was slow to make books, but after she learned from the Phœnicians (800 B.C.) how to use an alphabet she made up for lost time. In 600 B.C. there was a public library at Athens, and 200 years later the Greeks had written more good books than all the other countries in the world combined.

But the most productive of ancient book-makers were the Romans. In Rome publishing houses were flourishing in the time of Cicero (50 B.C.). Atticus, one of Cicero's best friends, was a publisher. Let us see how a book was made in his establishment. Of course, there were no type-setters or printing-presses. Every book was a manuscript; every word of every copy had to be written with a pen. The writing was sometimes done by slaves trained to write neatly and rapidly. We may imagine 50 or 100 slaves sitting at desks in a room writing to the dictation of the reader. Now if Atticus had ten readers each of whom dictated to 100 slaves it took only two or three days for the publication of 1,000 copies of one of his friend Cicero's books. Of course every copy would not be perfect. The slave would sometimes make blunders and write what the reader did not dictate. But books in our own time are not free of errors. An English poet recently wrote:

"Like dew-drops upon fresh blown roses."

In print the first letter of the last word in the line appeared as *n* instead of *r*. This mistake disfigured thousands of copies. In the Roman publishing house such a blunder marred only one copy.

You can readily see that by methods just described books could be made in great numbers. And so they were. Slaves were cheap and numerous and the cost of publication was small. It is estimated that a good sized volume in Nero's time (50 A.D.) would sell for a shilling. Books were cheaper in those days than they had ever been before and almost as cheap as they are to-day, perhaps. The Roman world became satiated with reading matter. The poet Martial exclaimed, "Every one has me in his pocket, every one has me in his hand." Books became a drug on the market and could be sold only to grocers for "wrapping up pastry and spices."



FIG. 8.—BOOK-MAKING IN THE MIDDLE AGES.

But a time was to come when books would not be so plentiful and cheap. With the overthrow of Rome (476 A.D.) culture received a blow from which it did not recover for a thousand years. The barbarian invaders of Southern Europe destroyed all the books they could find and caused the writers of books to flee within the walls of the churches. Throughout the Middle Ages nearly all the writing in Europe was done in the religious houses of monks (Fig. 8), and nearly all the books written were of a religious nature. The monks worked with the greatest patience and care upon their manuscripts. They often wrote on vellum (calf-skin parchment) and illuminated the page with beautiful colors and adorned it with artistic figures.

The manuscript volumes of the dark ages were beautiful and magnificent, but their cost was so great that only the most wealthy could buy. A Bible would sometimes cost thousands of dollars. Along in the 14th

and 15th centuries Europe began to thirst for knowledge and there arose a demand for cheap books. How could the demand be met? There were now no hordes of intelligent slaves who could be put to work with their pens, and without slave labor the cost of the written book could not be greatly reduced. Invention, as always, came to the rescue and gave the world what it wanted.

In the first place, writing material was made cheaper by the invention of paper-making. The wasp in making its nest had given a hint for paper-making, but man was extremely slow to take the hint. The Chinese had done something in the way of making paper from the bark of trees as early as the first century, but it was not until the middle of the 13th century that paper began to be manufactured in Europe from hemp, rags, linen, and cotton.



FIG. 9.

In the second place, *printing* was invented. On a strip of transparent paper write the word *post*. Now turn the strip over from right to left and trace the letters on the smooth surface of a block of wood. Remove the paper and you will have the result shown in Figure 9. With a sharp knife cut out the wood from around the letters. Ink the raised letters and press upon them a piece of paper. You have printed the word "post" in precisely the way the first books were printed. In the 13th century fancy designs were engraved on wood and by the aid of ink the figures were stamped on silk and linen. In the 14th century playing cards and books were printed on engraved blocks in the manner the word "post" was printed above. (Fig. 10.) The block-book was the first step in the art of printing.



FIG. 10.—A
BLOCK PRINT
CONTAINING
THE ALPHABET
USED BY
CHILDREN
WHEN
LEARNING TO
READ.

The block-book decreased the cost of a book, for when a page was once engraved as many impressions could be taken as were wanted, yet it did not meet the necessities of the time. In the middle of the 15th century the desire for reading began to resemble a frenzy and the books that could be got hold of "were as insufficient to slake the thirsty craving for religious and material knowledge as a few rain drops to quench the burning thirst of the traveler in the desert who seeks for long, deep-draughts at copious springs of living water." To meet the demand of the time book-makers everywhere were trying to improve on the block-making process and

by the end of the century the book as we have it to-day was being made throughout all Europe.

In what did the improvement consist? First let us call to mind what the book-maker in the early part of the 15th century had to begin with; he had paper, he had printing-ink, he had skill in engraving whole pages for block-books, and he had a rude kind of printing-press. The improvement consisted in this: Instead of engraving a whole page on a block, single letters were engraved on little blocks called types, and when a word or a line or a page was to be printed these types were set in the position desired; in other words, the improvement consisted in the invention of *moveable types*. The types were first made of wood and afterward of metal.

The great advantage of the moveable types over the block-book is easily seen. A block containing, say, the word "post" is useless except for printing the word *post*; but divide it into four blocks, each containing a letter: now you can print *post, spot, tops, stop, top, sop, sot, pot, so, to* and so forth.

The exact date of the invention of moveable types cannot be determined. We can only say that they were first used between 1450 and 1460. Nor can we tell who invented them. The Dutch claim that Lawrence Koster of Harlem (Holland) made some moveable types as early as 1430, and that John Faust, an employee, stole them and carried them to Mayence (Germany), where John Gutenberg learned the secret of printing with them. The Germans claim that Gutenberg was the real inventor. Much can be said in behalf of both claims. What we really know is that the earliest complete book printed on moveable types was a Bible which came from the press of John Gutenberg in 1455.

Since 1450 there has been no discovery that has changed the character of the printed volume. There have been wonderful improvements in the processes of making and setting type, and printing-presses (Fig. 11) have become marvels of mechanical skill, but the book of to-day is essentially like the book of four hundred years ago. The tablet of the memory, the knotted cord and notched stick, the uncanny picture-writing, the clumsy picture-sign, the alphabet, the manuscript volume, the printed block-book and the volume before you bring to an end the story of the book.

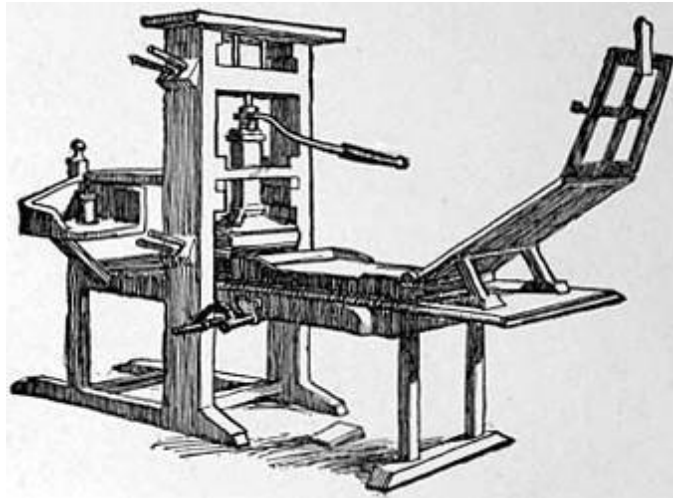


FIG. 11.—AN EARLY PRINTING PRESS.

THE MESSAGE

Men had not been living together long in a state of society before they found it necessary to communicate with their fellow-men at a distance and in order to do this the *message* was invented. We have seen (p. 205) that among certain tribes of savages notched sticks bearing messages were sent from one tribe to another. Among the ancient Peruvians the message took the form of the curious looking quipu. After the alphabet had been invented and papyrus had come into use as a writing material, the message took the form of a written document and resembled somewhat the modern *letter*.

The ancient Egyptians, as we would expect, were the first to make use of the letter in the sending of messages (Fig. 1). The ancient Hebrews were also familiar with the letter as a means of communication. We read in the book of Chronicles how the post went with the letters of the king and his princes throughout all Israel. The word *post*, as used here and elsewhere in the Bible, signifies a runner, that is, one specially trained to deliver letters or despatches speedily by running. Thus Jeremiah predicted that after the fall of Babylon "one post shall run to meet another and one messenger to meet another to show the King that his city is taken." Although we frequently read of the post in Biblical times we are nowhere told that the ordinary people enjoyed the privileges of the post. In olden times it was only kings and princes and persons of high degree that sent and received letters.



FIG. 1.—A
LETTER
CARRIER OF
ANCIENT
EGYPT.

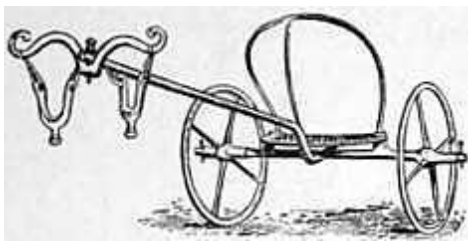


FIG. 2.—AN EGYPTIAN MAIL
CART.

In nearly all the countries of antiquity there was an organized postal system which was under the control of the government and which carried only government messages. In Egypt there were postal chariots (Fig. 2) of wonderful lightness designed especially for carrying the letters of the king at the greatest possible speed. In

ancient Judea messengers must have traveled very fast, for Job, in his old age, says: "Now my days are swifter than the post, they flee away." In ancient Persia the postal system awakened the admiration of Herodotus. "Nothing mortal," says this old Greek historian, "travels so fast as these Persian messengers. The entire plan is a Persian invention and this is the method of it. Along the whole line of road there are men stationed with horses, the number of stations being equal to the number of days which the journey takes, allowing a man and a horse to each day, and these men will not be hindered from accomplishing at their best speed the distance they will have to go either by snow, or rain, or heat, or by the darkness of night. The first rider delivers the message to the second and the second to the third, and so it is borne from hand to hand along the whole line."

The postal system which Herodotus found in Persia was better than the system which existed in his own country for the reason that the Greeks relied upon human messengers rather than upon horses to carry their messages. Young Greeks were specially trained (Fig. 3) as runners for the postal service and Greek history contains accounts of the marvelous endurance and swiftness of those employed to carry messages. After the defeat of the Persians by the Greeks at Marathon (490 B. C.) a runner carried the news southward and did not pause for rest until he reached Athens when he shouted the word "Victory!" and expired, being overcome by fatigue. Another Greek, Phillipides by name, was despatched from Athens to Sparta to ask the Spartans for aid in the war which the Athenians were carrying on against Persia, and the distance between the two cities—about 140 miles—was accomplished by the runner in less than two days.



FIG. 3.—A LETTER CARRIER OF ANCIENT GREECE.



FIG. 4.—A LETTER

But the best postal system of ancient times was the one which was organized by the Romans. As one country after another was brought under the dominion of Rome it became more and more necessary for the Roman government to keep in close touch with all the parts of the vast empire. Accordingly, by the time of Augustus (14 A.D.), there was established throughout the Roman world

CARRIER OF ANCIENT ROME.

a fully organized and well-equipped system of posts. Along the magnificent roads which led out from Rome there were built at regular distances stations, or post-houses, where horses and riders were stationed for the purpose of receiving the messages of the government and hurrying them along to the place of their destination. The stations were only five or six miles apart and each station was provided with a large number of horses and riders. By the frequent changes of horses a letter could be hurried along with considerable speed (Fig. 4). "By the help of the relays," says Gibbon, "it was easy to travel a hundred miles in a day."

When Rome fell (476 A.D.) before the attacks of barbarous tribes her excellent postal system fell with her and many centuries passed before messages could again be regularly and quickly despatched between widely separated points. Charles the Great, the emperor of the Franks, established (800 A.D.) a postal system in his empire but the service did not long survive the great ruler. In the 13th century the merchants of the Hanse towns of Northern Germany could communicate with each other somewhat regularly by letter, but the ordinary people of these towns did not enjoy the privileges of a postal service. In the Middle Ages, as in the ancient times, the public post was established solely for the benefit of the government. Private messages had to be sent as best they could be by private messengers and at private expense. As late as the reign of Henry VIII (1509-1547) the only regular post route in England was one which was established for the exclusive use of the king.

But the time was soon to come when ordinary citizens as well as officers of state were to share in the benefits of a postal system. In 1635 Charles I of England gave orders that a post should run night and day between Edinburgh and London and that postmen should take with them all such letters as might be directed to towns on or near the road which connected the two cities. The rate of postage²¹ was fixed at two pence for a single letter when the distance was under sixty miles; four pence when the distance was between 60 and 140 miles; six pence for any longer distance in England; and eight pence from London to any place in Scotland. It was ordered that only messengers of the king should be allowed to carry letters for profit unless to places to which the king's post did not go. Here was the beginning of the modern postal system and the modern post-office.

Henceforth the post was to carry not only the king's messages, but the messages of all people who would pay the required postage.

The example set by England in throwing the post open to the public was followed by other nations, and before a hundred years had passed nearly all the civilized countries of the world were enjoying the privilege and blessings of a well-organized postal system. It is true that the post for a long time moved very slowly—a hundred miles a day was regarded as a flying rate—and postage for a long time was very high, but the service grew constantly better and by the close of the nineteenth century trains were dashing along with the mails at the rate of a thousand miles a day and postage within a country had been reduced to two cents,²² while for a nickel a letter could be sent to the most distant parts of the globe.

Thus far we have traced the history of only one kind of message, the kind that has the form of a written document and that is conveyed by a human carrier over land and water from one place to another. But there is a kind of message which is not borne along by human hands and which does not travel on land or water. This is the *telegraph*,²³ the message which darts through space and is delivered at a distant point almost at the very instant at which it is sent.

The first telegraph was an aerial message and consisted of a signal made by a flash of light. From the earliest times men have used fire signals as a means of sending messages to distant points. When the city of Troy in Asia Minor was captured by the Greeks (about 1100 B.C.) torches flashing their light from one mountain top to another quickly carried the news to the far-off cities of Greece. The ancient Greeks gave a great deal of attention to the art of signaling by fire and they invented several very ingenious systems of aerial telegraphy. The most interesting of these systems is one invented and described by the Greek historian Polybius, who flourished about 150 B.C. When signaling with fire Polybius arranged for using two groups of torches

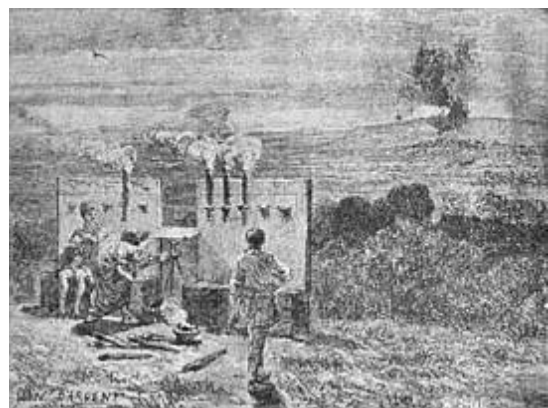


FIG. 5.—TELEGRAPHING BY MEANS OF FIRE, 150 B. C.

with five torches in each group, and for the purpose of understanding the signals he divided the letters of the alphabet into five groups of five letters each.²⁴ The torches were raised according to a plan that made it possible to flash a signal that would indicate any letter of the alphabet that might be desired. Thus if the desired letter was the third one of the first group—that is, the letter *k*—one torch would show which group was meant and three torches would show which letter was meant (Fig. 5). In theory this system was perfect, for it provided for sending any kind of message whatever. But in practice it had little value, for it required so many torches and signals that an entire night was consumed in spelling out a few words.

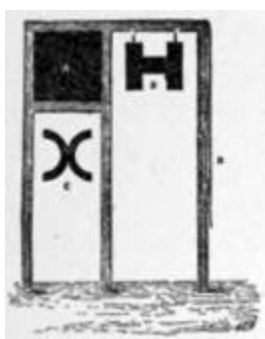


FIG. 6.—HOOKE'S
AERIAL
TELEGRAPH, 1684.

Although the elaborate system of aerial telegraph proposed by Polybius was not generally adopted, nevertheless for centuries, both in ancient times and during the middle ages, the fire signal was everywhere used for the quick despatch of important news. In the seventeenth century inventors began to devise new systems of aerial telegraphy. In 1663, the Marquis of Worcester, who was always busy with some great invention (p. 178), announced to the world that he had discovered a plan by which one could talk with another as far as the eye could distinguish between black and white, and that this conversation could be carried on by night as well as by day, even though the night were as dark and as black as pitch. But the telegraph of the Marquis was like many of his other inventions—it was chiefly on paper. In 1684, Dr. Robert Hooke of England invented a method by which aerial messages could be sent a distance of thirty or forty miles. His plan was to erect on hill tops a series of high poles connected above by cross-pieces and by means of pulleys suspend from the cross-pieces the letters of the alphabet which would spell out the message (Fig. 6). In order to read the letters at such great distances the eye was assisted by the telescope, an instrument which had recently been invented.

But the greatest improvement in aerial telegraphy was made during the French Revolution by Claude Chappe, a Frenchman living in Paris. In 1793, Chappe erected on the roof of the palace of the Louvre a post at the top of which was a cross-beam which moved on a pivot about the center like a scale beam (Fig. 7). The cross-beam could be moved horizontally, vertically

or at almost any angle by means of cords. Chappe invented a number of positions for these arms and each position stood for a certain letter of the alphabet. Machines of this kind were erected on towers at places from nine to twelve miles apart and soon Chappe was sending messages from Paris to the city of Lille, 130 miles away. The messages were sent with great rapidity, for they passed from one tower to another with the velocity of light—about 185,000 miles a second—and it was possible for the operator to spell out about 100 words in an hour. And Chappe's messages could be sent at any time, day or night, for the arms of the machine were furnished with Argand lamps for night work.

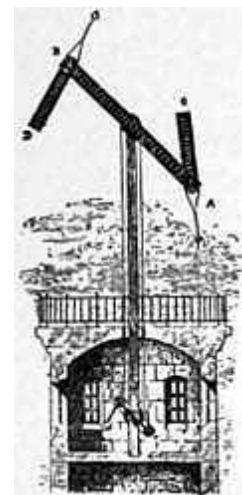


FIG. 7.—
CHAPPE'S
AERIAL
TELEGRAPH,
1793.

Chappe's invention was the greatest which had thus far been made in the history of the message. The new system of telegraphy proved to be entirely successful and practical and it was not long before machines similar to those invented by Chappe were in use in England and other countries. In 1828, an English writer had the following words of praise for aerial telegraphy: "Telegraphs have now been brought to so great a degree of perfection that they carry information so speedily and distinctly and are so much simplified that they can be constructed and maintained at little expense. The advantages, too, which result from their use are almost inconceivable. Not to speak of the speed with which information is communicated and orders given in time of war, by means of these aerial signals the whole kingdom could be prepared in an instant to oppose an invading enemy."

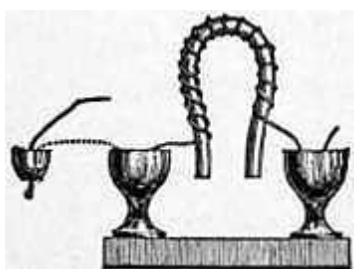


FIG. 8.—STURGEON
ELECTRO-MAGNET, 1825.

But the aerial telegraph was soon to have a most dangerous rival. This rival was *the electric telegraph*. Many years before the invention of Chappe men had been experimenting with electricity with a view of sending messages by means of an electric current. These experiments began in 1728 when an Englishman named Gray caused electricity to produce motion in light bodies located at a distance of more than 600 feet. In 1748, the great Benjamin Franklin, who conducted so many wonderful

experiments in electricity, sent an electric current through a wire which was stretched across the Schuylkill River and set fire to some alcohol which was at the opposite end of the wire. We may regard the flash of alcohol as a telegraph, for it could have been used as a signal. In 1819, Professor Oersted of Copenhagen brought a magnetic needle close to a body through which an electric current was passing and he observed that the needle had a tendency to place itself at right angles to the electrified body. In 1825, William Sturgeon of England coiled a copper wire around a bar of soft iron and found that when a current of electricity was sent through the wire the bar of iron became a temporary magnet; that is, the bar of iron attracted a needle when the current was passing through the wire and ceased to attract it when the current was broken (Fig. 8). These discoveries of Oersted and Sturgeon led to the invention known as the *electro-magnet* and the electro-magnet led rapidly to the invention of the electric telegraph, for by means of the electro-magnet a signal can be sent to a distance as far as a current of electricity can be sent along a wire. In 1831, Professor Joseph Henry, one of America's most distinguished scientists, discovered a method by which an electric current could be sent along a wire for a very great distance. The next year Henry constructed and operated an apparatus which was essentially an electric telegraph (Fig. 9). "I arranged," he said, "around one of the upper rooms of the Albany Academy a wire of more than a mile in length through which I was enabled to make signals by sounding a bell. The mechanical arrangement for effecting this object was simply a steel bar permanently magnetized, supported on a pivot and placed with its north end between the two arms of a horse-shoe magnet. When the latter was excited by the current the end of the bar thus placed was attracted by one arm of the horse-shoe and repelled by the other and was thus caused to move in a horizontal plane and its further extremity to strike a bell suitably adjusted." Thus by 1832 the electric current had been used for sending signals at a distance and the electric telegraph had been invented.

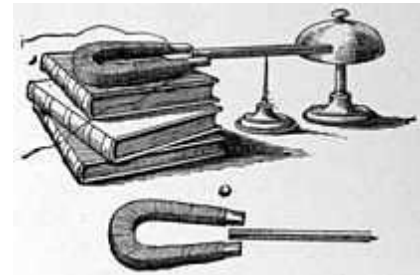


FIG. 9.—PROFESSOR
HENRY'S ELECTRO-MAGNET,
1832.

But the electric telegraph was still only a toy. How could it be made a practical machine? How could it be used for sending messages in a satisfactory manner? Inventors everywhere worked diligently to discover a

satisfactory method of signaling and many ingenious systems were invented. As early as 1837 a telegraph line was established between Paddington, England and Drayton—a distance of 13 miles—and messages were sent over the wire. But the line failed to give satisfaction and its use was discontinued. The honor of inventing the first really practical and useful system of electrical telegraphy was at last won by an American, S. F. B. Morse, a painter and professor of literature in the University of the City of New York. In 1832 Morse began to think about a plan for recording signals sent by electricity and by 1837 he was about ready to take out a patent for making signals "by the mechanical force of electro-magnetic motion." Morse was a poor man and he lacked the means of conducting his experiments. He was fortunate, however, in making the acquaintance and gaining the confidence of Alfred Vail, a student of the University. Vail furnished the money for the experiments and assisted Morse in perfecting his system. Indeed some of the most original and valuable features of Morse's system were invented by young Vail and not by Morse. In the face of much discouragement and bad luck Morse and Vail worked patiently on together and by 1843 their invention was completed.

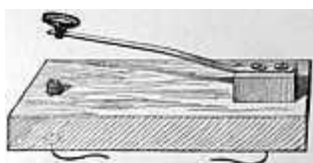


FIG. 10.—THE KEY
USED BY MORSE.

The main feature of Morse's system was to use the electric current for sending an alphabetical code consisting of certain combinations of "dots and dashes." The "dots" were simply clicking sounds and the "dashes" were simply intervals between the clicking sounds. The sounds were made by closing and breaking the current by means of a key or button (Fig. 10). If the sender of the message pressed upon the key and immediately released it he made at the other end of the line a sharp click which was called a "dot," and a single dot according to the code was the letter E. If the sender of the message pressed upon the key and held it down for a moment he made what was called a "dash," and a single dash according to the code was the letter T. Thus by means of "dots and dashes" any letter of the alphabet could be speedily sent.

Morse applied to Congress to aid him in his plans and in 1843 he secured an appropriation of \$30,000 for establishing a telegraph line between Baltimore and Washington. Morse and Vail now hurried the great work on and by May, 1844, the wires had been stretched between the two

cities and the instruments were ready for trial. And such heavy, clumsy affairs the instruments (Fig. 11) were! "The receiving apparatus weighed 185 pounds and it required the strength of two strong men to handle it. At the present day an equally effective magnet need not weigh more than four ounces and might be carried in the vest pocket." But, awkward and clumsy as it was, the new telegraph did its work well. On May 24, 1844, Morse sent from Washington the historic message, "What hath God wrought?" (Fig. 12) and in the twinkling of an eye it was received by Vail at Baltimore, forty miles away.

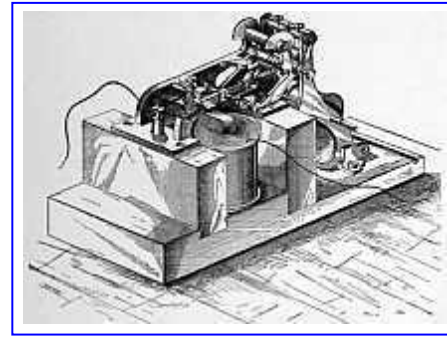


FIG. 11.—MORSE'S TELEGRAPHIC INSTRUMENT.

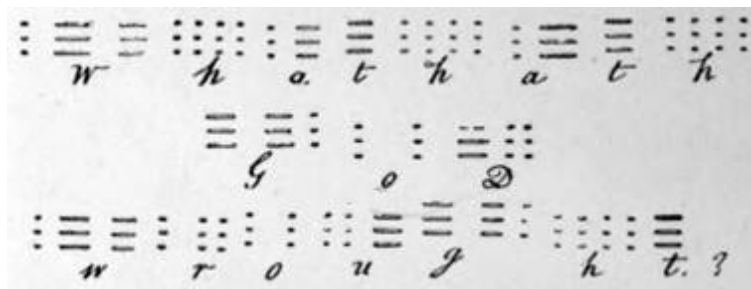


FIG. 12.—THE FIRST TELEGRAPHIC MESSAGE SENT FROM WASHINGTON TO BALTIMORE, MAY 24, 1844.

The Morse system proved to be profitable as well as successful and after 1844 the electric telegraph was soon in general use in all parts of the world. In the United States cities were rapidly connected by wire and by 1860 all the principal places in the country could communicate with each other by telegraph. In 1861, a telegraph line extended across the continent and connected New York and San Francisco. Five years later, thanks to the perseverance and energy of Cyrus W. Field, of New York, the Old World and the New were joined together by a telegraphic cable passing through the waters of the Atlantic from a point on the coast of Ireland to a point on the coast of Newfoundland. With the laying of this cable, in 1866, all parts of the world were brought into telegraphic communication and it seemed that the last step in the development of the message had been taken.

But the story of the Message did not end with the invention of the telegraph and the laying of the Atlantic cable. Almost as soon as inventors had learned how to send a current along a wire and make signals at a distance they began trying experiments to see if they could not also send sounds, especially the sound of the human voice, along a wire; as soon as they had made the *telegraph* they began to try to make the *telephone*.²⁵ In 1855 Professor Wheatstone of England invented an instrument by means of which musical sounds made in one part of a building were carried noiselessly along a wire through several intervening halls and reproduced at the other end of the wire in a distant part of the building. About the same time a Frenchman named Bourseul produced a device by which a disk vibrating under the influence of the human voice would, by means of an electric current, produce similar vibrations of a disk located at a distance.

About 1874 Professor Alexander Graham Bell, of Boston, seized upon an idea similar to that of Bourseul's. Bell saw in the vibrating disk a resemblance to the drum of the human ear. In imagination he beheld "two iron disks, or ear drums, far apart and connected by an electrified wire, catching vibrations of sound at one end and reproducing them at the other." With this conception in mind he went to work to construct an apparatus that would actually catch the sounds of the voice and reproduce them at a distance. Bell, like Morse, was without means to conduct his experiments, but friends came to his aid and furnished him with the necessary money and by 1876 his labors had resulted in making a machine that would carry the human voice; he had invented the telephone. At first the telephone was only a toy and would operate at only short distances, but as improvements were made the distances grew greater and greater until at last one could talk in Boston and be heard in Denver, or talk in New York and be heard in London. The telephone grew rapidly into favor as a means of communication and in a short time it was used more than the telegraph. It is estimated that in the entire world about ten billion conversations are held over the telephone in the course of a single year.



FIG. 13.—PROFESSOR ALEXANDER GRAHAM BELL SPEAKING OVER THE FIRST LONG DISTANCE TELEPHONE BETWEEN NEW YORK AND CHICAGO.

As wonderful as the telephone was it was quickly followed by an invention even more wonderful. Almost as soon as men had thoroughly mastered the art of sending messages by the aid of wires they set about trying to find a way by which messages could be sent long distances without any wires at all. In 1889, Heinrich Hertz, a German scientist, showed that electric waves could be sent out in all directions just as light waves go out in all directions. He also showed how these waves might be produced and how they might be detected or caught as they passed through space. In 1896, William Marconi, an Italian electrician, making use of the facts discovered by Hertz, sent a message a distance of 300 feet without the use of wires. This was the first *wireless telegraph*. Marconi continued his experiments, sending wireless messages between places further and further apart, and by 1911 he was able to signal without cables across the Atlantic Ocean.

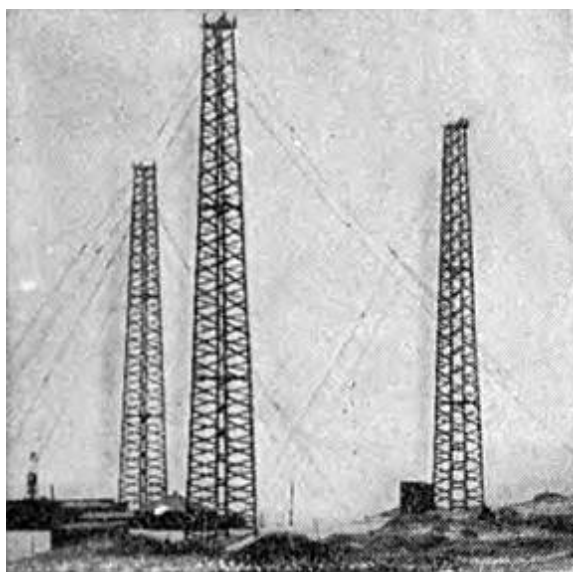


FIG. 14.—A WIRELESS TELEGRAPH STATION.

And now it seems that the wireless telegraph is to be followed by an invention still more wonderful. Men are now working upon a *wireless telephone*. Already it is possible to talk without the aid of wires between places so far apart as Newark and Philadelphia, and many inventors believe that it is only a matter of time when the wireless telephone will be used side by side with the wireless telegraph.

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FOOTNOTES

¹ Where readers are quite young the Foreword had better be postponed until the stories themselves are read.

² Mr. Walter Hough of the National Museum, himself a wizard in the art of fire-making, tells me that a blaze cannot be produced simply by rubbing sticks together. All that can be done by rubbing is to make them glow.

³ A narrow strip of leather.

⁴ The ancient Greeks used a burning-glass or -lens for kindling fire. The lens focused the sun's rays upon a substance that would burn easily and set it afire. The burning-glass was not connected in any way with the development of the match.

⁵ Several of the illustrations in this chapter are reproduced through the courtesy of the Boston Stove Co.

⁶ Hold the end of a dry towel in a basin of water and watch the water rise in the towel. It rises by capillary action.

⁷ Light a short piece of candle and place it in a tumbler, and cover the top of the tumbler. The experiment teaches that a flame must have a constant supply of fresh air and will go out if the air is shut off.

⁸ J. R. Smith, "The Story of Iron and Steel," p. 3.

⁹ From "Five Black Arts," p. 311.

¹⁰ The old forge continued to be used by the side of the blast furnace for centuries, and of course where it was used it was still called a forge. Thus we are told that in Maryland in 1761, there were eight furnaces and ten forges. It is said that as late as twenty-five years ago in certain parts of the Appalachian regions the American mountaineer still worked the little primitive forge to make his iron.

¹¹ It was given the name of *pig* iron because when the molten metal ran into the impressions made for it upon the sanded floor and cooled, it assumed a shape resembling a family of little pigs.

[12](#) Daniel Webster was another great statesman who turned his attention to the making of plows. He planned a plow ([Fig. 11](#)) and had it made in his workshop on his farm at Marshfield. When the plow was ready for use, Webster himself was the first man to take hold of the handles and try it. The plow worked well and the great man is said to have been as much delighted with his achievement as he was with any of his triumphs in public life at Washington.

[13](#) To winnow grain is to separate it from the chaff by a fanning process.

[14](#) Matthew xxiv, 41. In ancient times nearly all the grinding was done by women.

[15](#) Ceres was the goddess of grain.

[16](#) In the thirteenth century wind-power began to be used for turning mills, and in some countries windmills were as common as water-mills.

[17](#) About the year 1700 elliptical springs were invented, but they did not find their way into general use until more than a hundred years later.

[18](#) A spirited account of life on a Roman galley is found in Wallace's "Ben Hur."

[19](#) Wood, "Curiosities of Clocks and Watches."

[20](#) The illustration is taken from Keary's "Dawn of History."

[21](#) In the payment of the postage no stamps were as yet used. Indeed the postage stamp is a late invention. Postage stamps were not used in England until the year 1840, while in the United States they were not regularly used until 1847.

[22](#) In 1840, the English government following the recommendations of Sir Rowland Hill, adopted throughout the United Kingdom a uniform rate of one penny for letters not exceeding half an ounce in weight, and after this cheap postage became the rule in all countries.

[23](#) The verb telegraph means to write at a distance afar off.

[24](#) As there are only 24 letters in the Greek alphabet, the last group was one letter short, but this did not interfere with the working of the system.

[25](#) Just as the word telegraph means to "write afar off," so the word telephone means to "sound afar off."

Transcriber's Notes:

Hyphenation, punctuation, and spelling standardized when a predominant choice was available; otherwise unchanged.

Illustrations have been repositioned closer to the text that refers to them, and in many cases, their relative sizes have been changed. On some display devices, clicking a blue-bordered illustration will show a larger, higher-resolution version in the current window.

On some display devices, page numbers of the original book are shown as [123] in the right margin. The ones for some full-page illustrations have been omitted.

Page [81](#): illustration captioned "[FIG. 11.—DANIEL WEBSTER'S PLOW.](#)" is referenced in footnote [12](#).

Page [147](#): text apparently omitted after "of one piece"

Page [198](#): reference to "Fig. 1" is incorrect.

Index entry for "[Iron, history of, 41-63.](#)" probably should be "41-53."

The referenced terms on pages 81, 147, and 198 are underlined in gray dots. On devices that support it, holding the mouse cursor over those terms will show these explanations.

"B. C." and "A. D." were split across two lines of the original text on pages [55](#) and [86](#). Since they normally were printed in small-caps, that convention was applied to these two instances as well.

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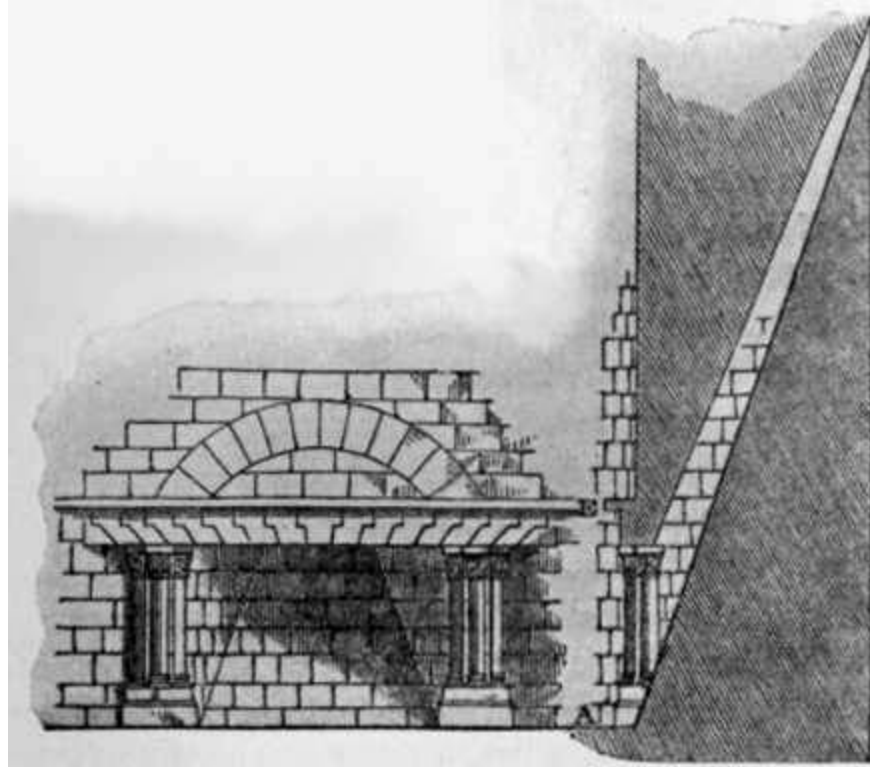
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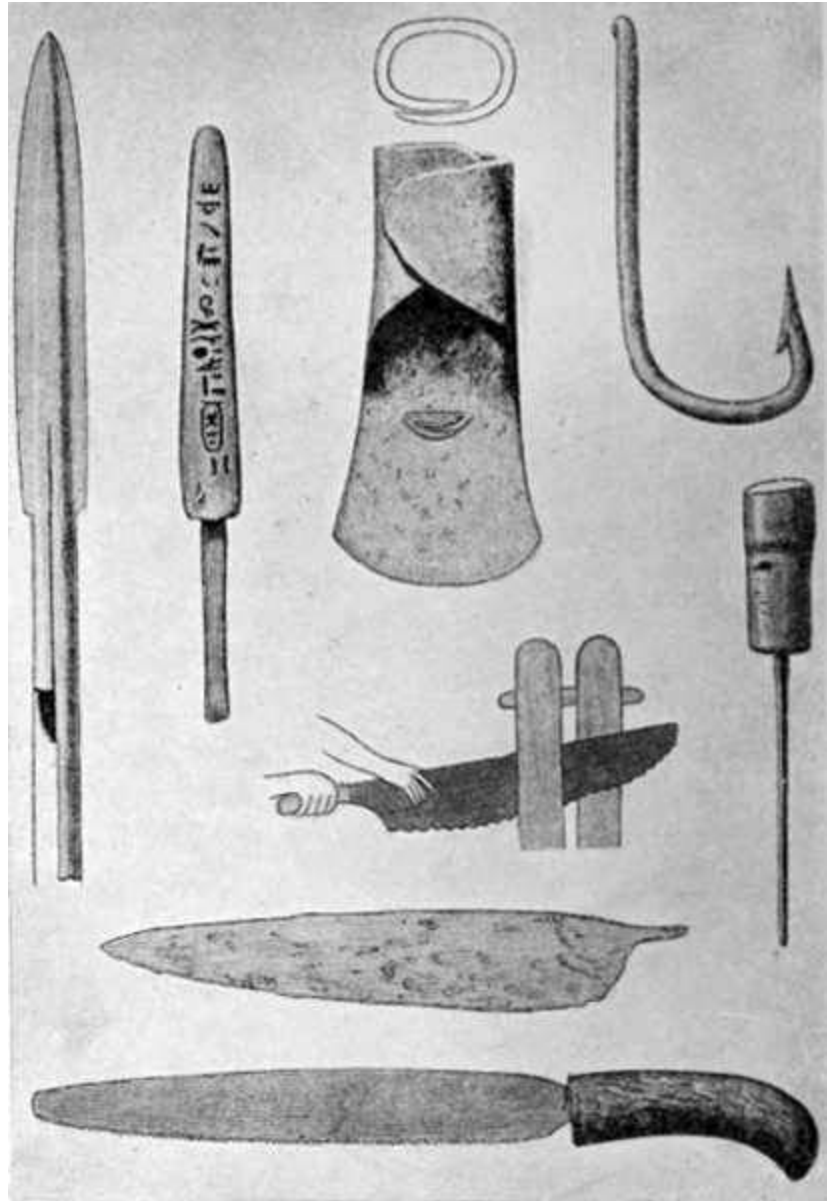
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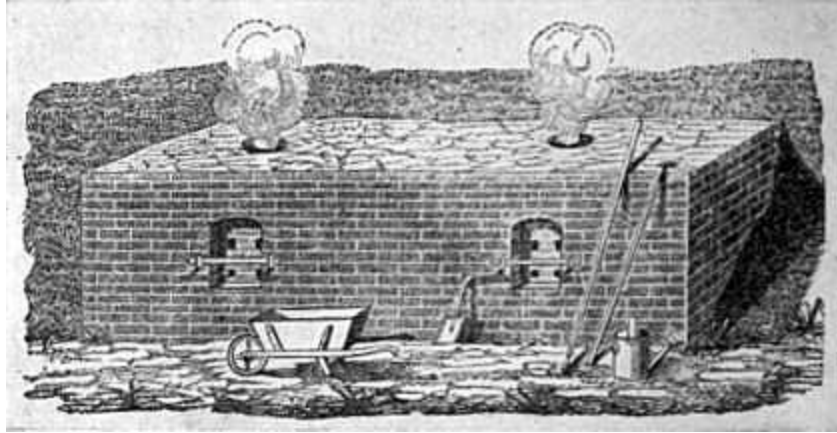
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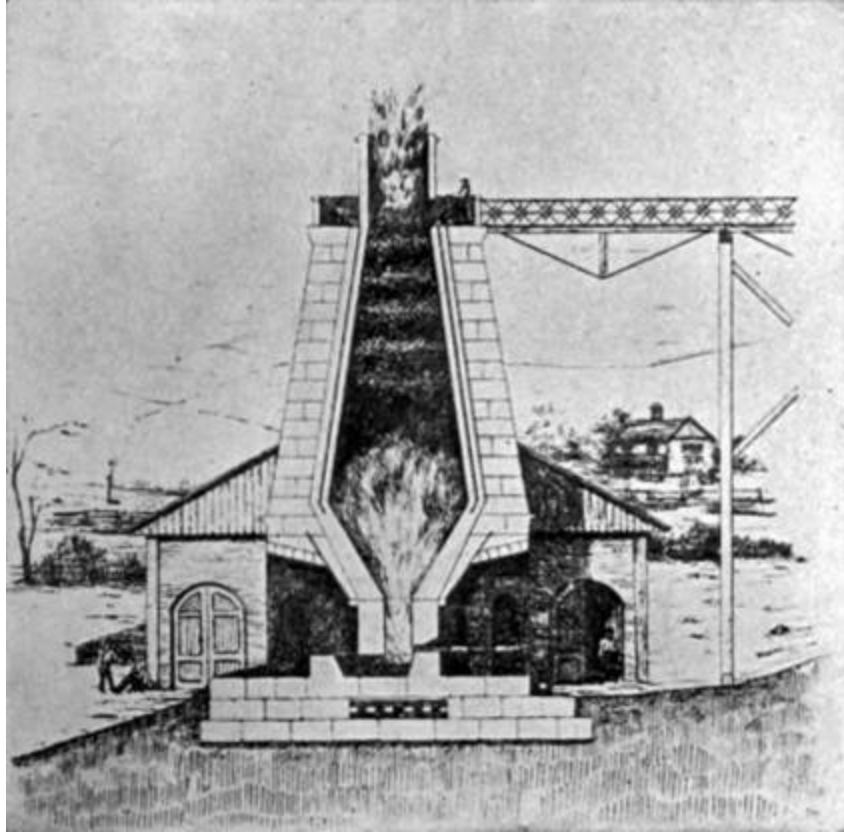
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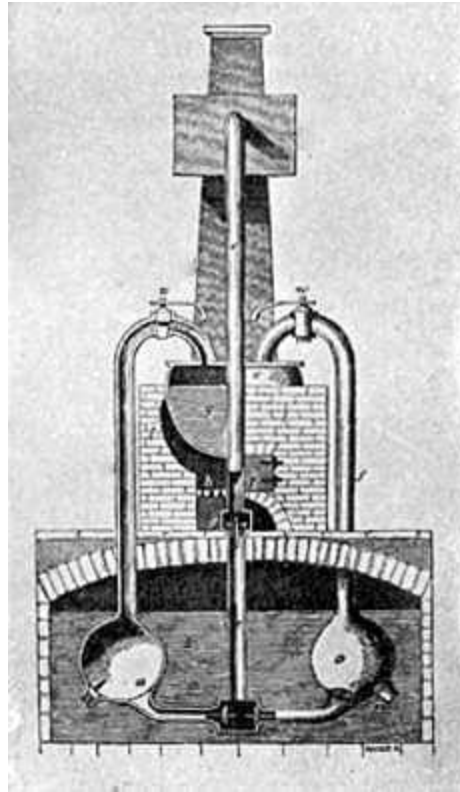
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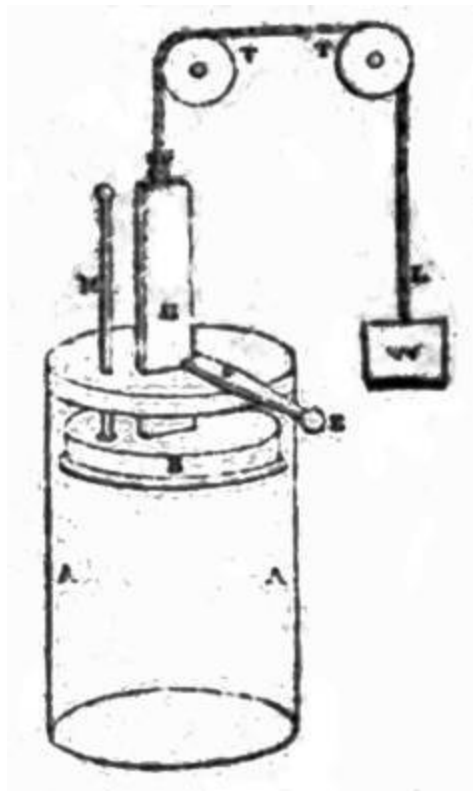
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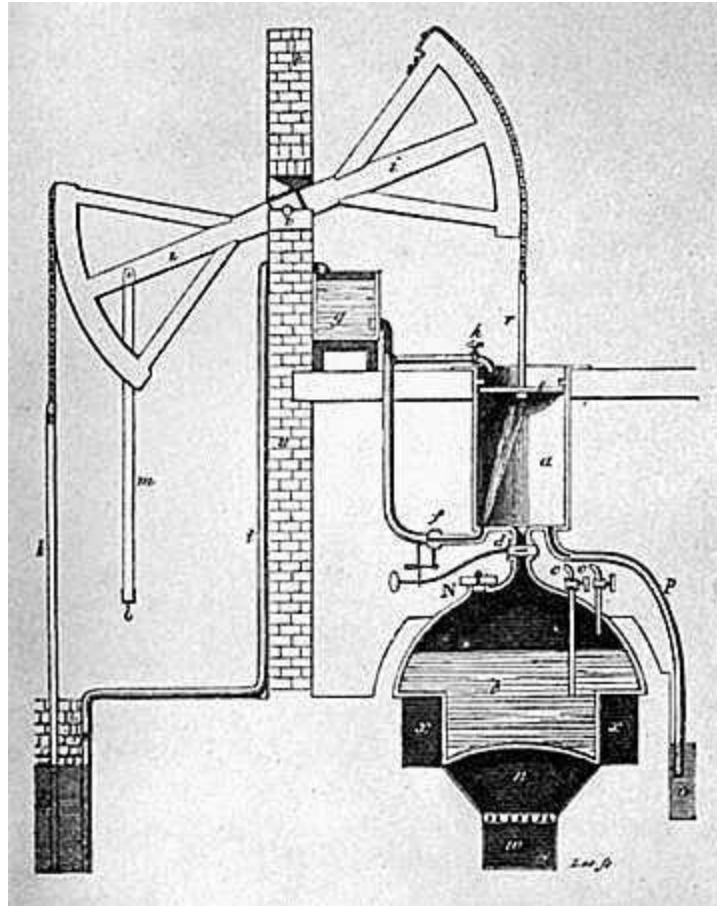
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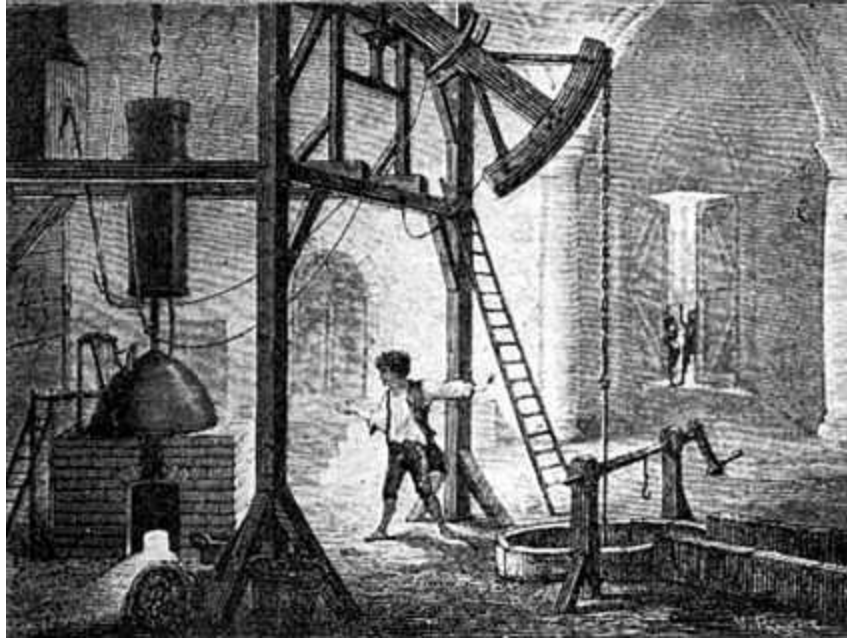
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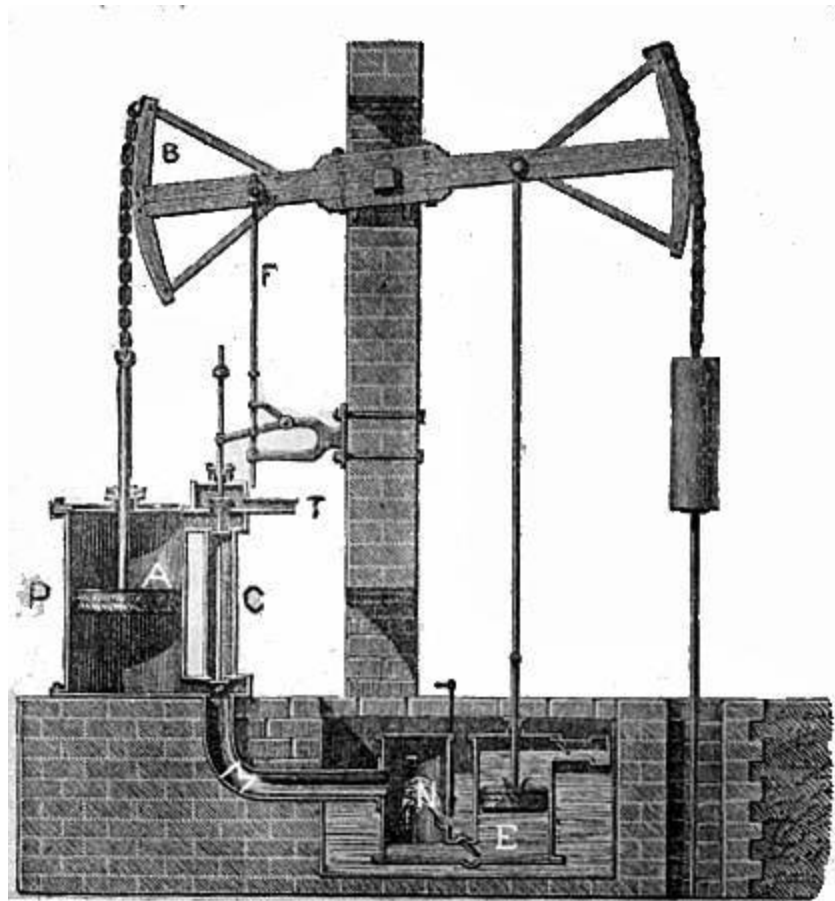
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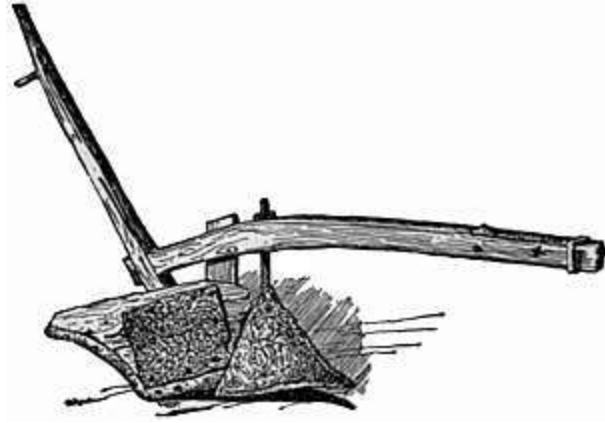
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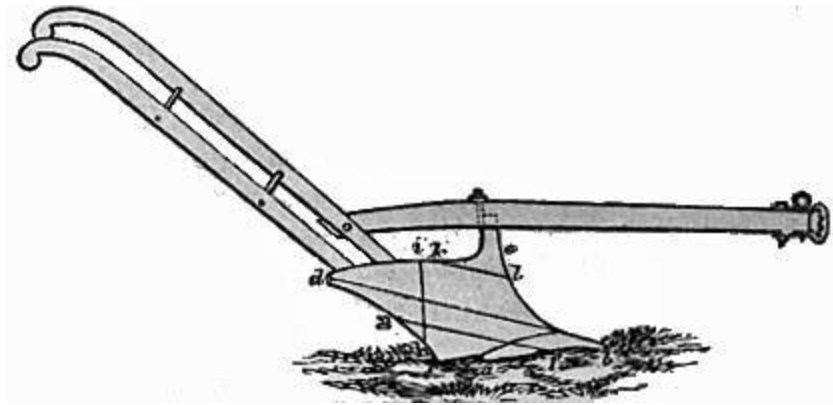
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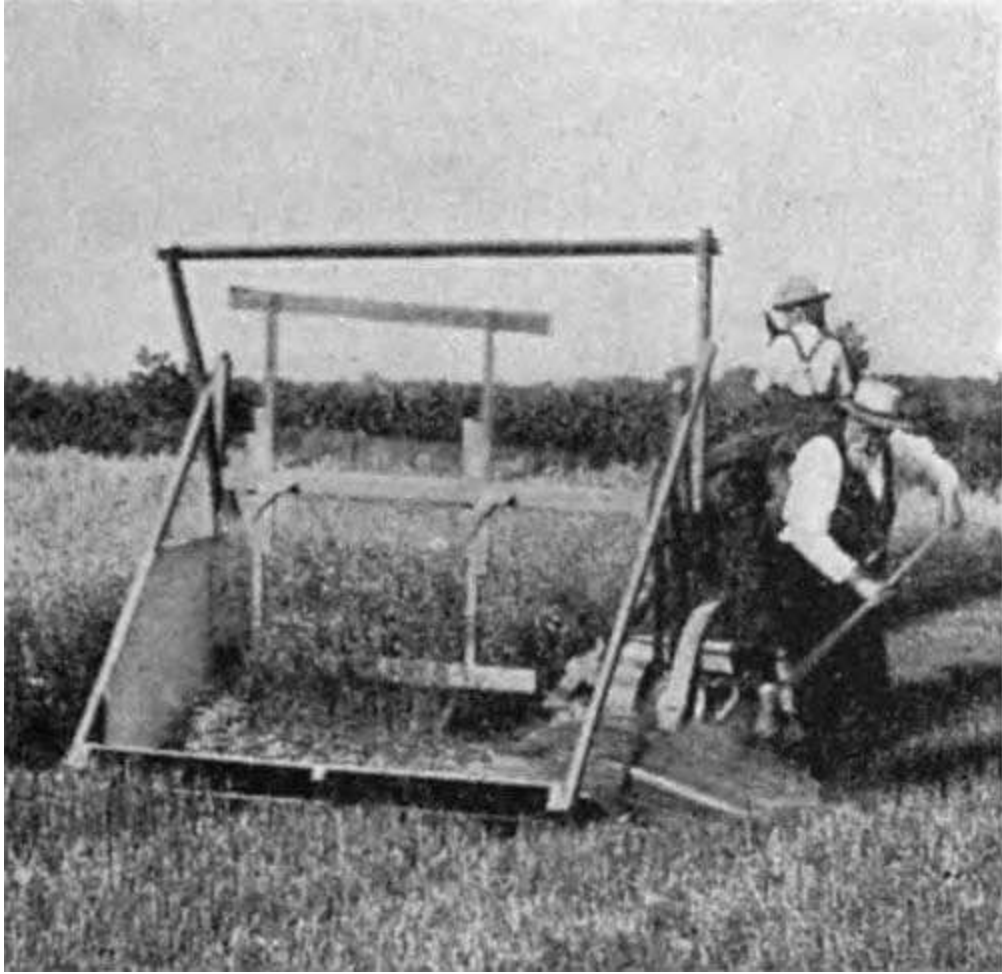
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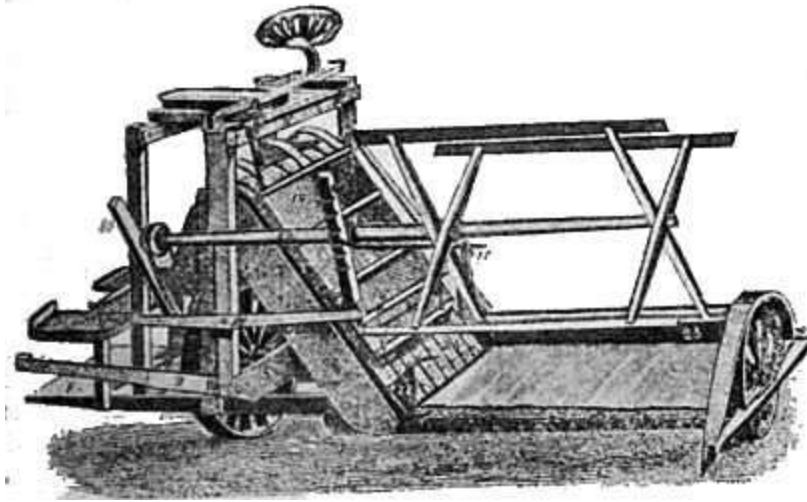
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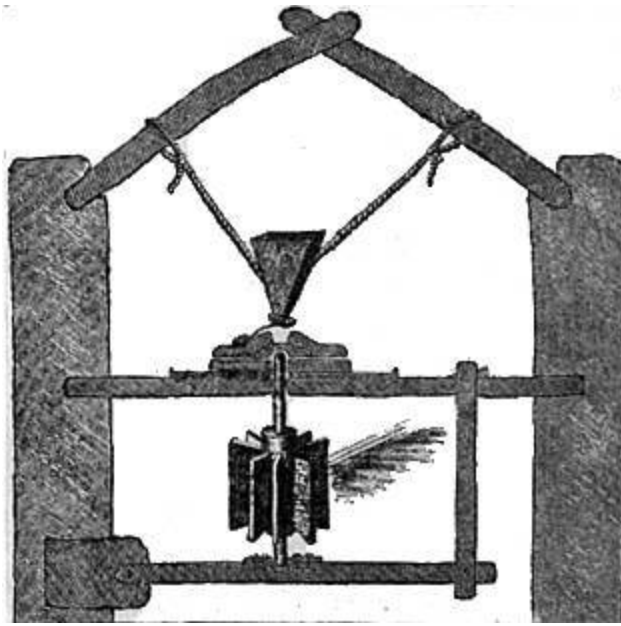
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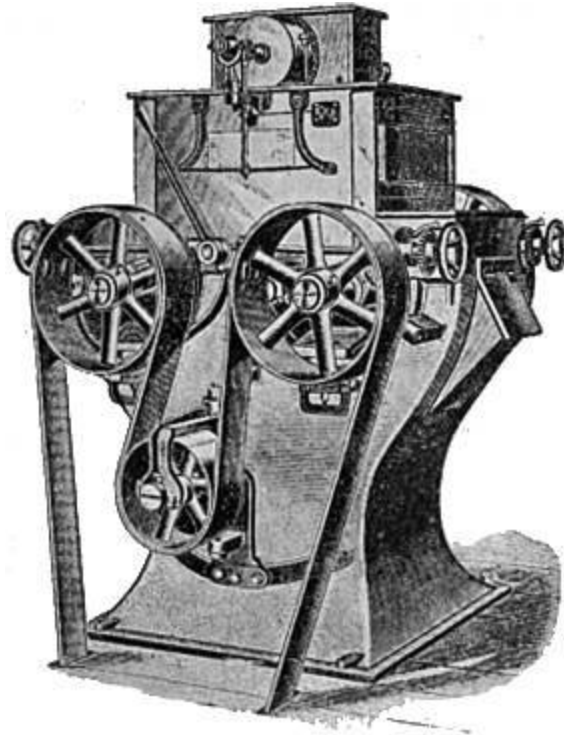
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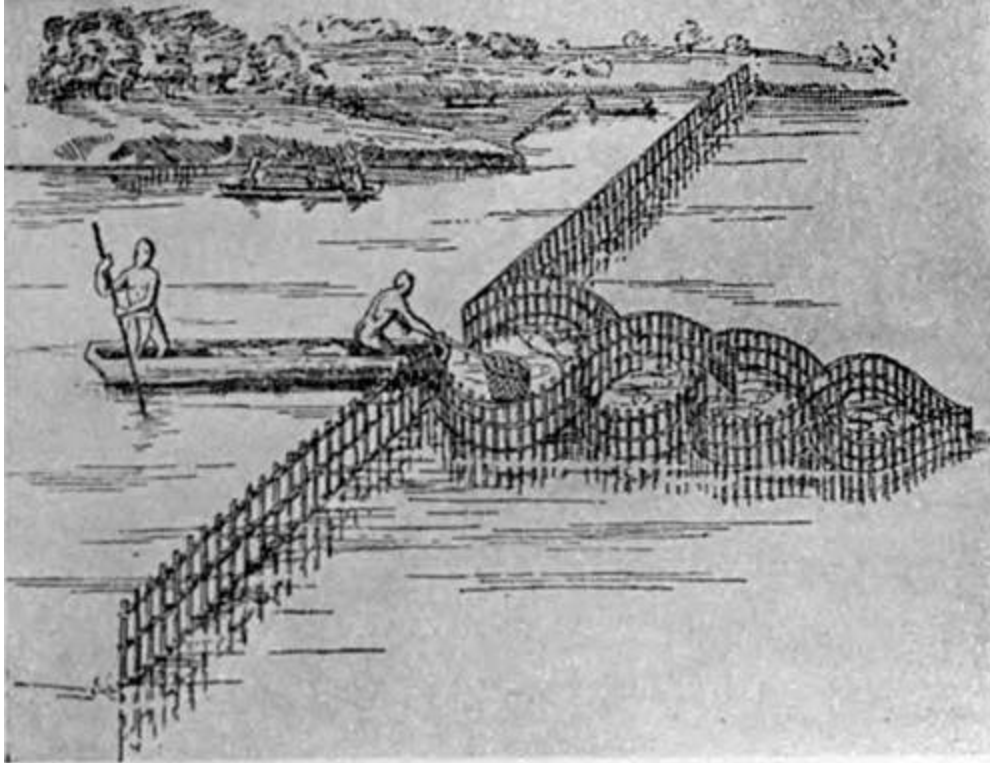
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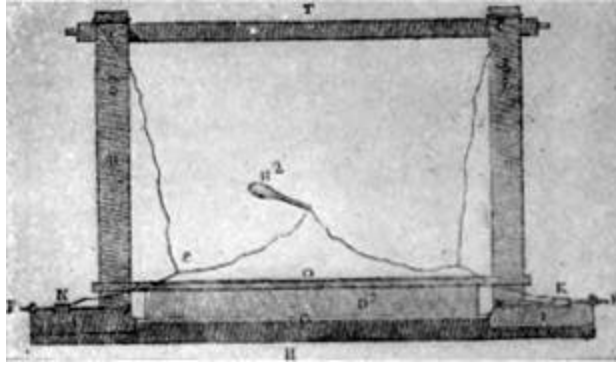
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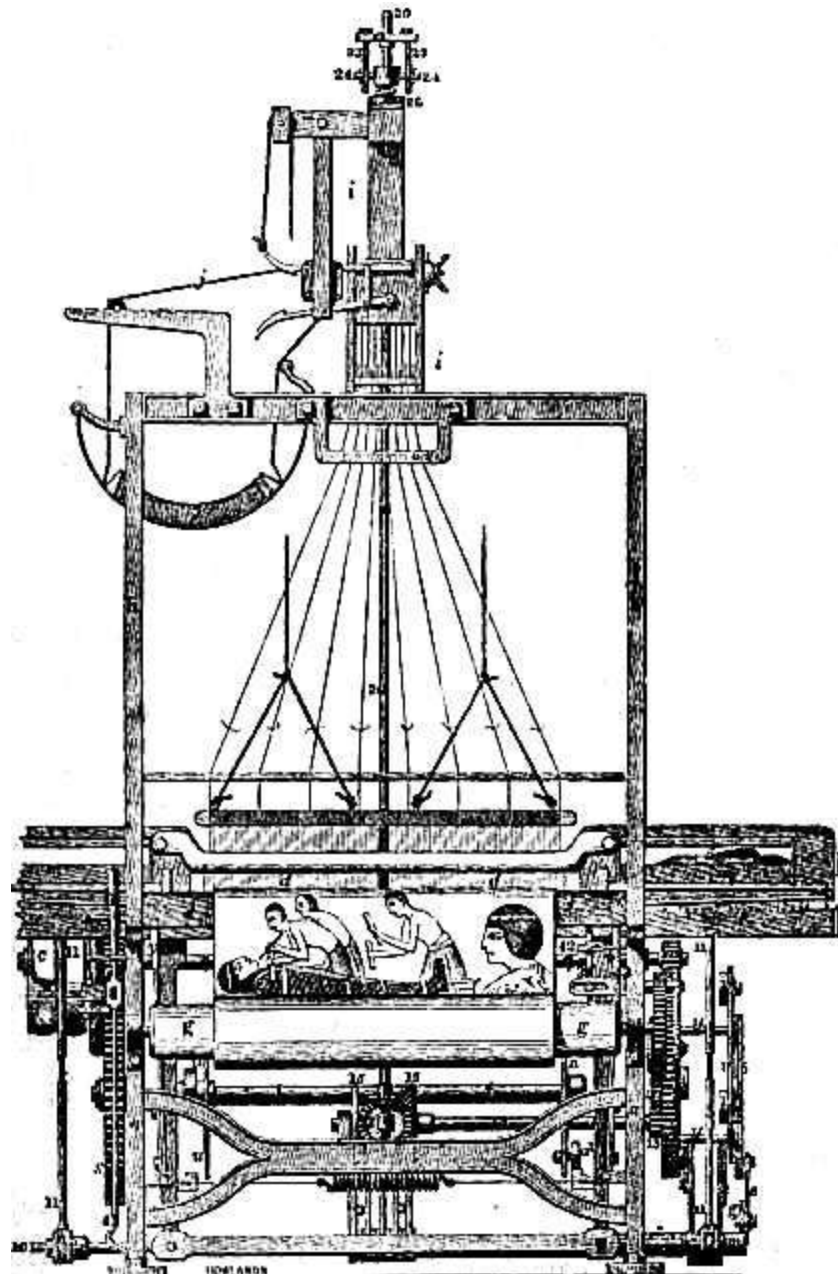
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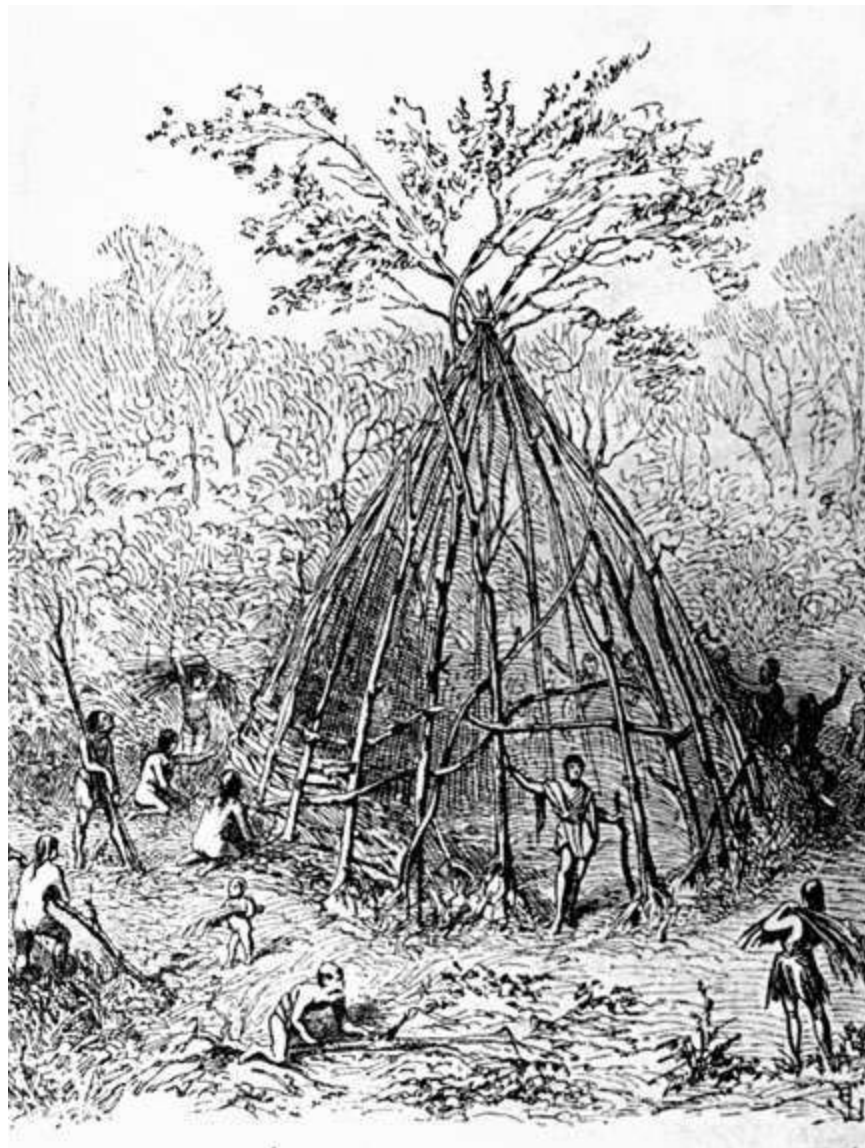
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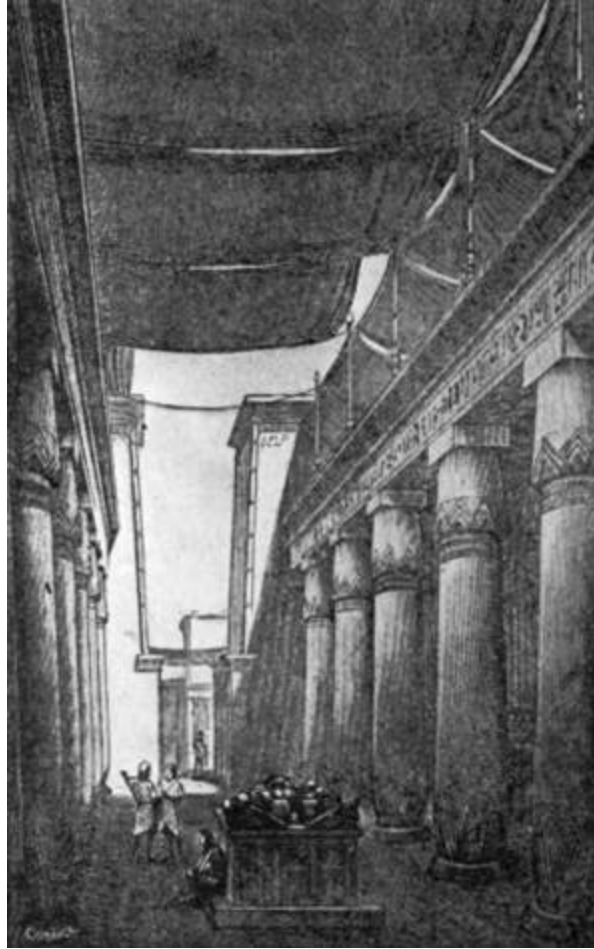
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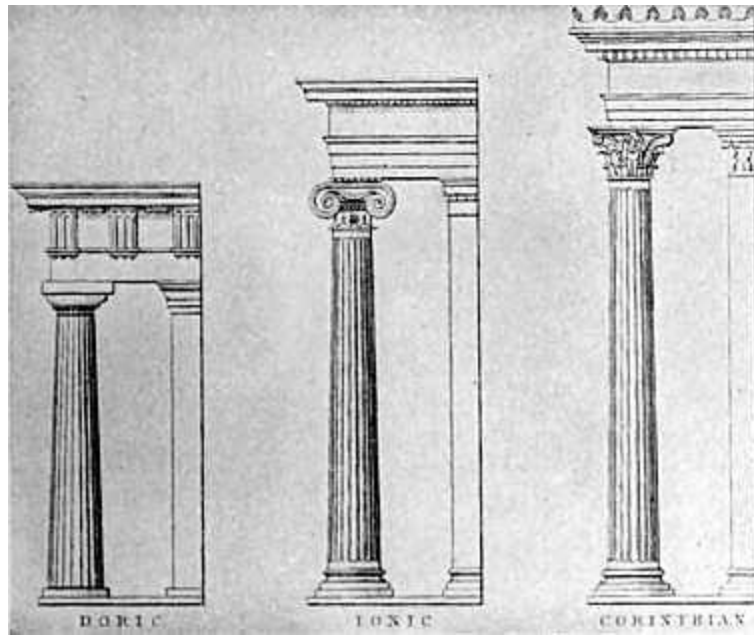
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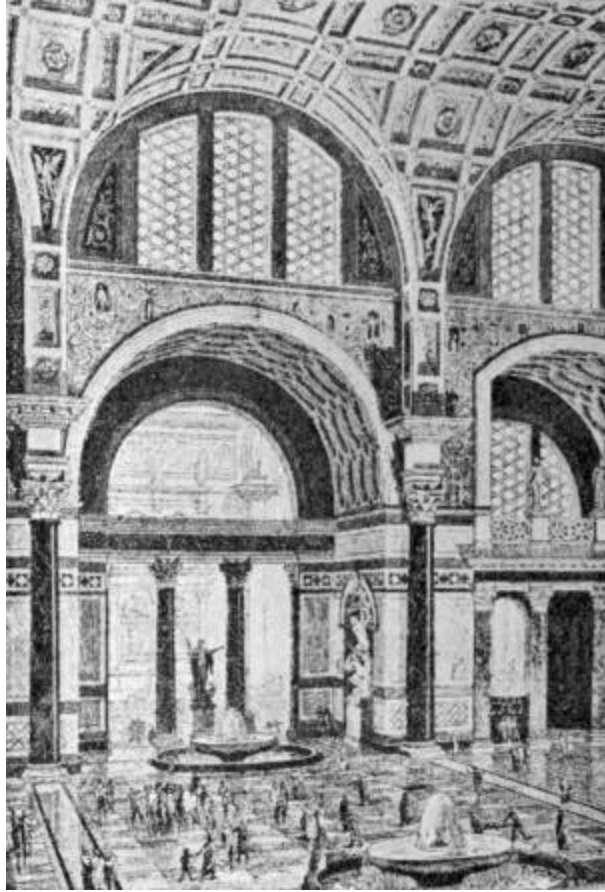
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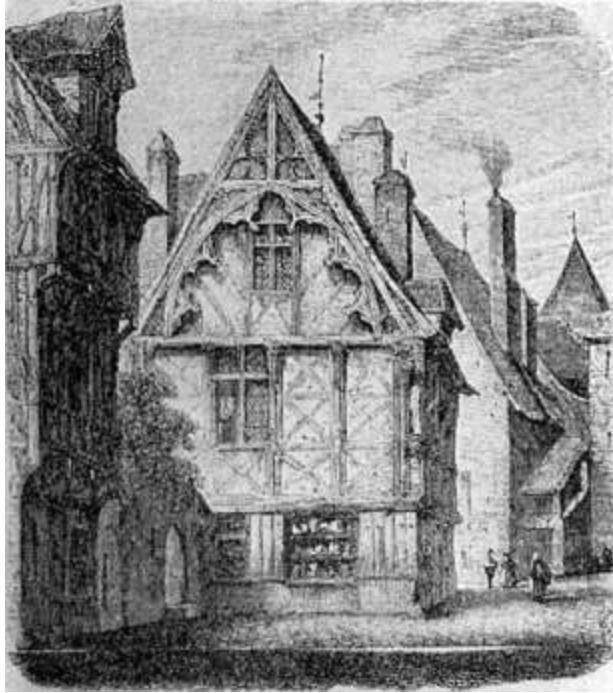
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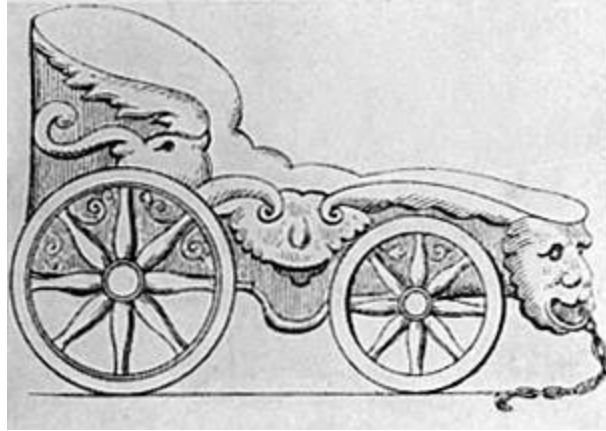
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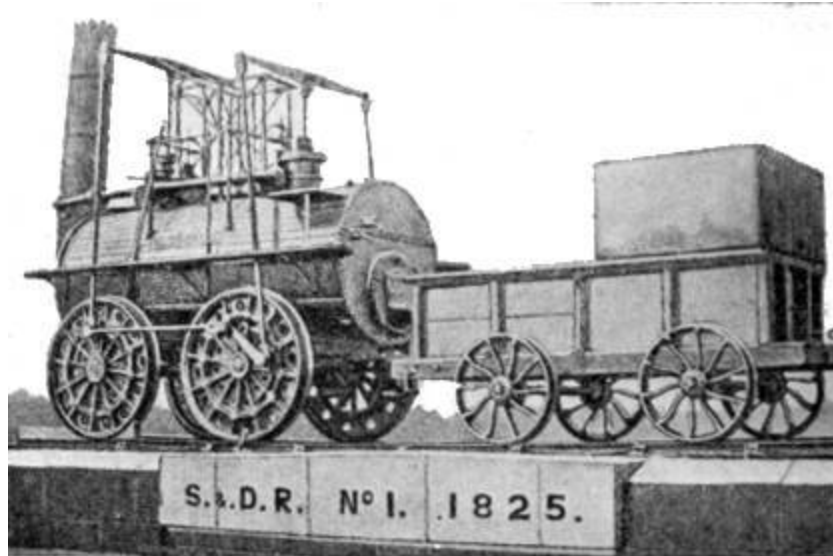
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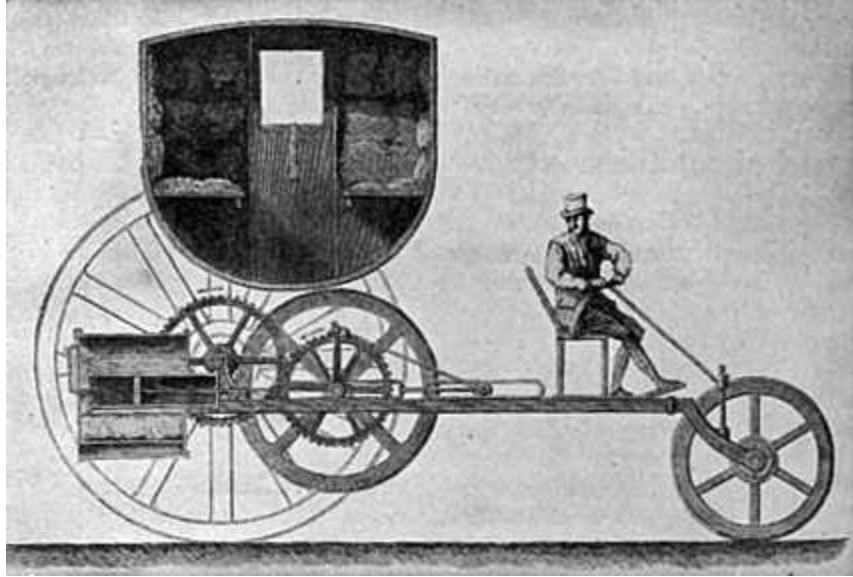
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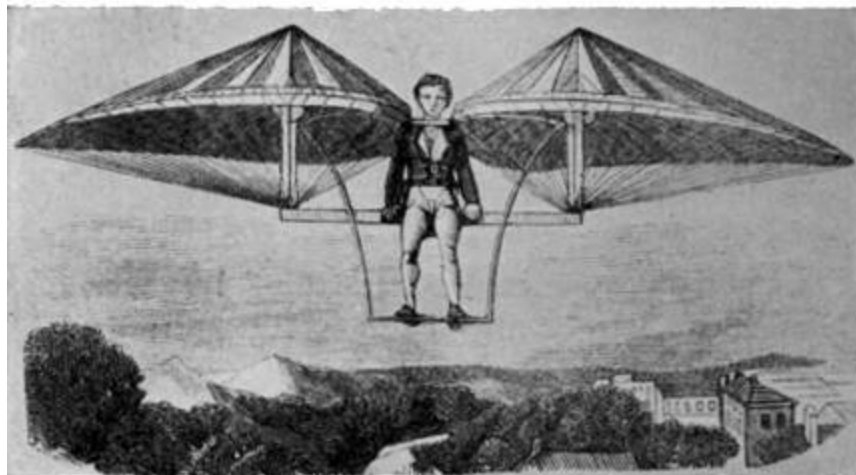
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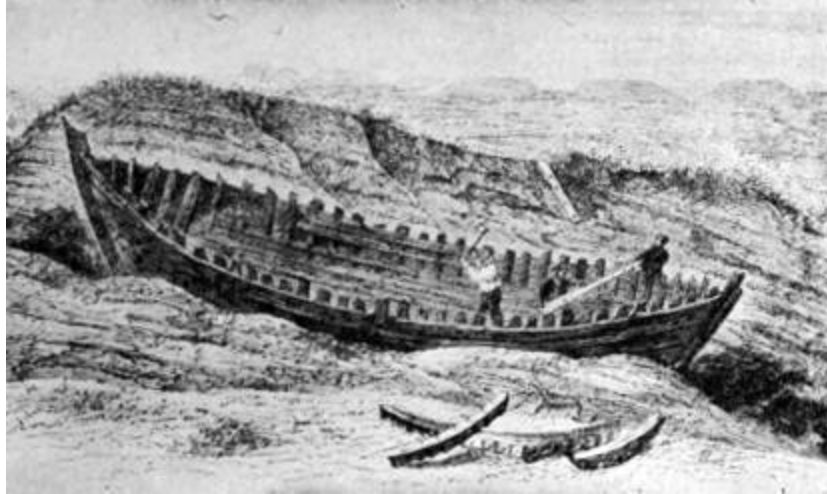
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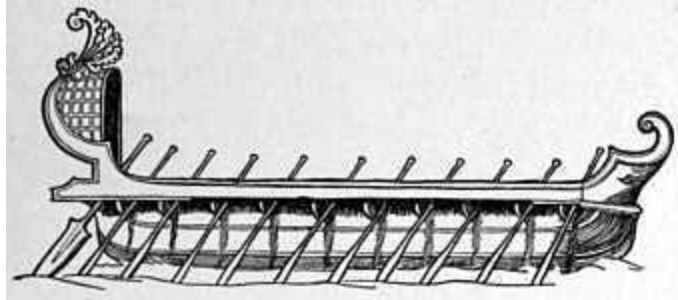
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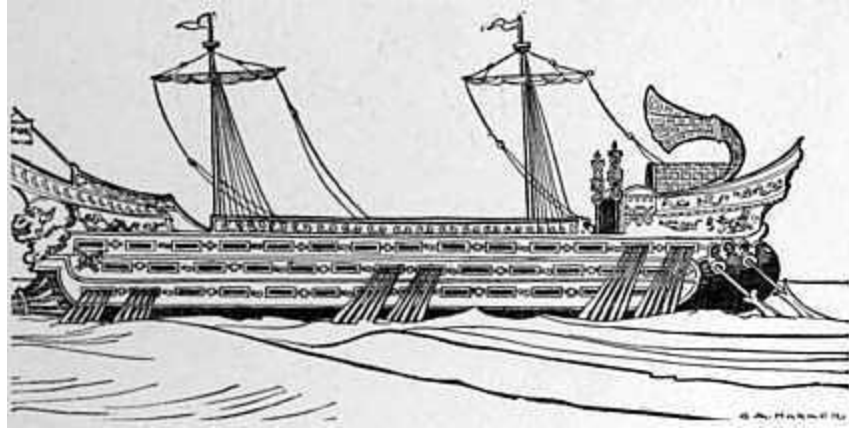
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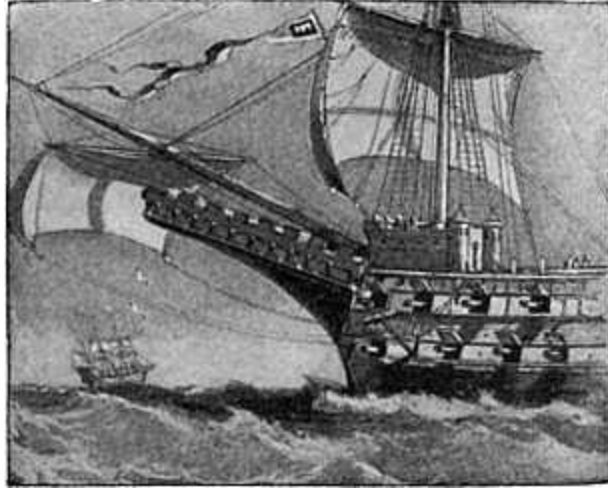
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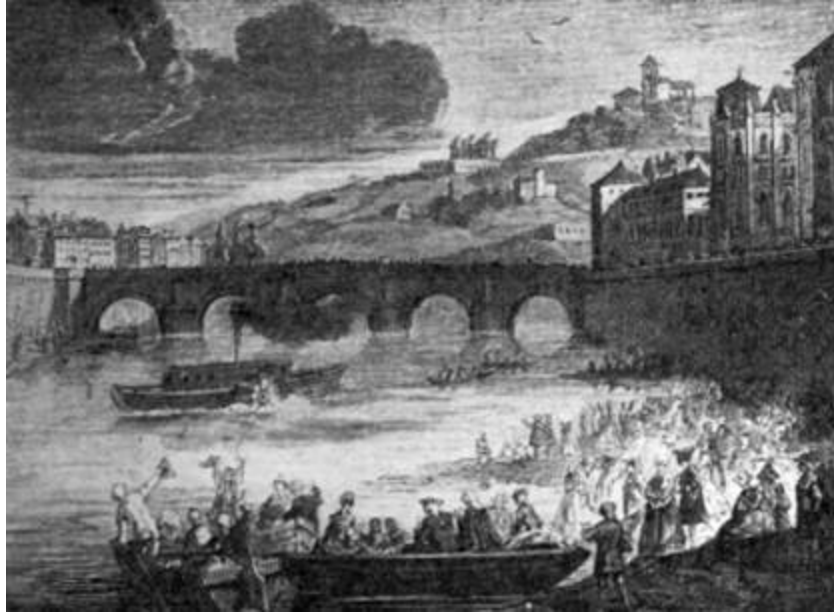
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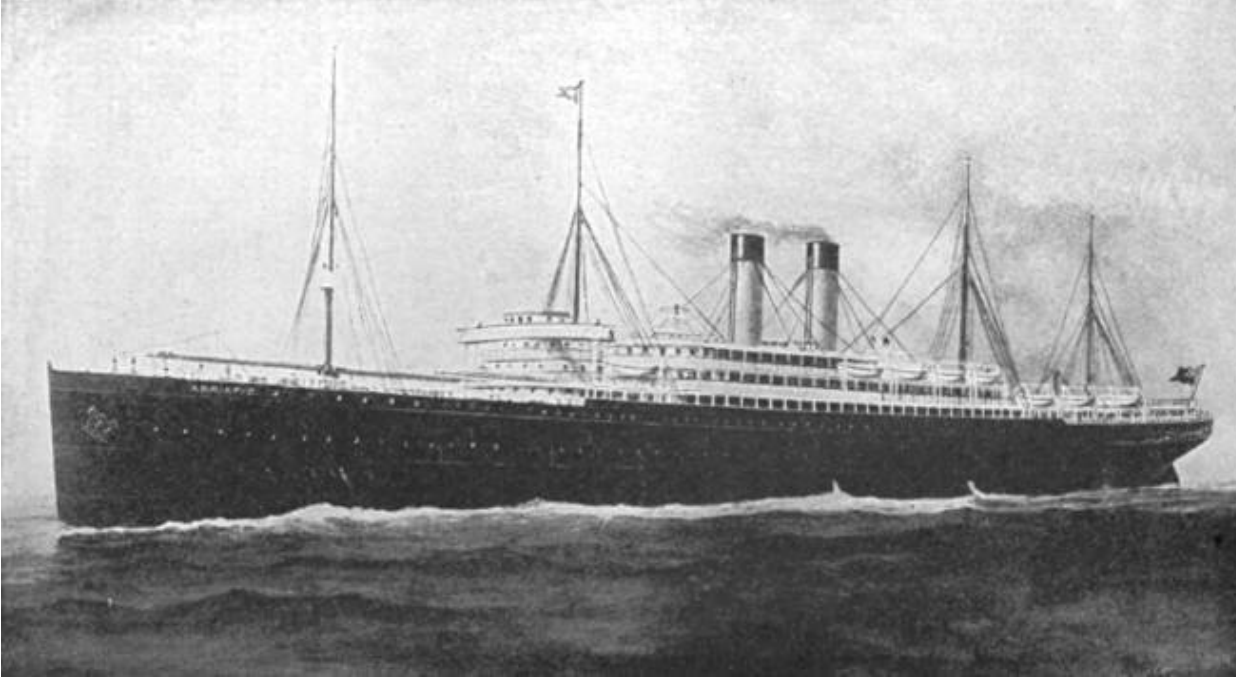
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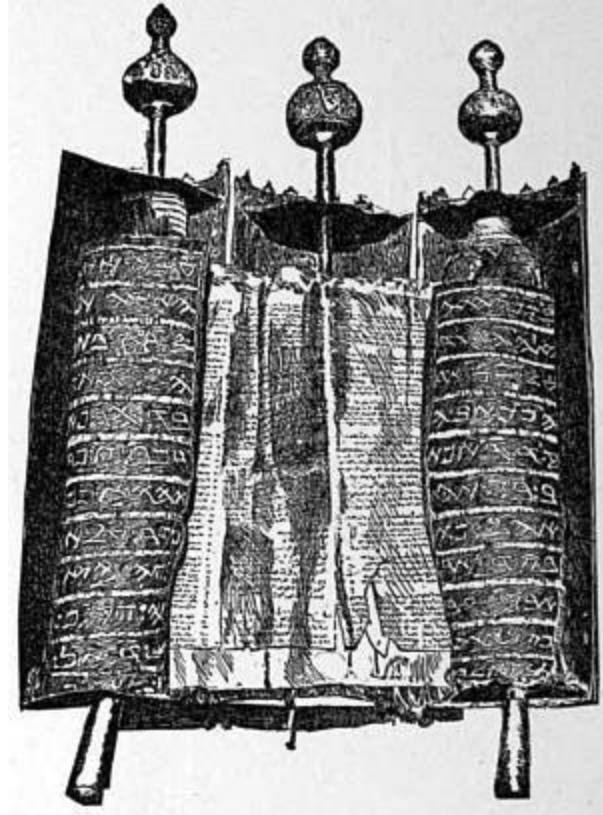
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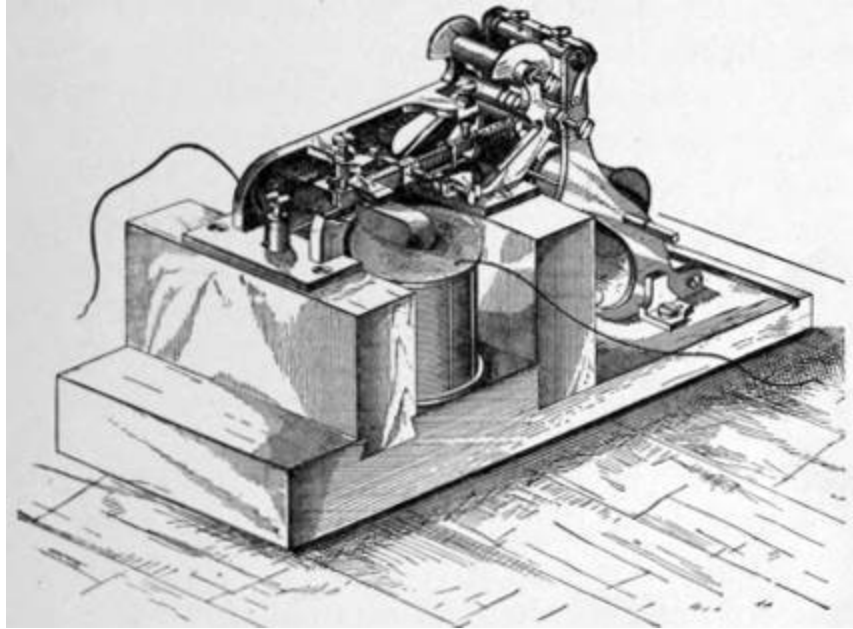
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